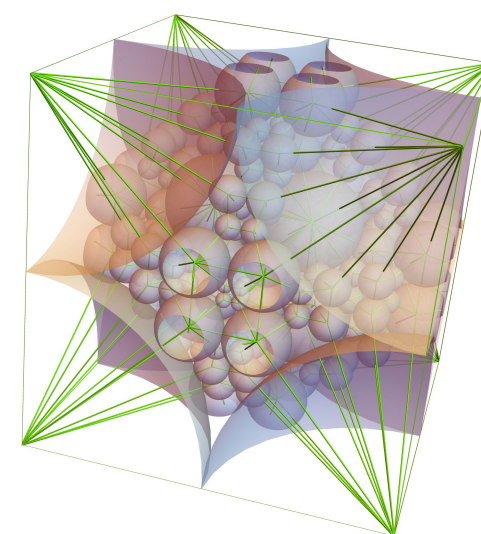
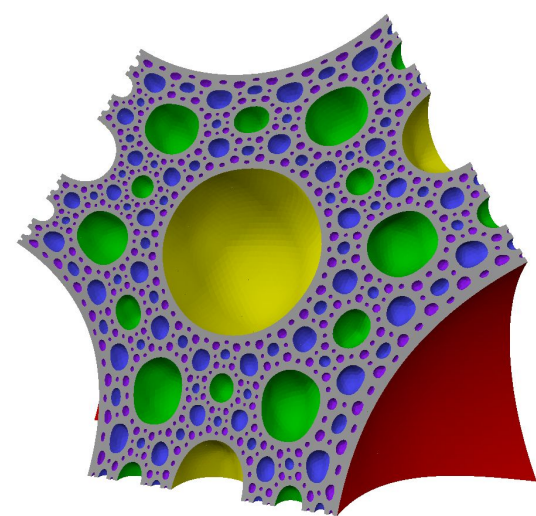


Représentation des empilements polytopaux de sphères avec le modèle BC-IFS



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Boris Bordeaux¹, Iván Rasskin², Christian Gentil¹, Jorge Luis Ramírez Alfonsín³, Lionel Garnier¹

¹*LIB – Université de Bourgogne*
²*AZT – Université Technique de Graz*
³*IMAG – Université de Montpellier*

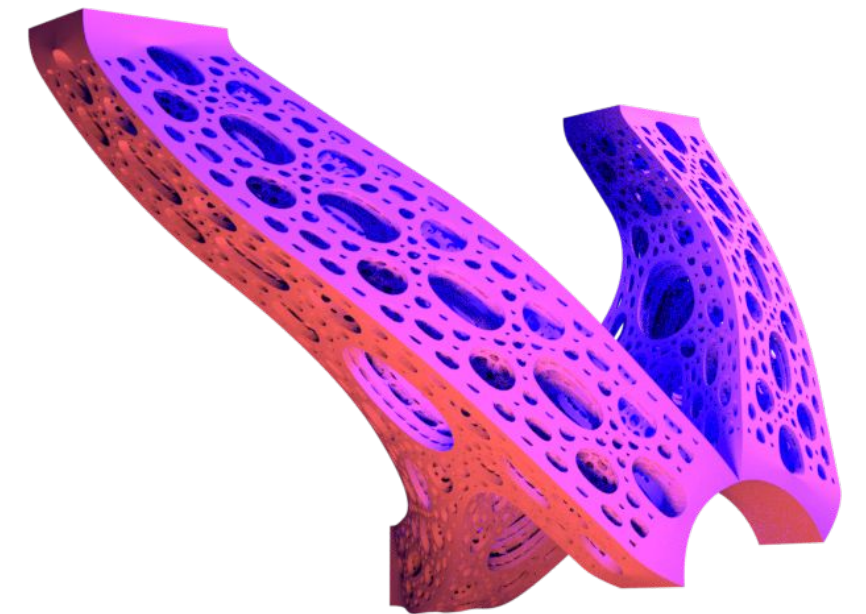
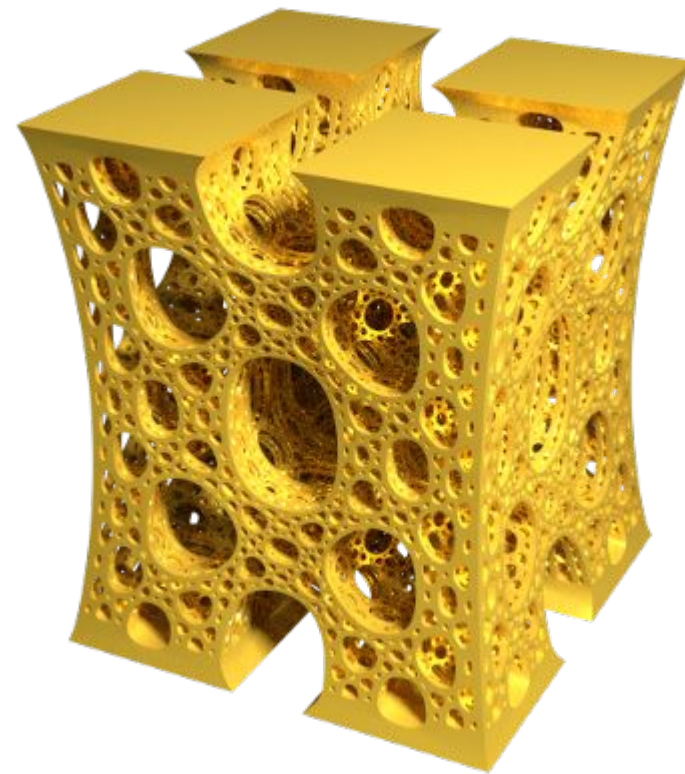
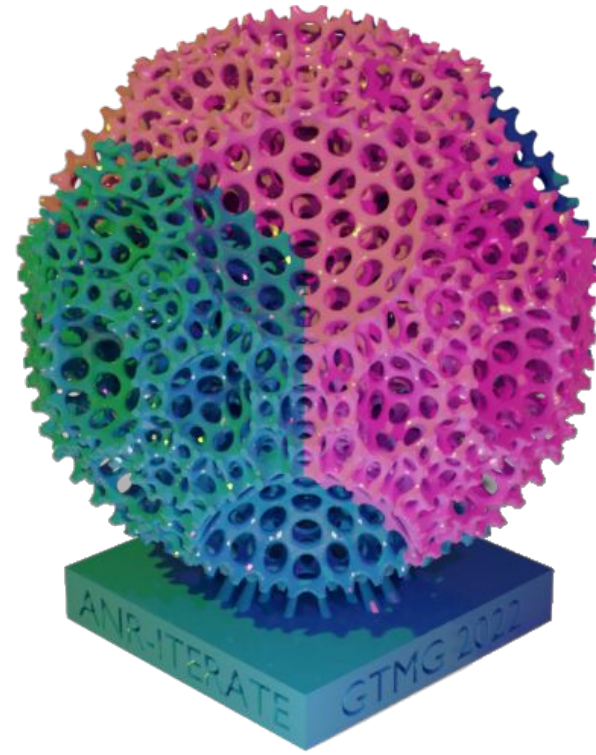
Projet ANR ITERATE – ANR-20-CE46-0011 (2021-2025)

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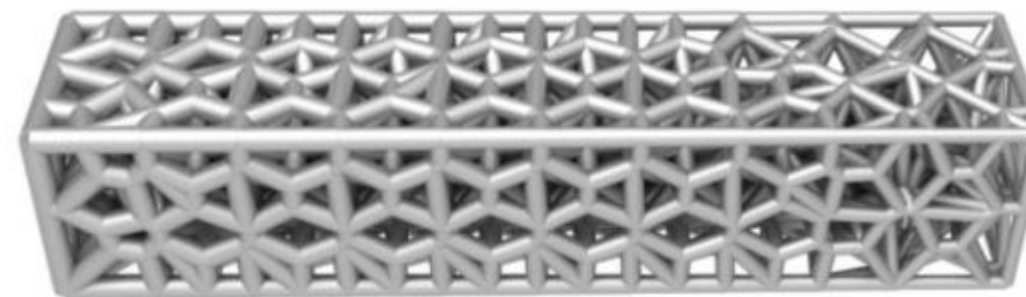
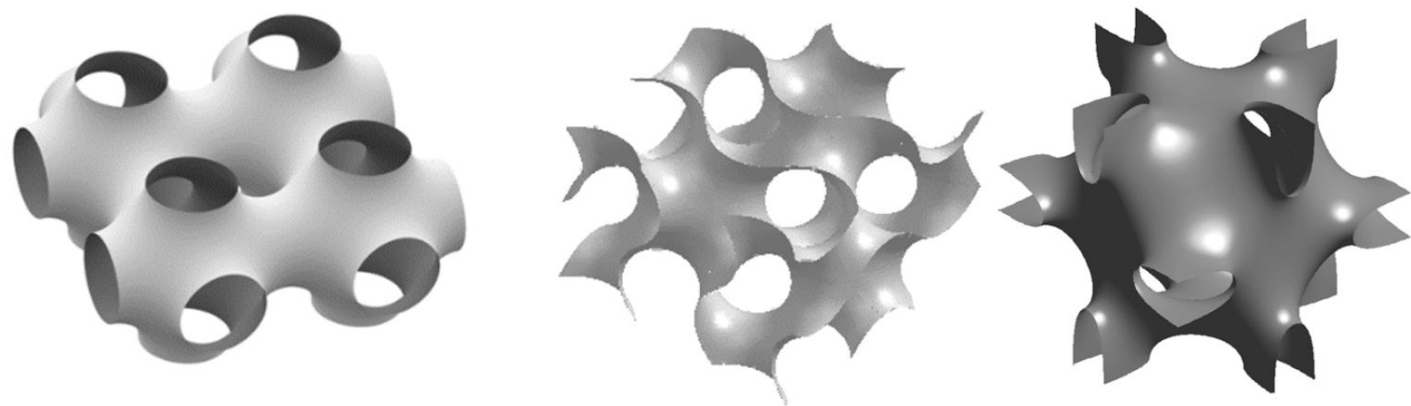
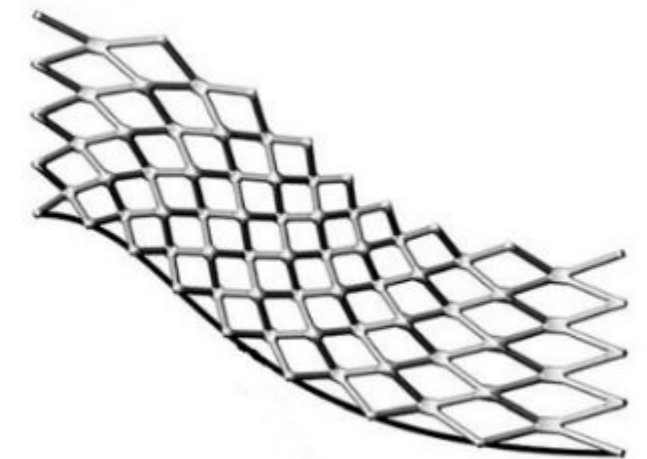
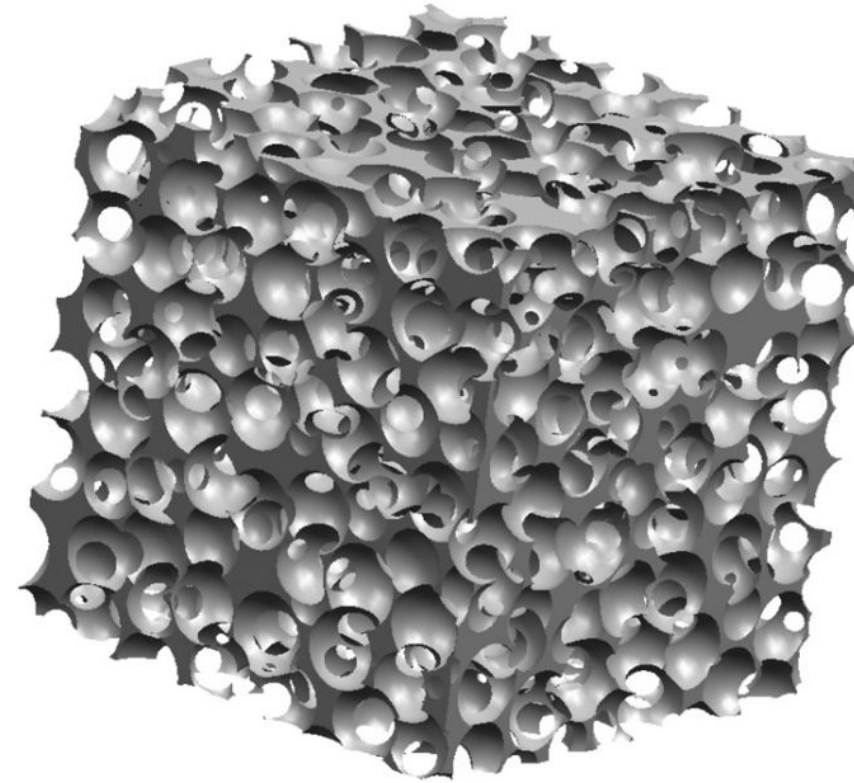
27 Octobre 2023, JCGéo - IHP Paris

COLLÈGE DOCTORAL
VOTRE DOCTORAT
EN BOURGOGNE-
FRANCHE-COMTÉ

UBFC
UNIVERSITÉ
BOURGOGNE FRANCHE-COMTÉ



- Mousses
 - Aléatoire
- Structures lattices
 - Topologie régulière
 - Variation de géométrie
- Surfaces minimales
 - Triplement périodique
 - Géométrie contrainte



- Multi échelle
- Déterministe
- Répond à plusieurs propriétés physiques
 - Absorption mécanique
 - Transferts thermiques
 - Isolation acoustique
- Généralise les structures lattices



- Modèle BC-IFS
- Empilements polytopaux de sphères
- Représentation avec le modèle BC-IFS

Le modèle BC-IFS



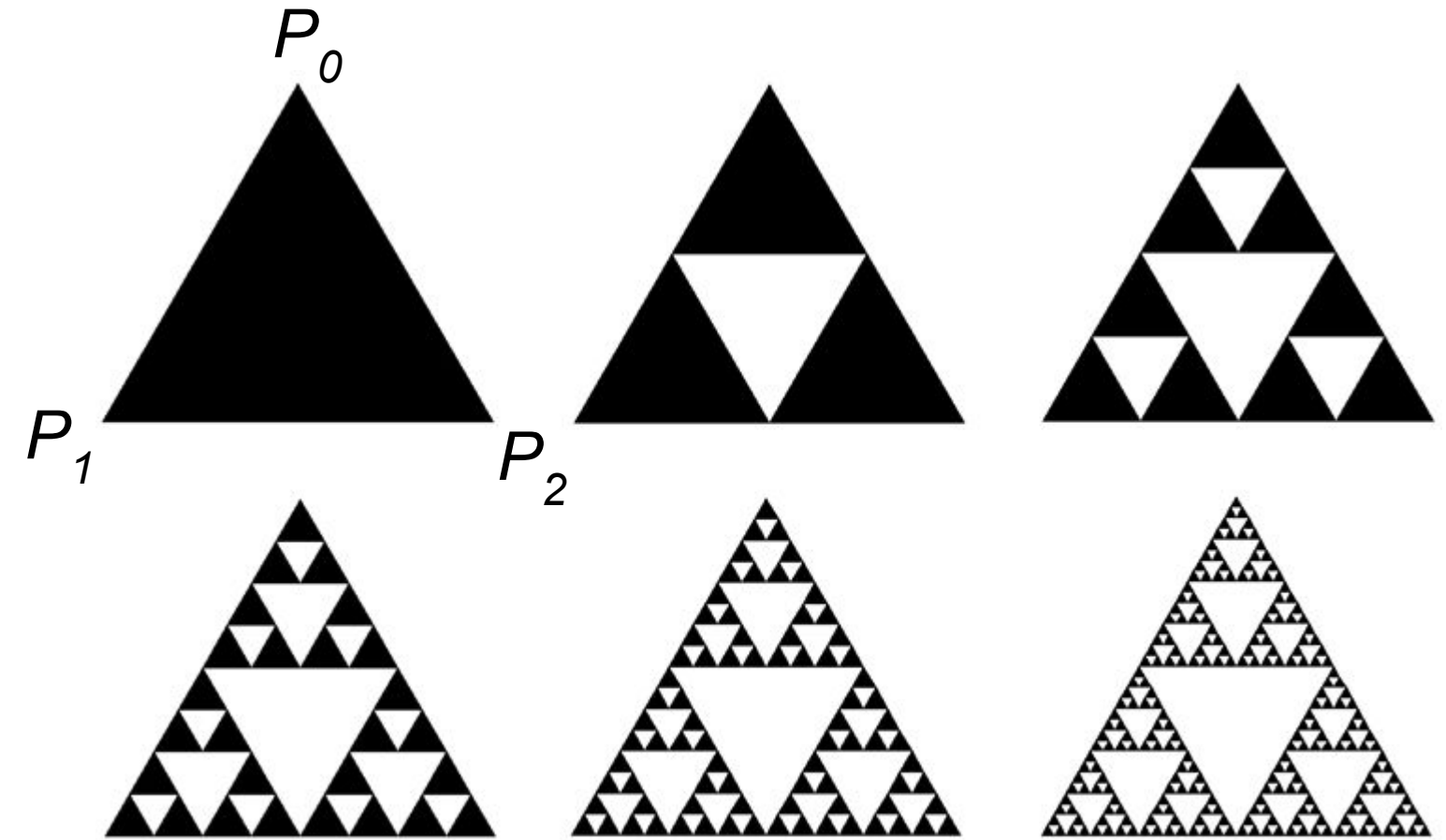
■ Iterated Functions System (IFS)

- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$



$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$

■ Iterated Functions System (IFS)

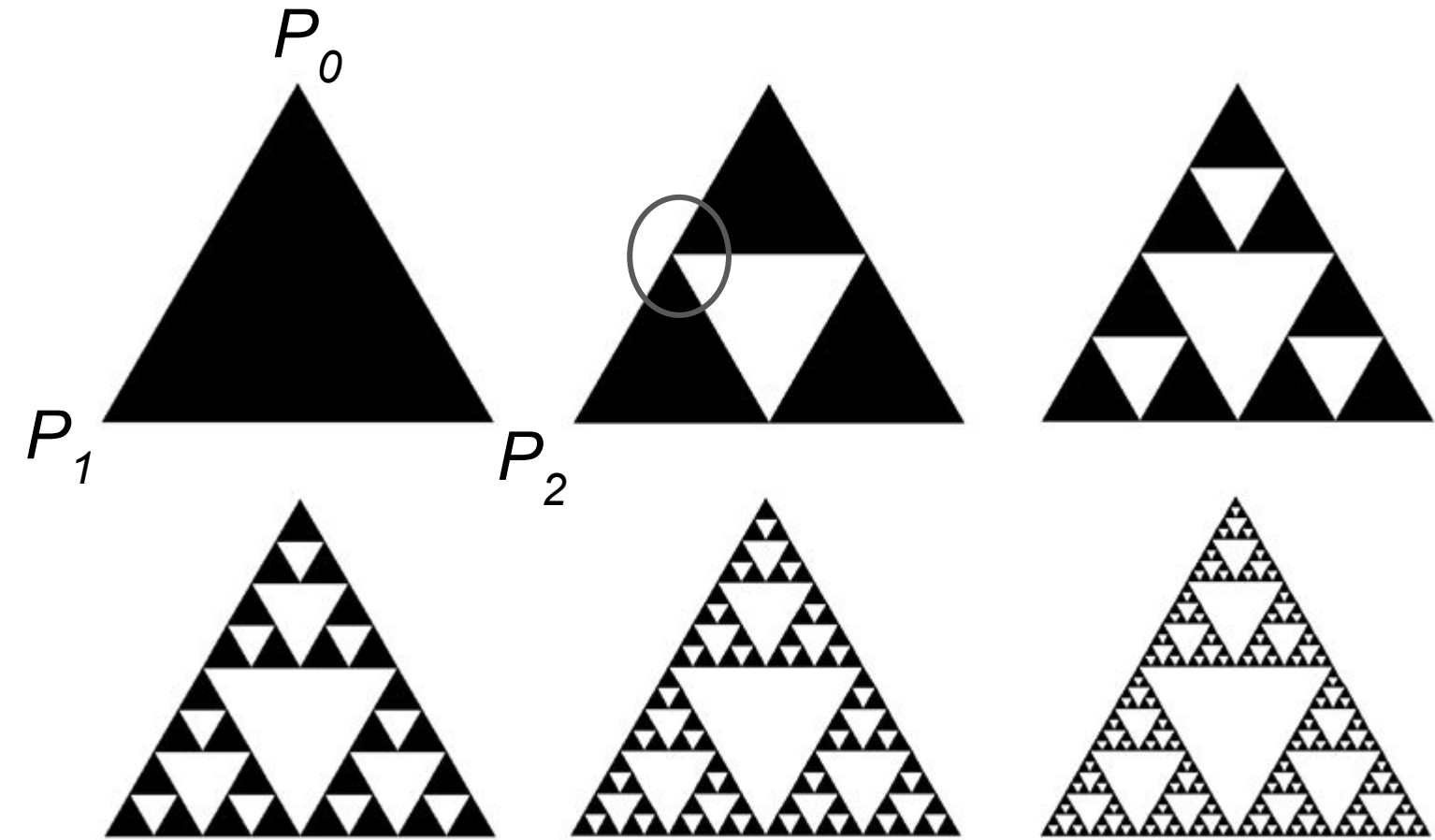
- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \boxed{\frac{1}{2}} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

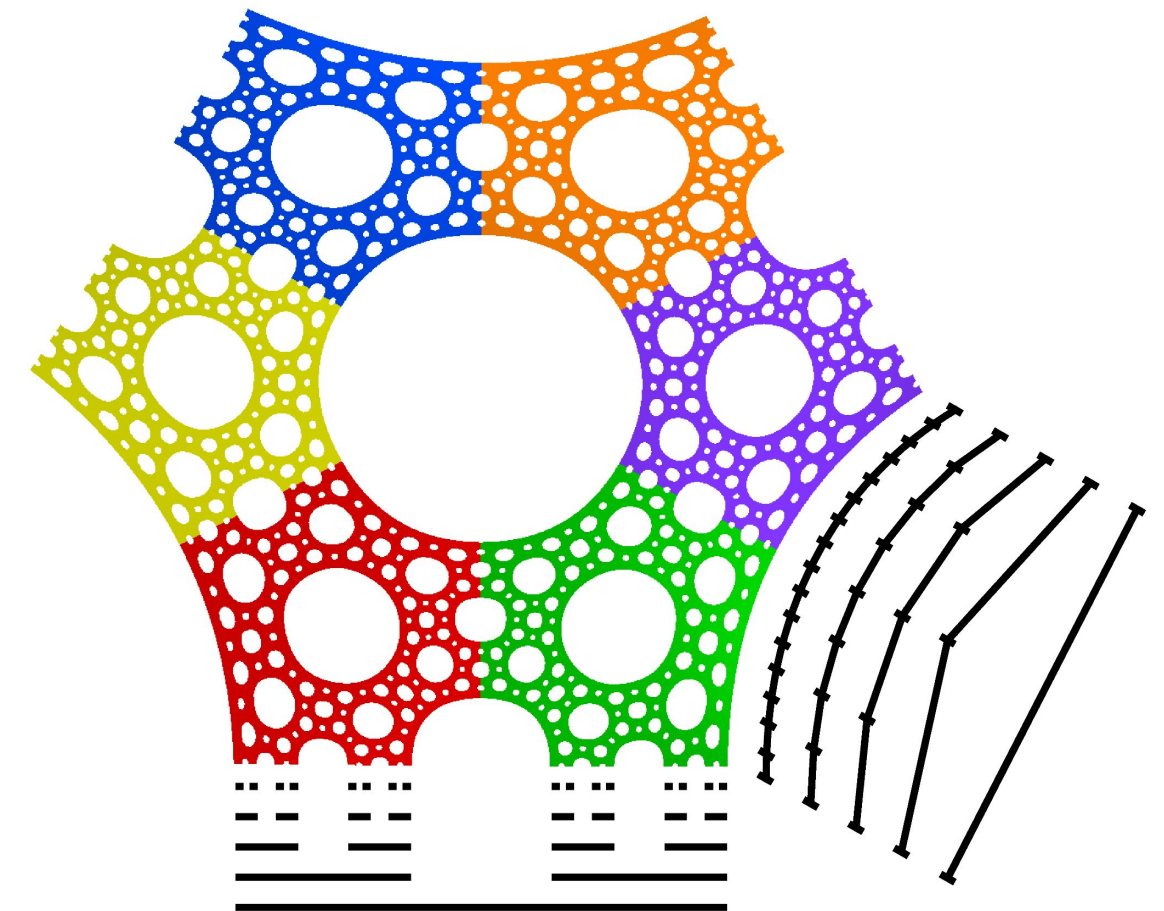
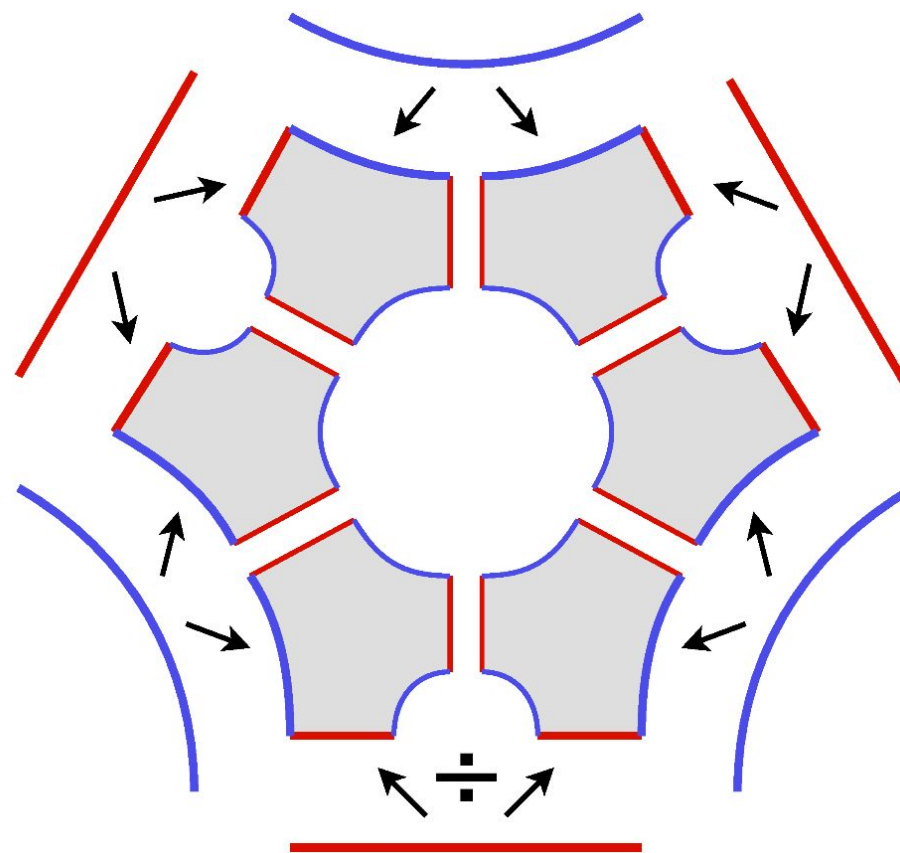
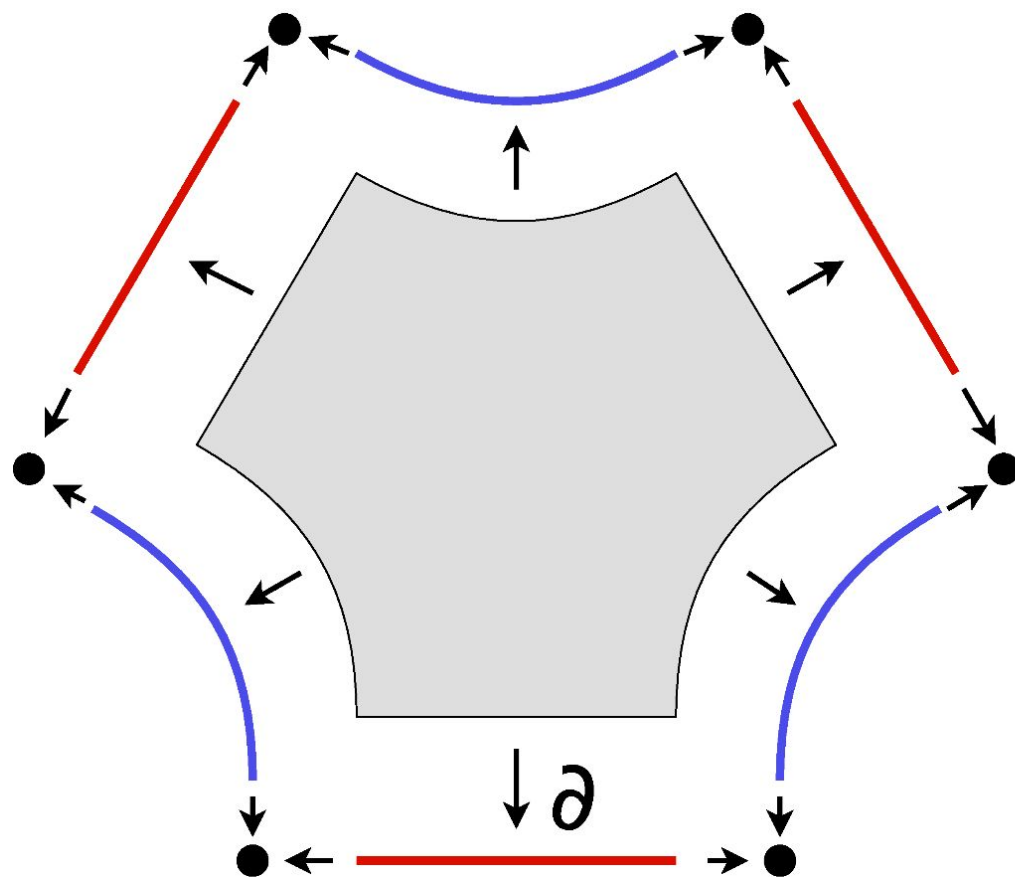
$$T_2 = \begin{pmatrix} 1 & \boxed{\frac{1}{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

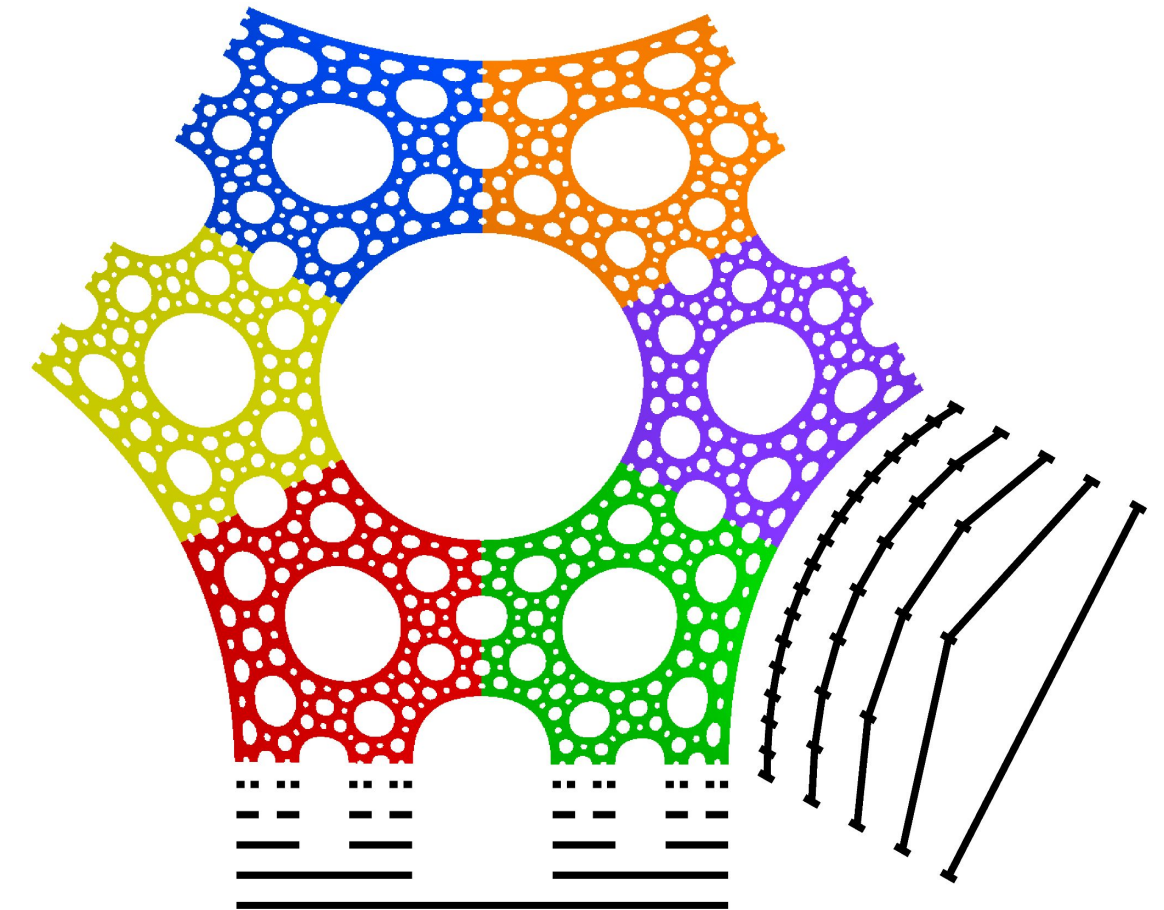
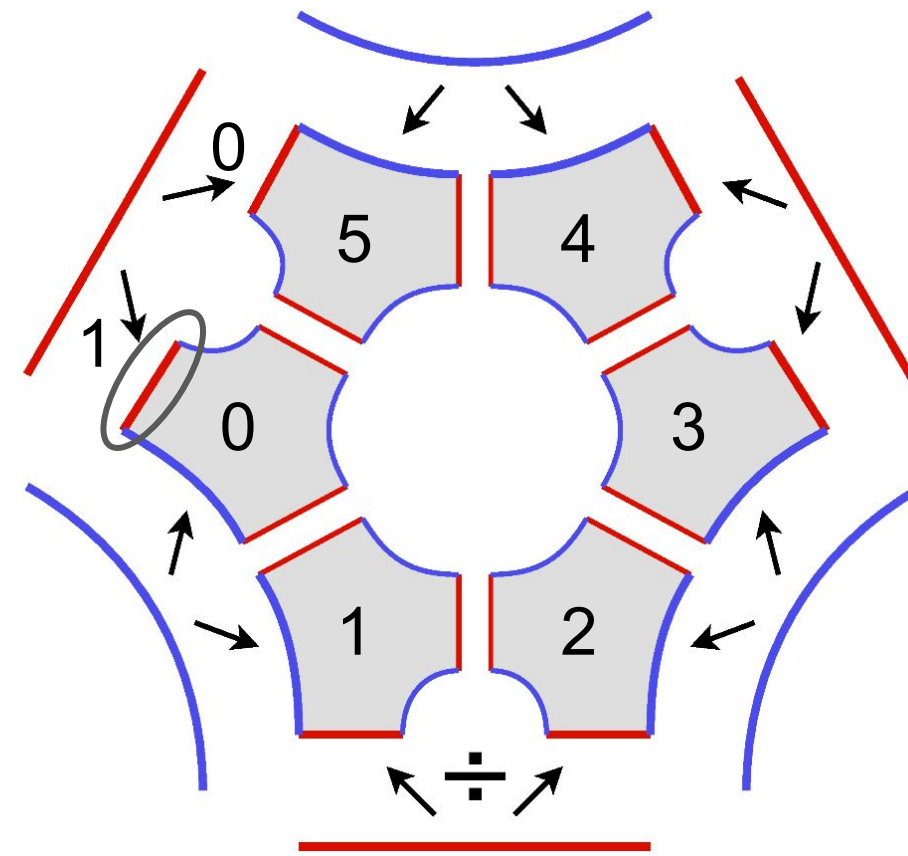
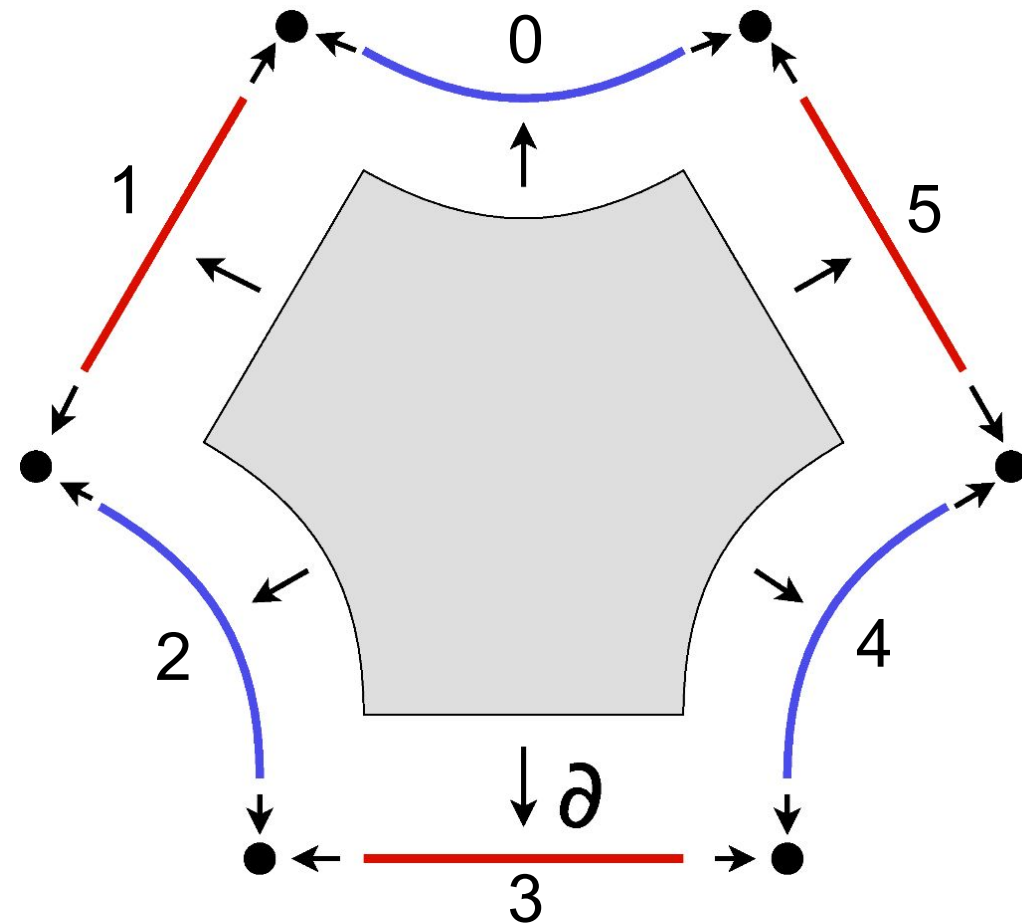
$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$



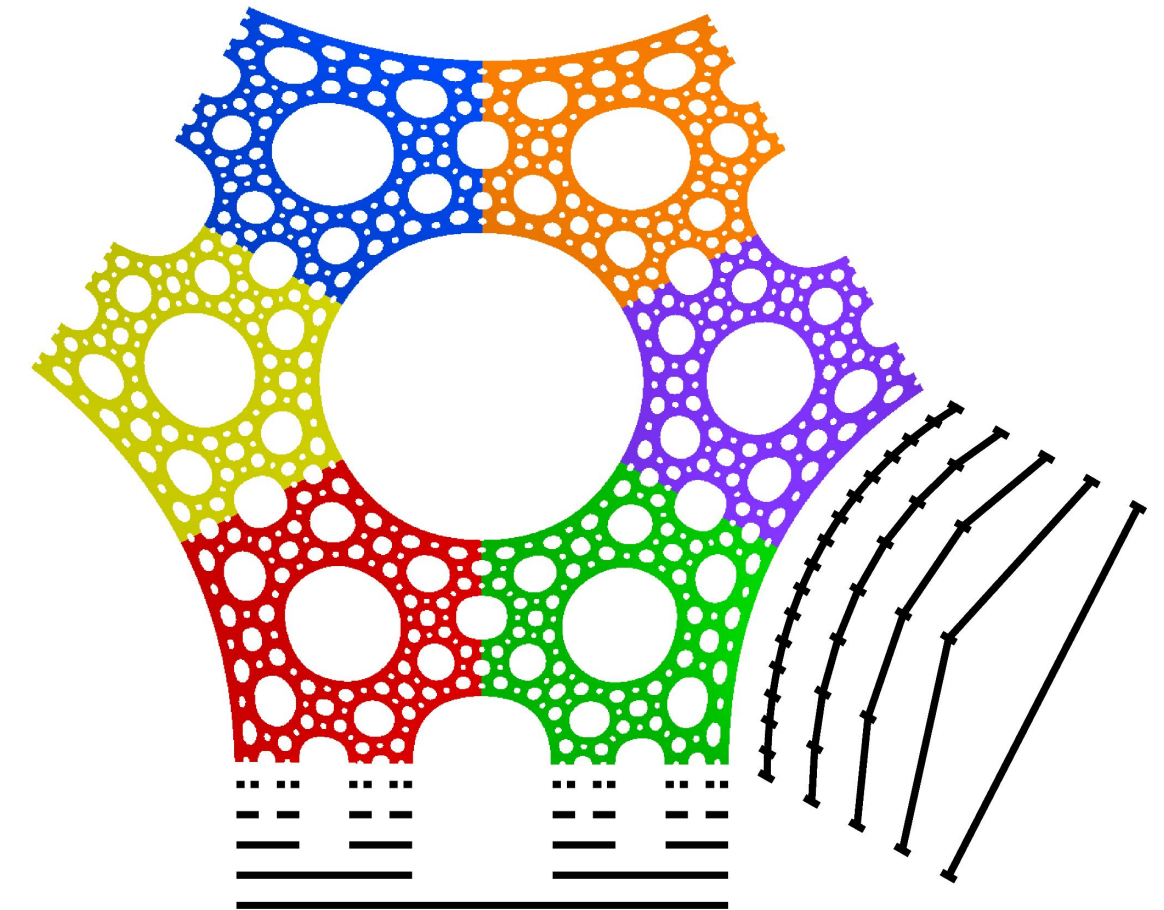
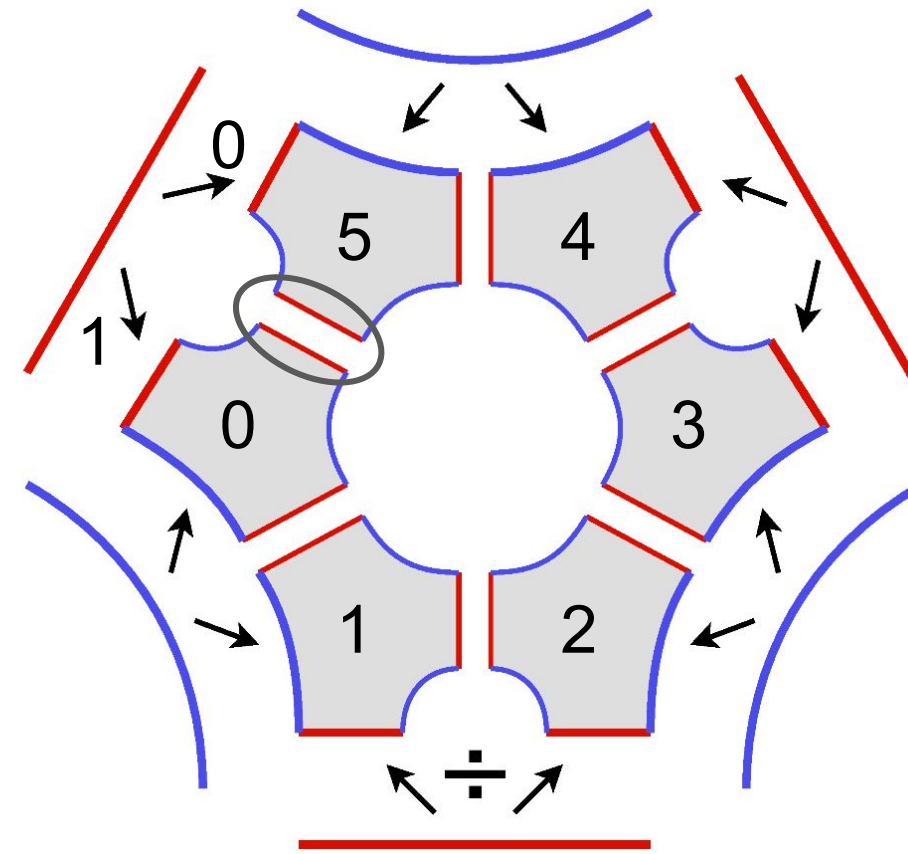
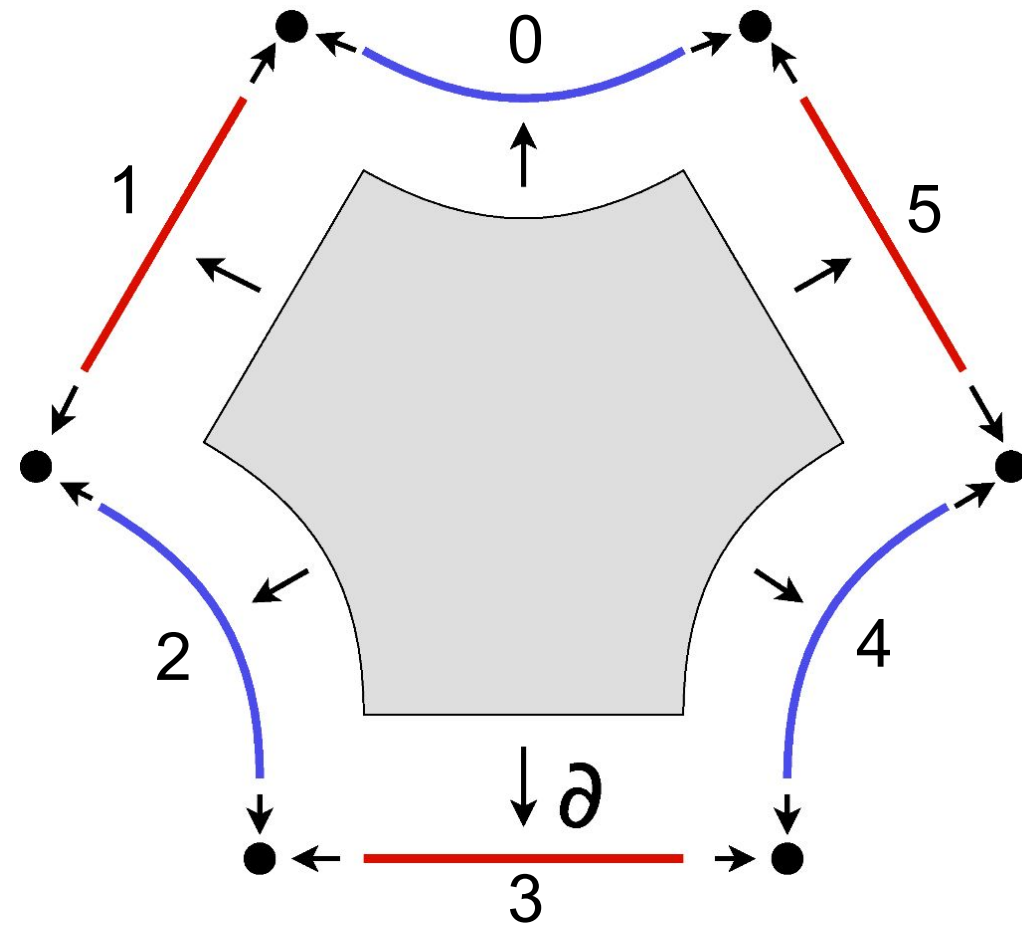
■ BC-IFS

- Décomposition cellulaire
- Contrôle de la topologie fractale
- Topologie cohérente





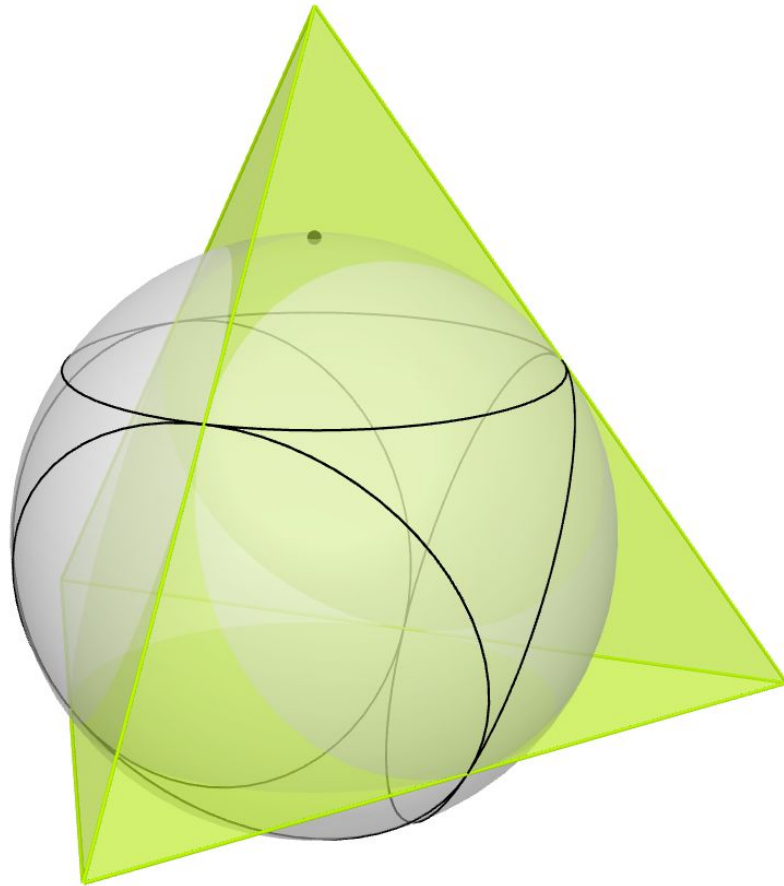
Incidence : $\text{Sub}(1) \text{ de } \text{Bord}(1) = \text{Bord}(1) \text{ de } \text{Sub}(0)$

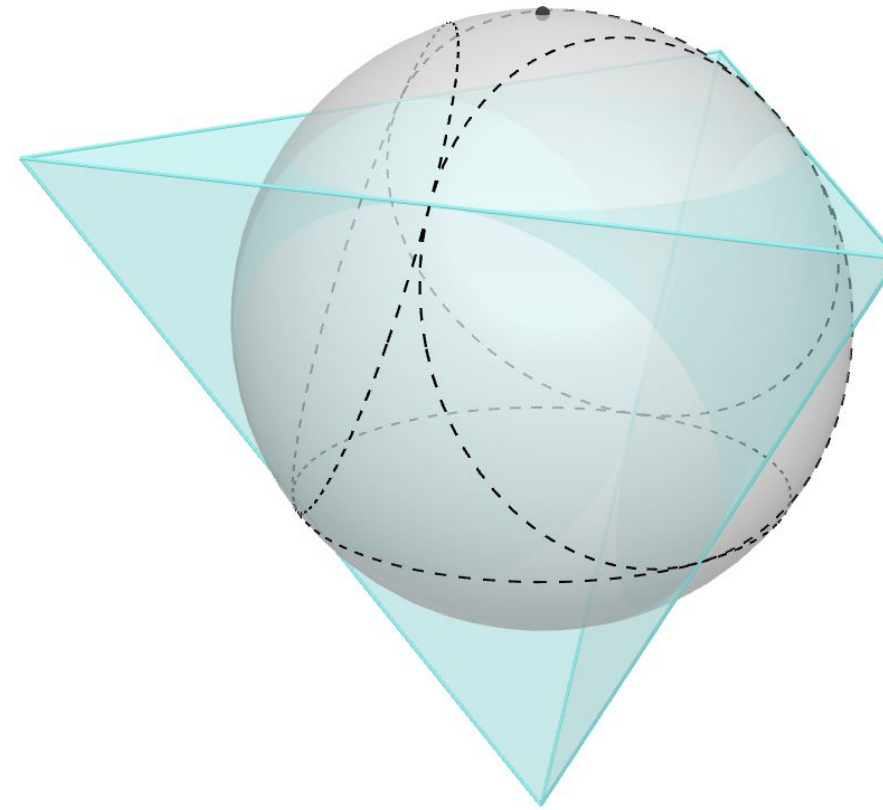
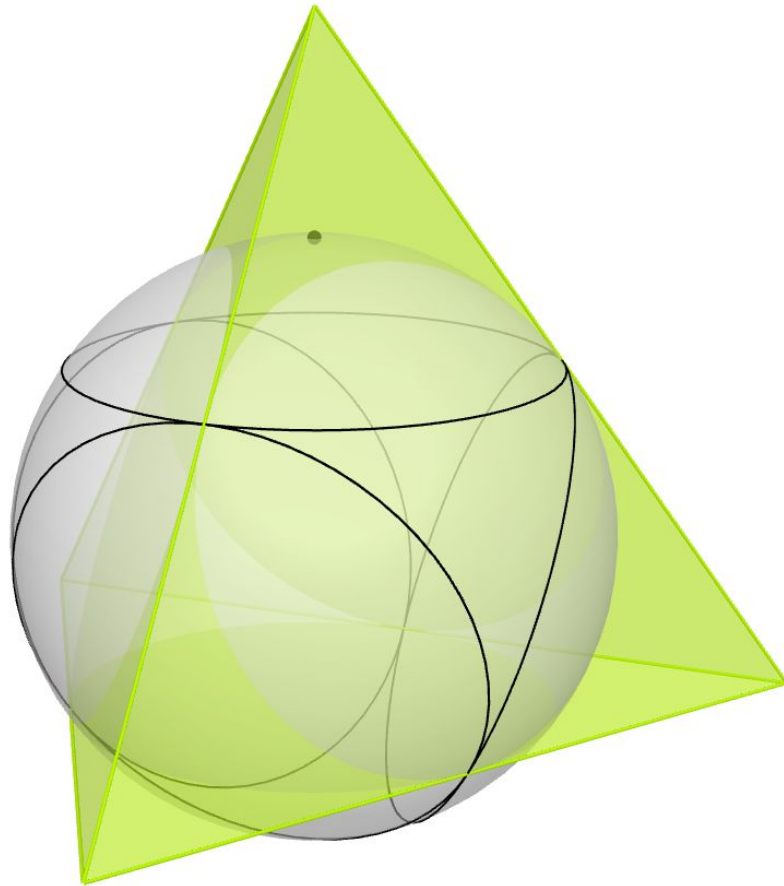


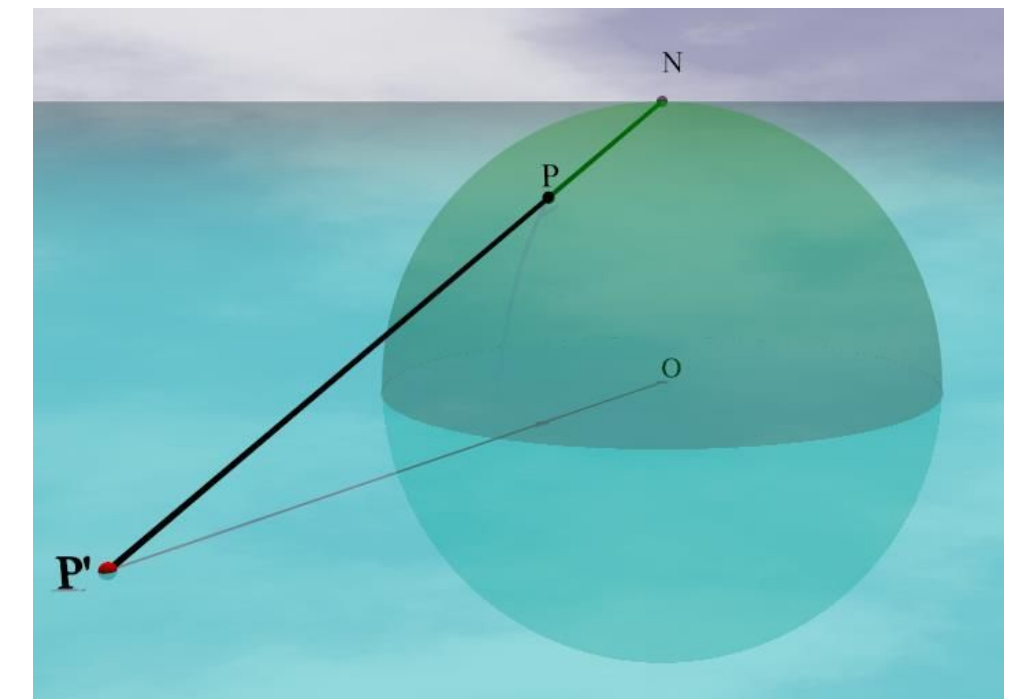
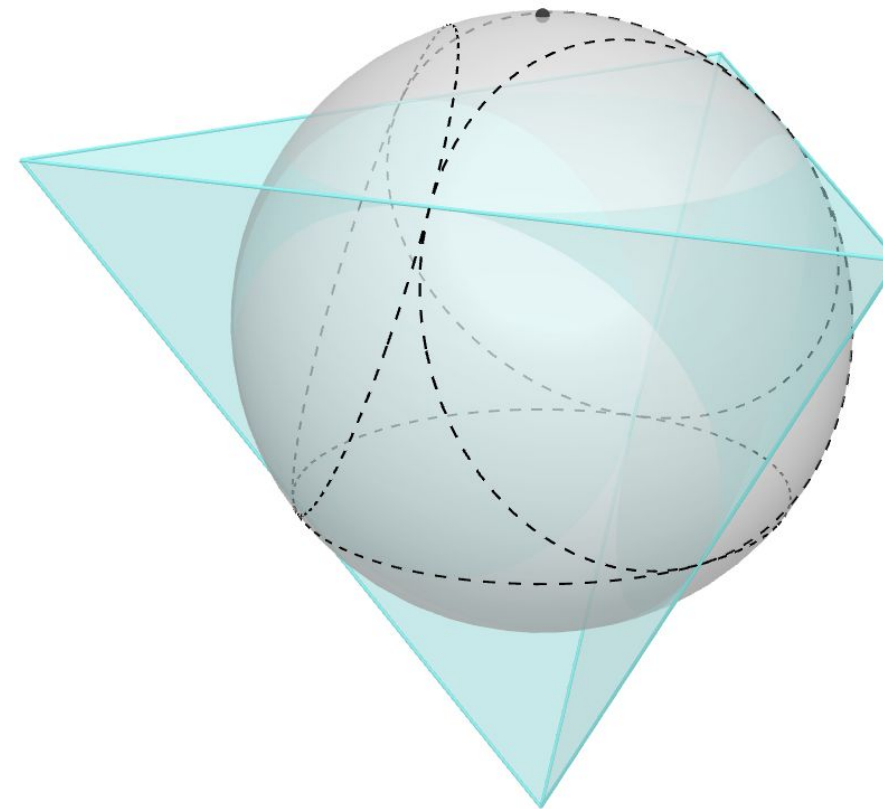
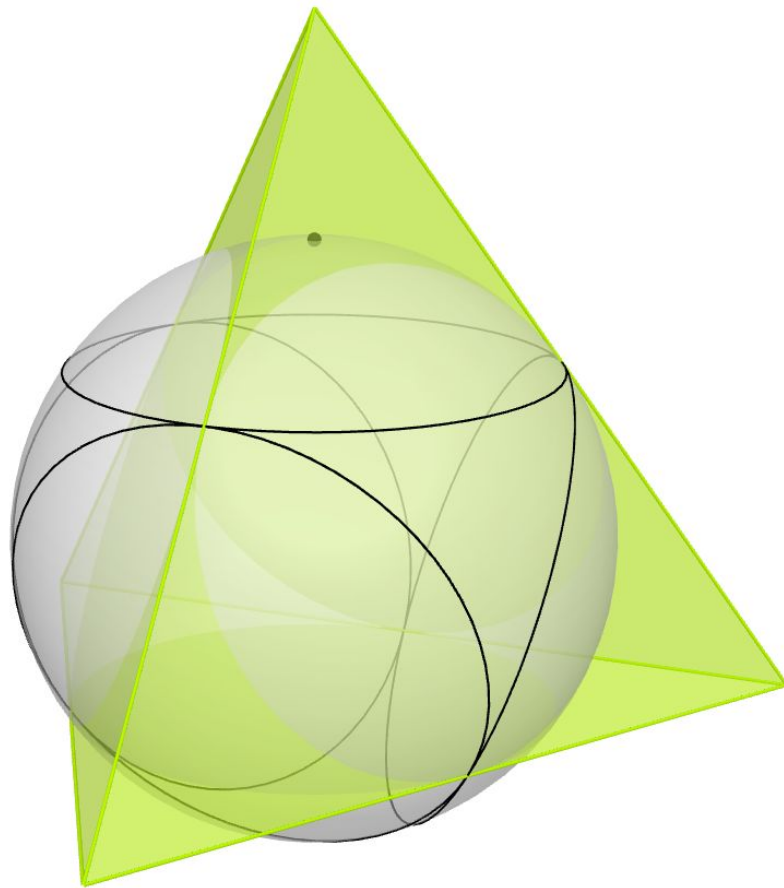
Adjacence : $\text{Bord}(5) \text{ de Sub}(0) = \text{Bord}(3) \text{ de Sub}(5)$

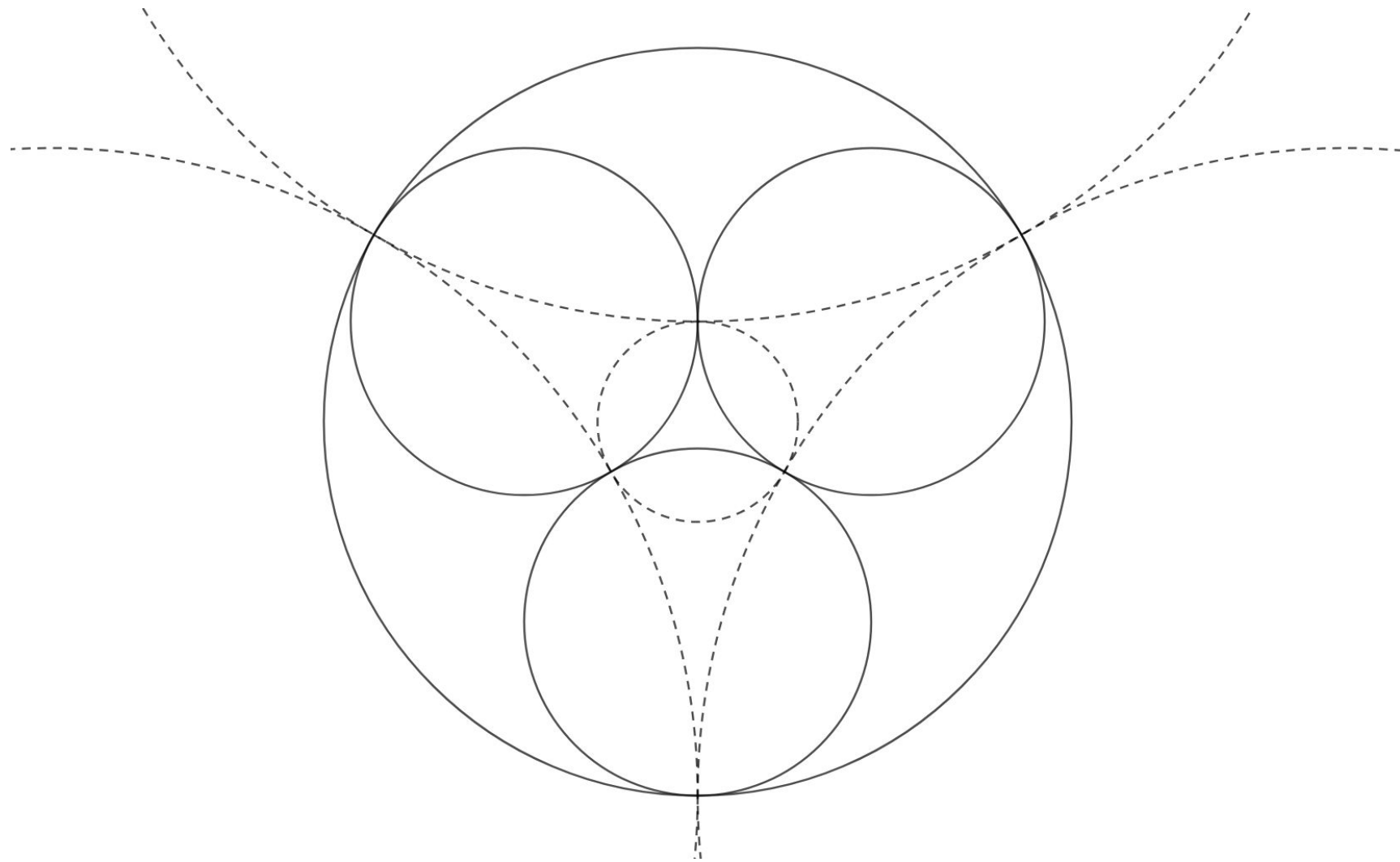
Empilements polytopaux de sphères

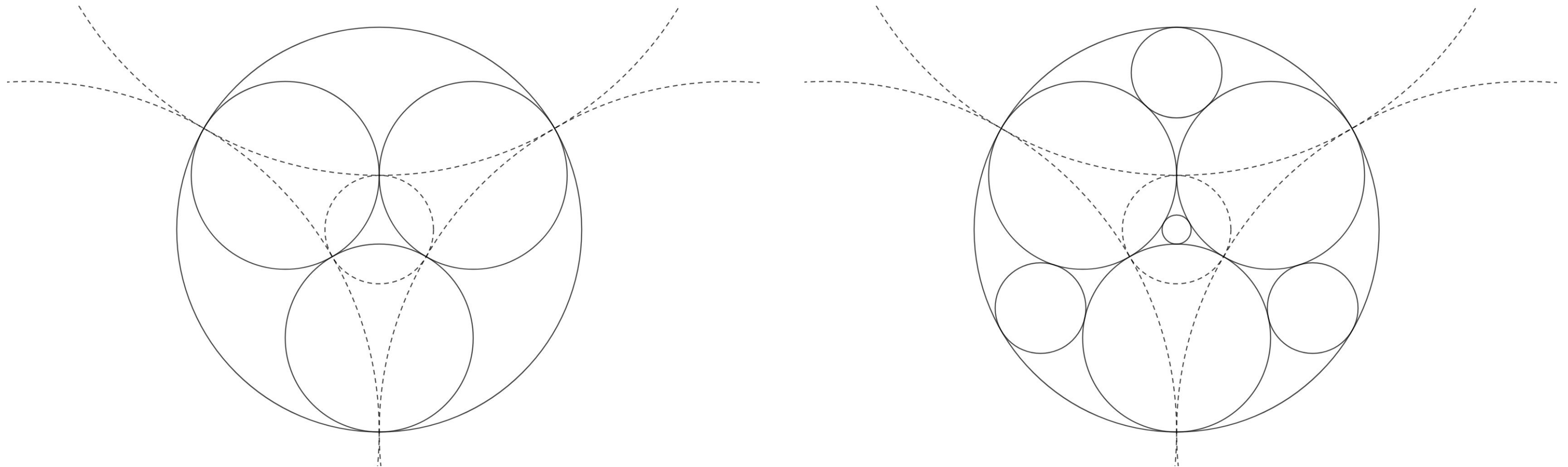


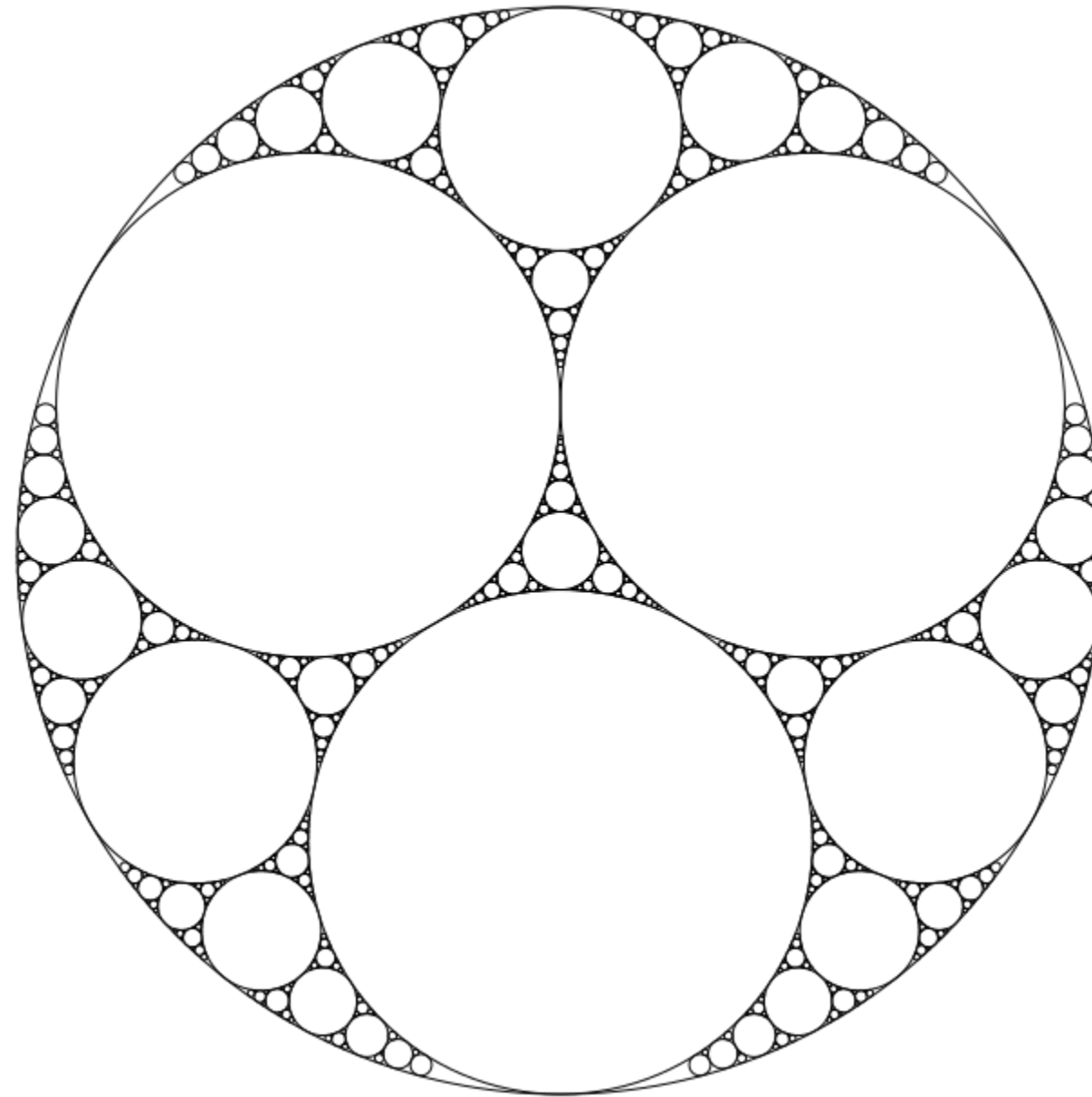


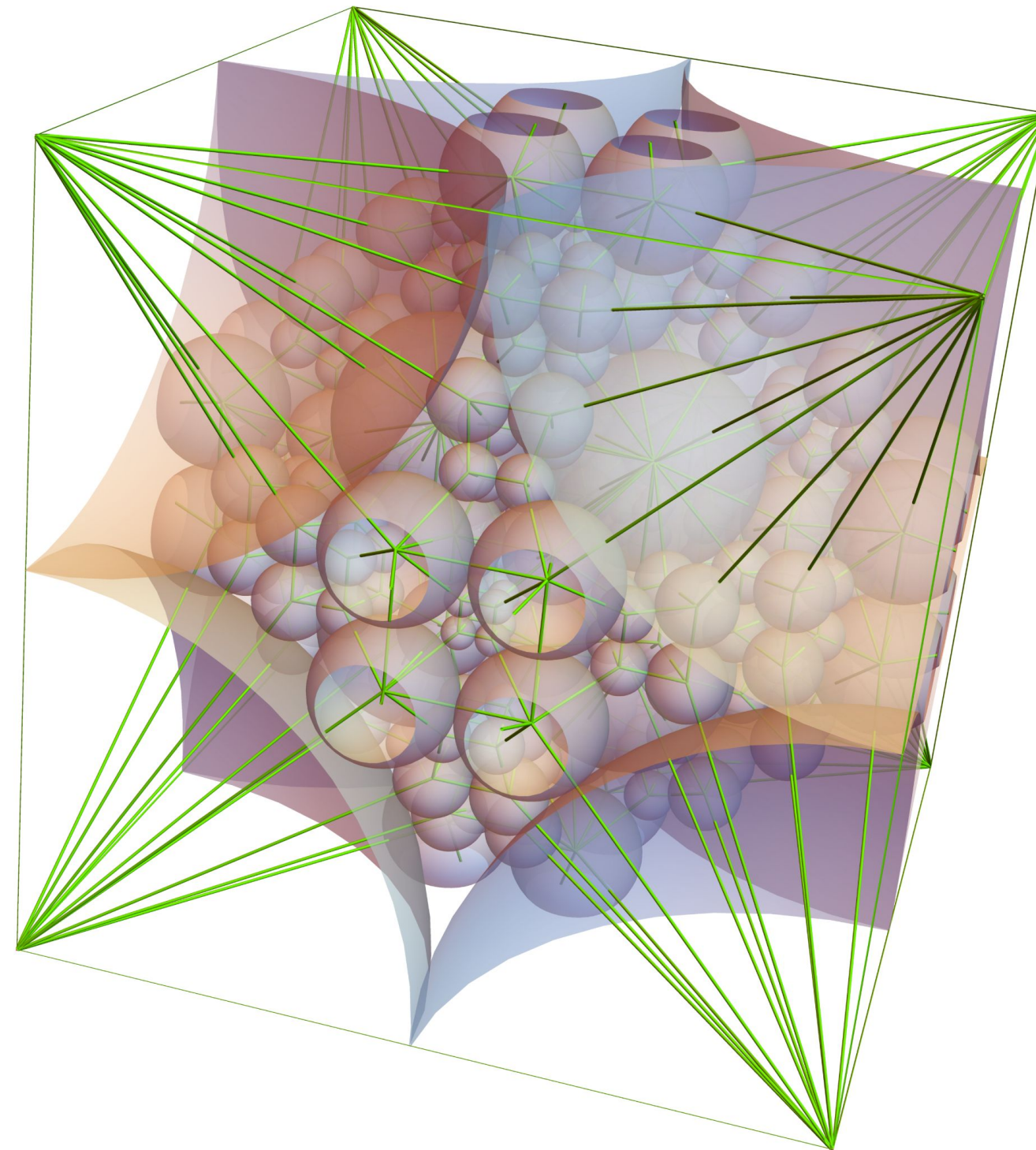






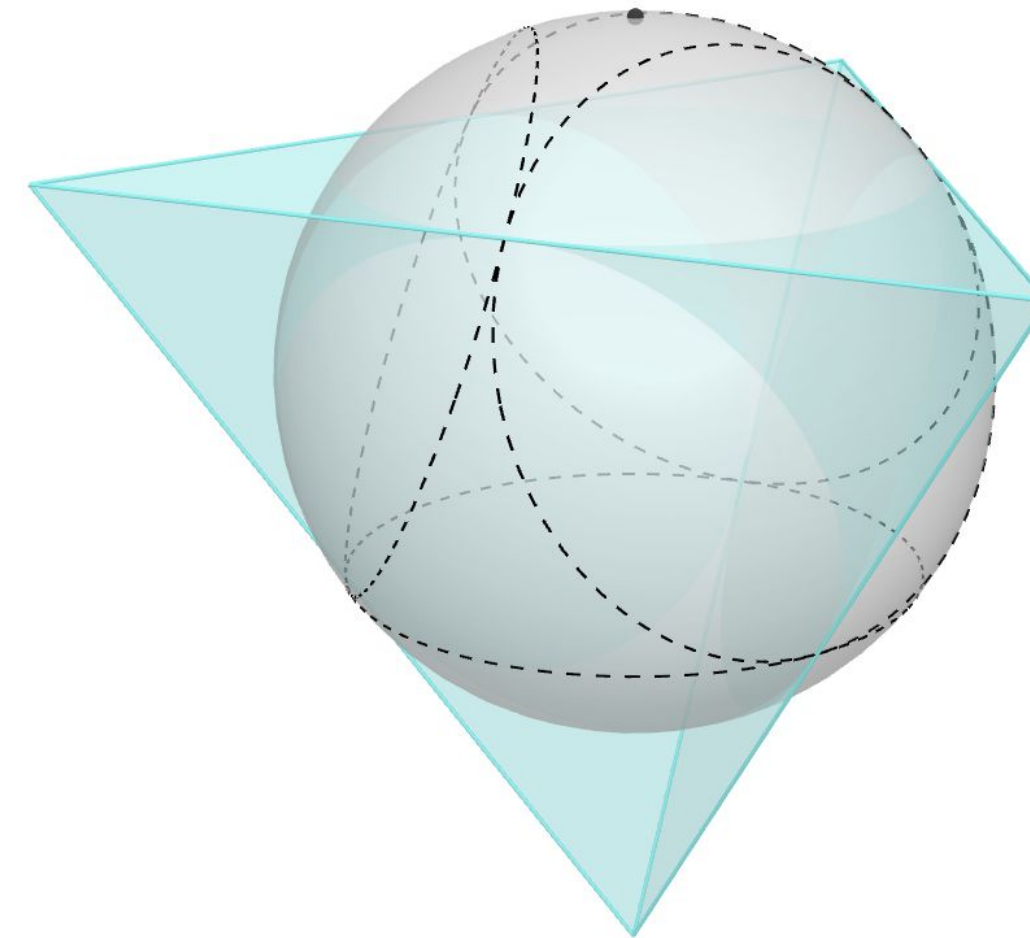
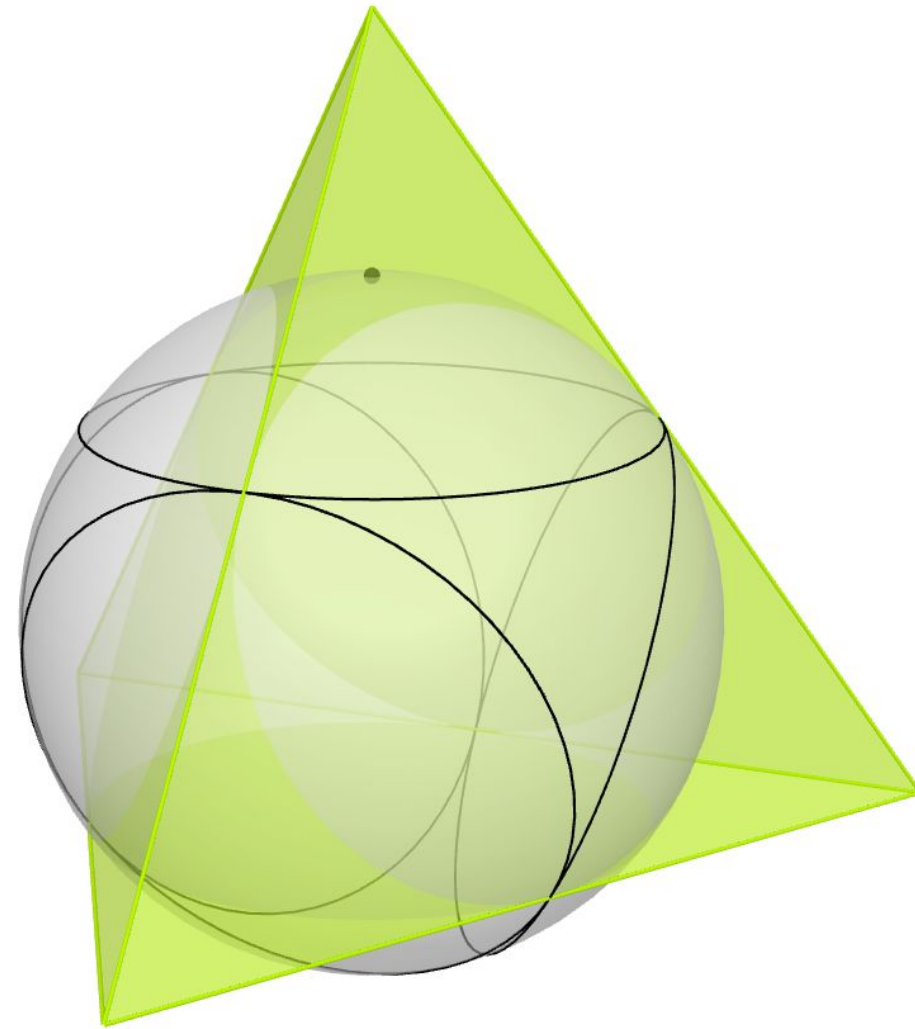


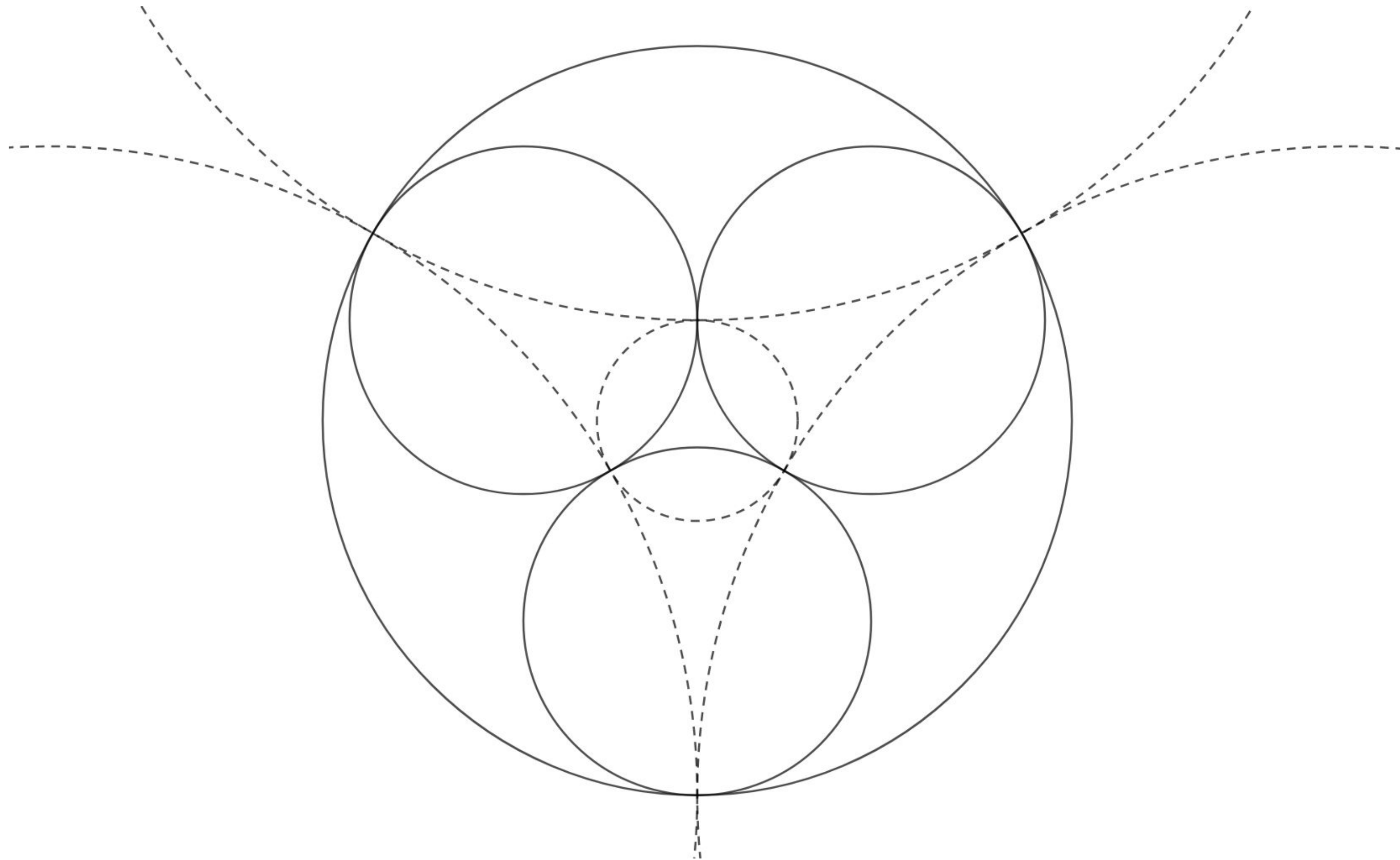


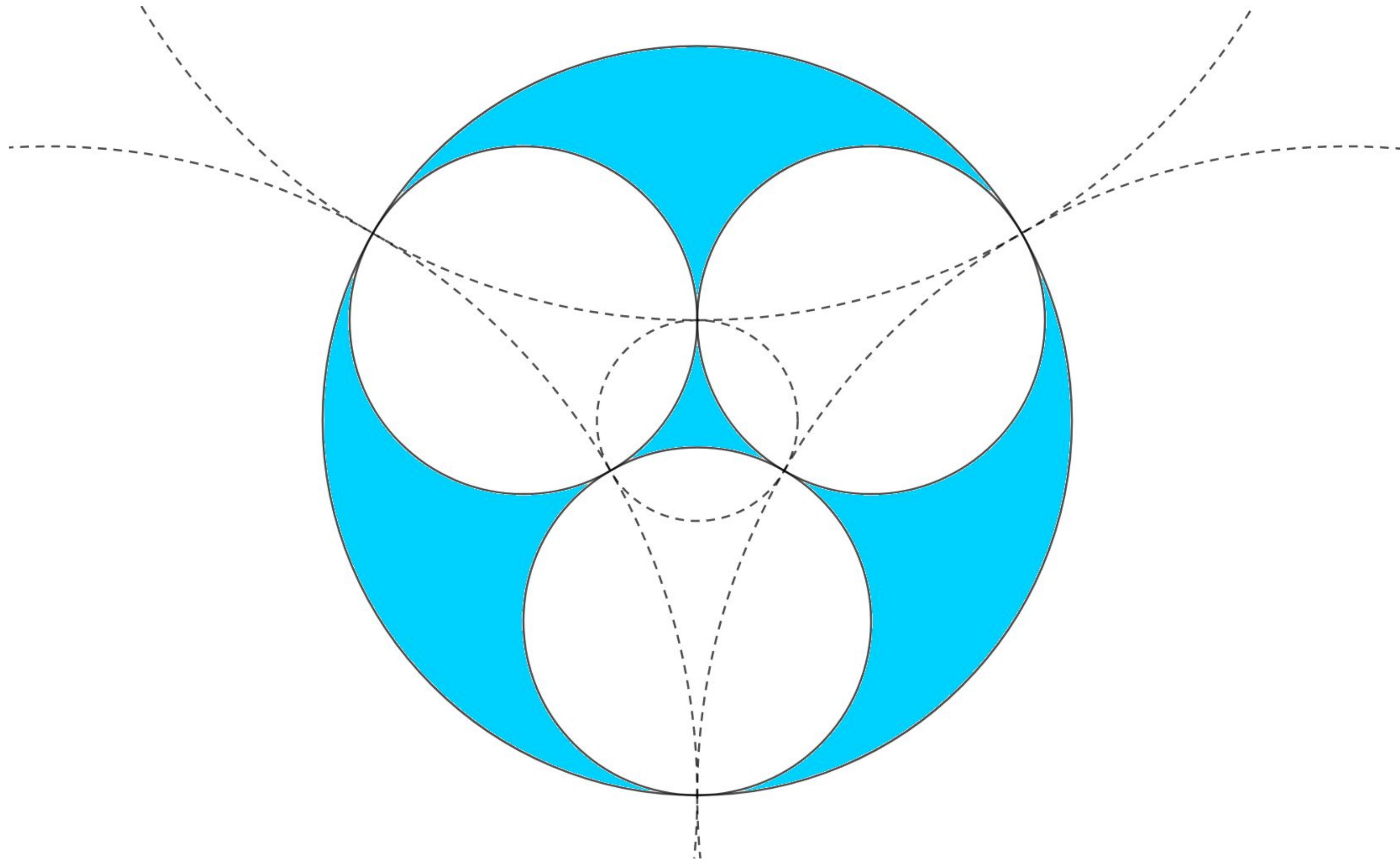


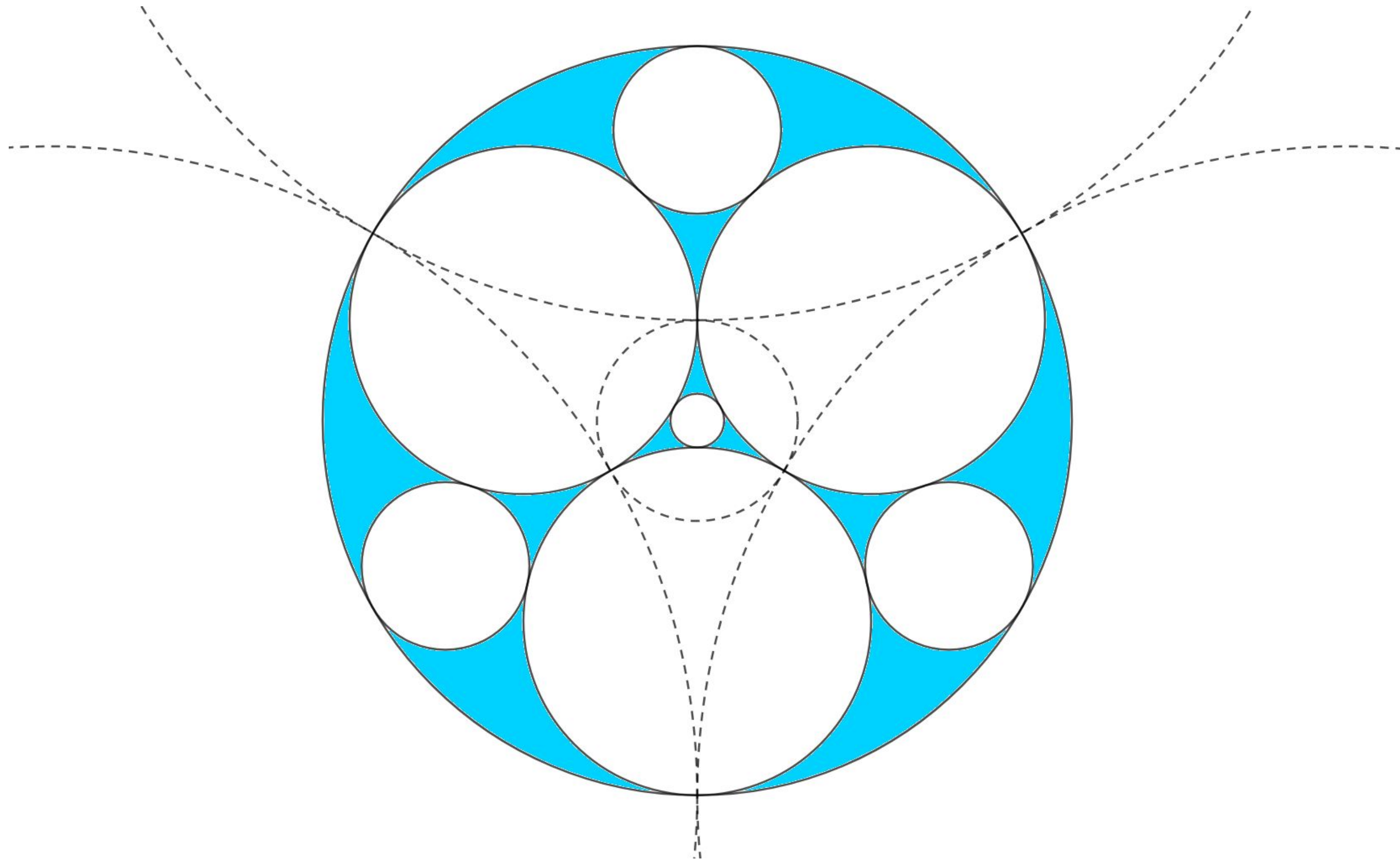
Représentation avec le modèle BC-IFS

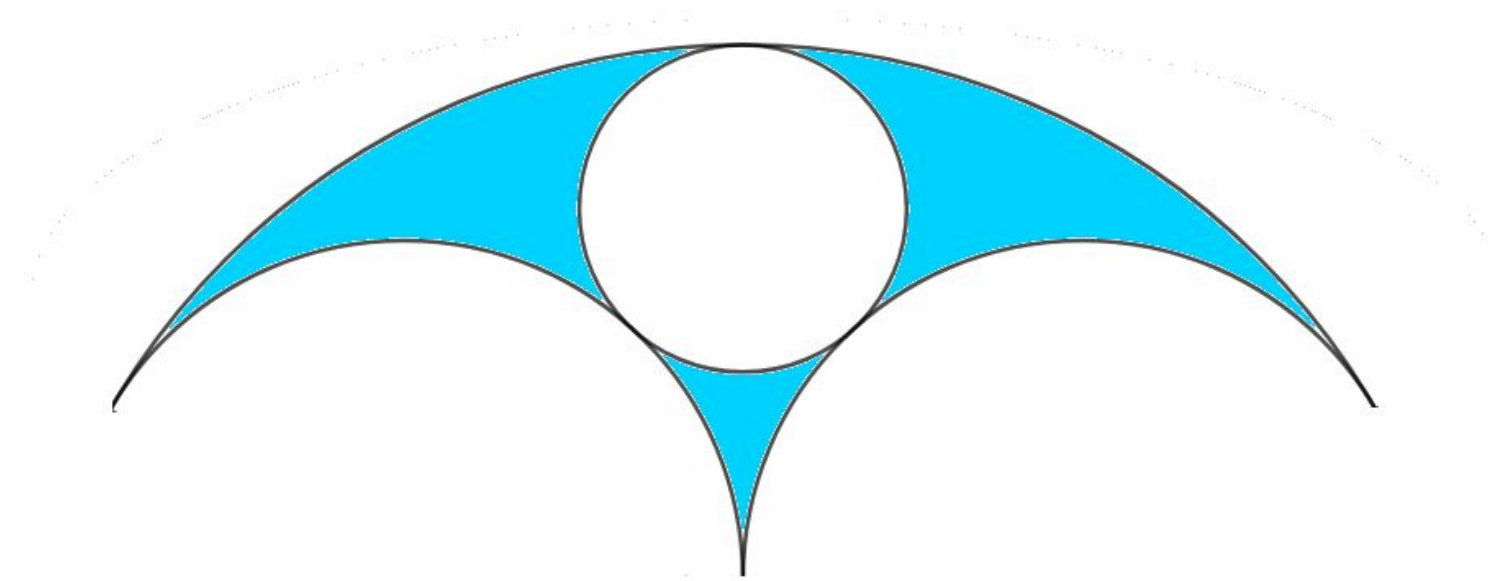
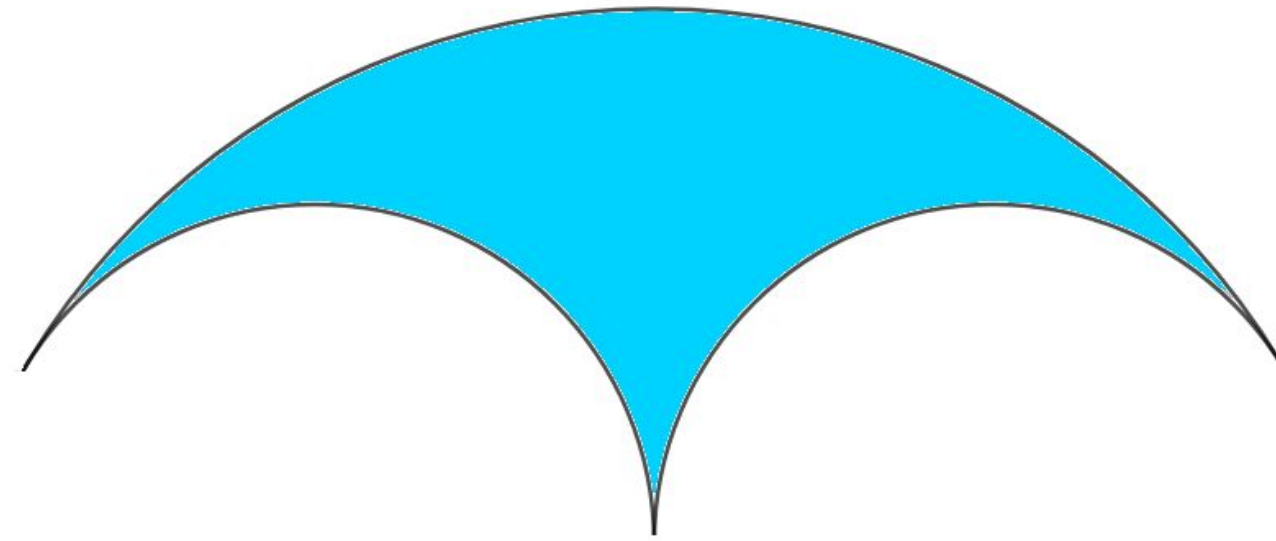


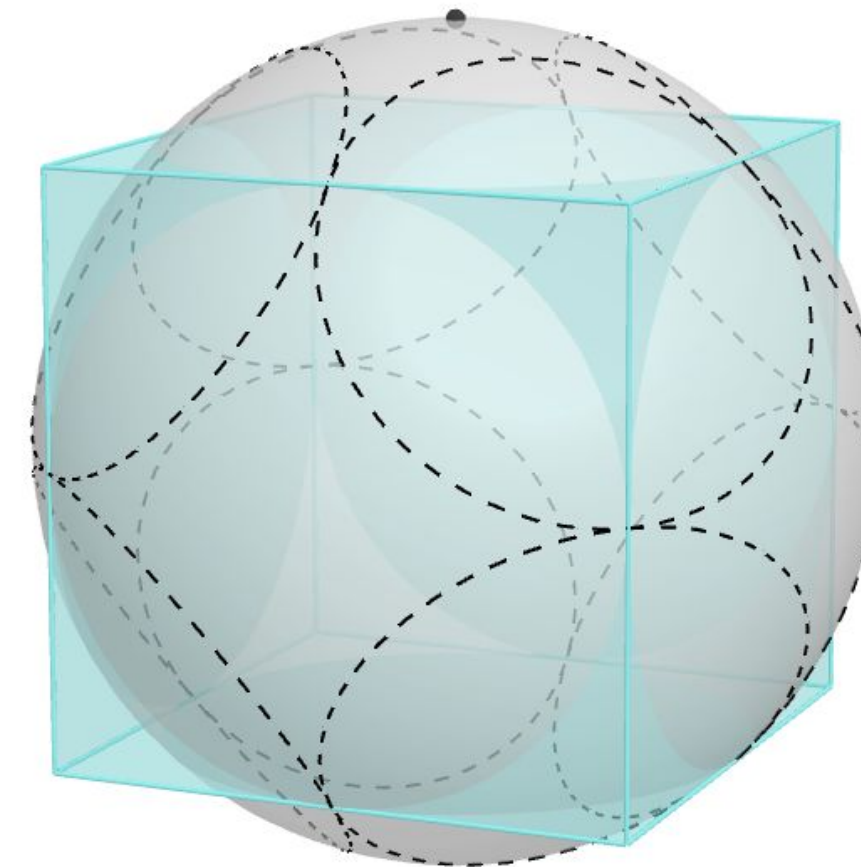
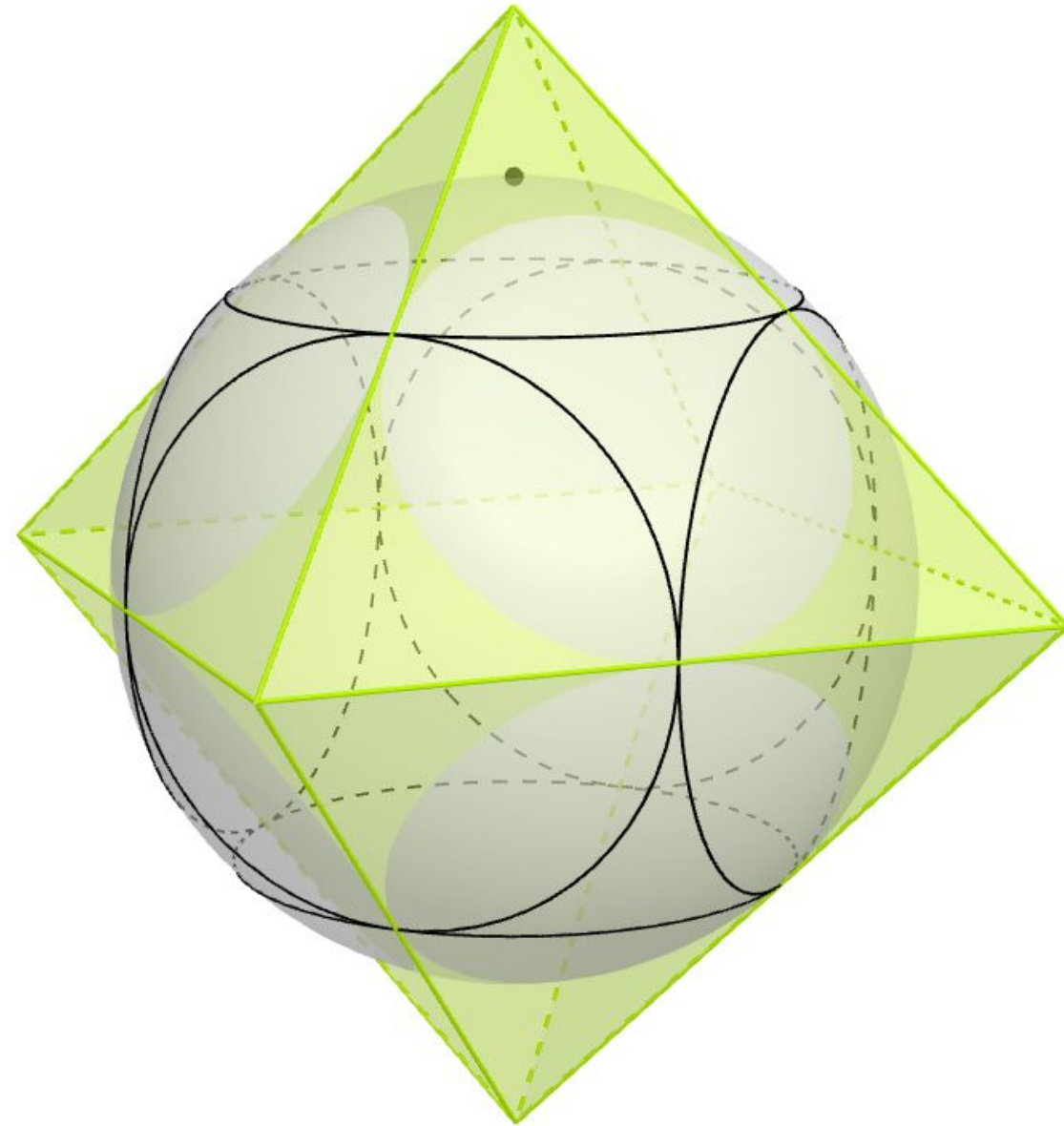


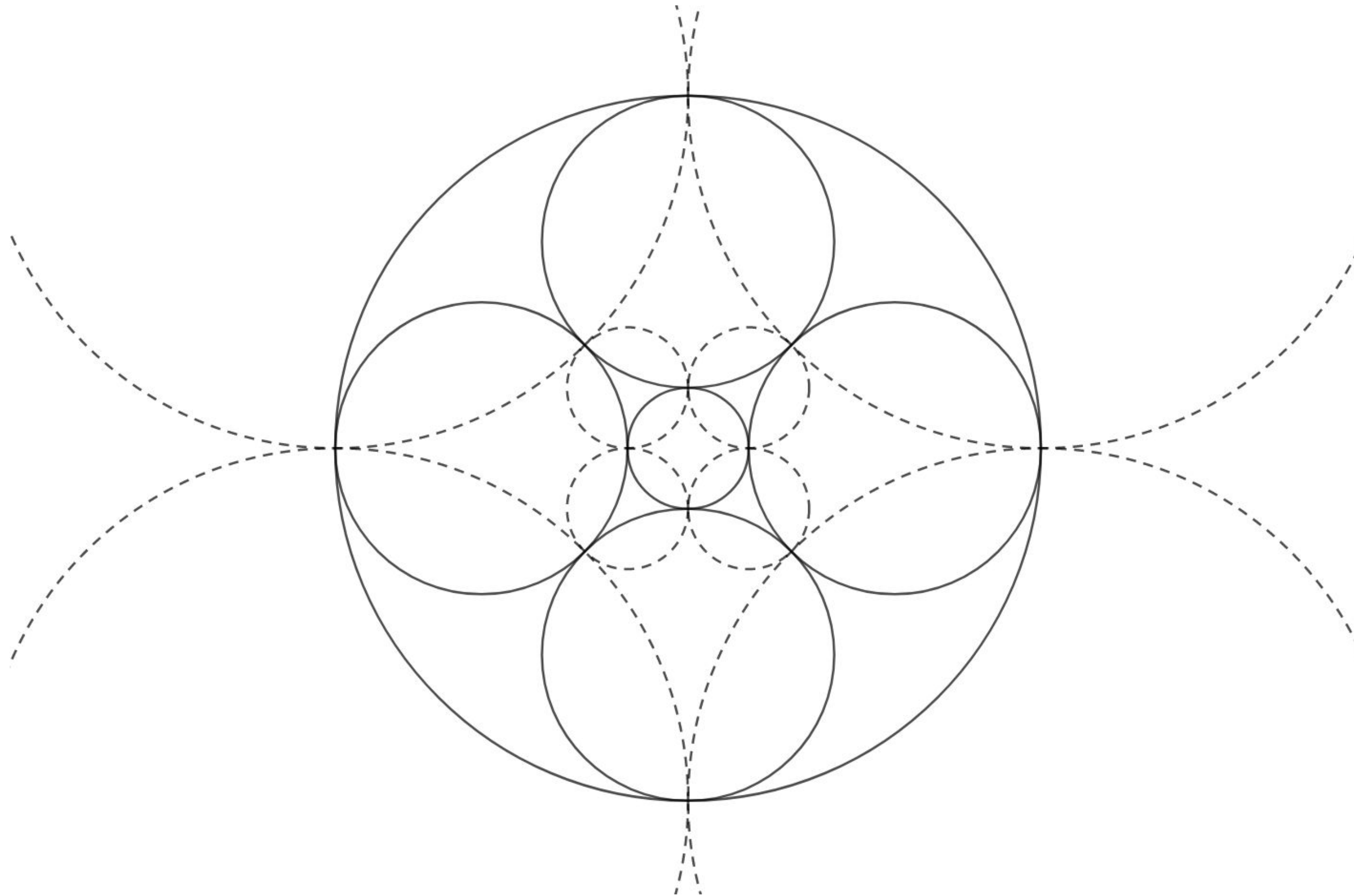


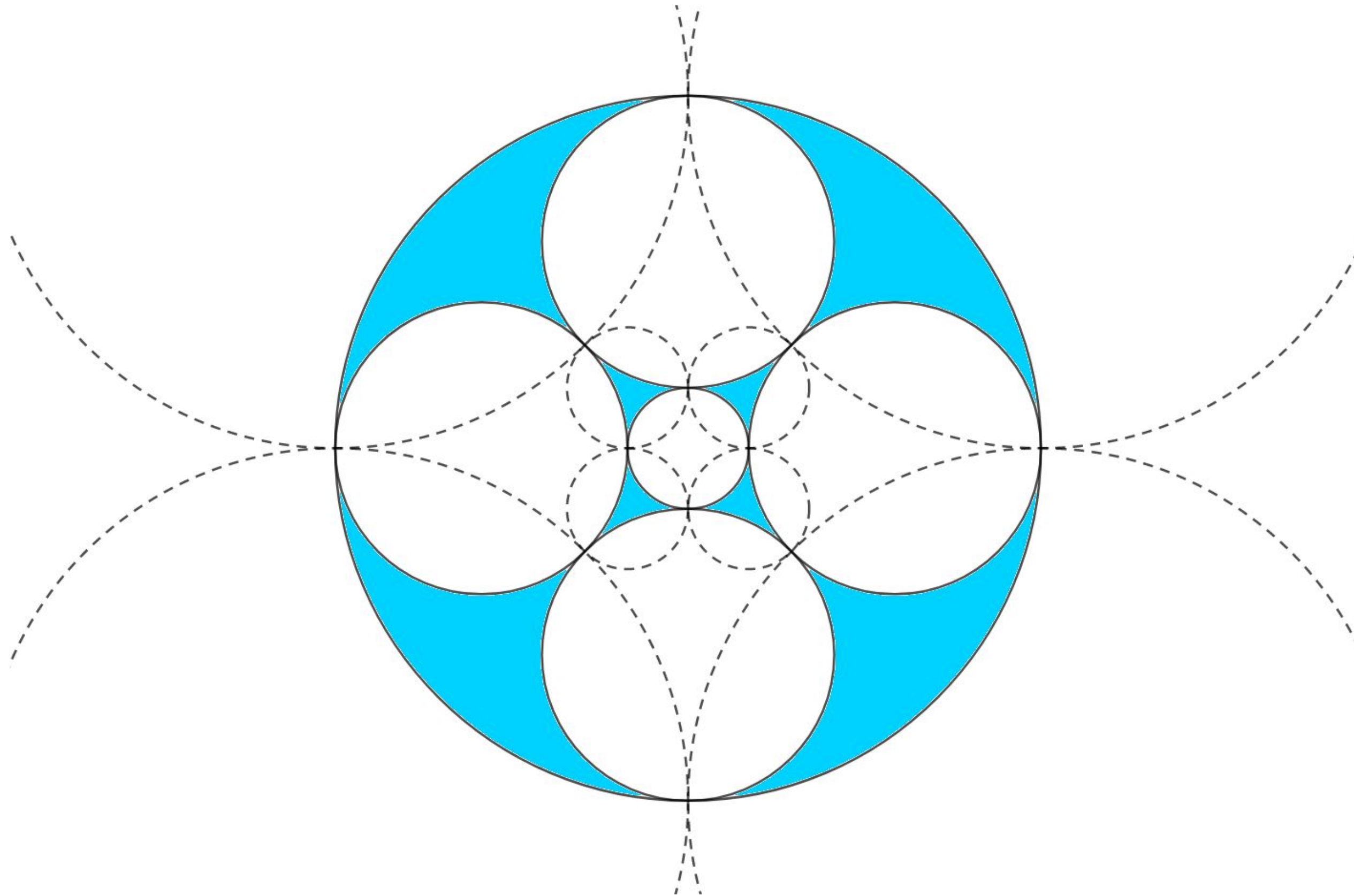


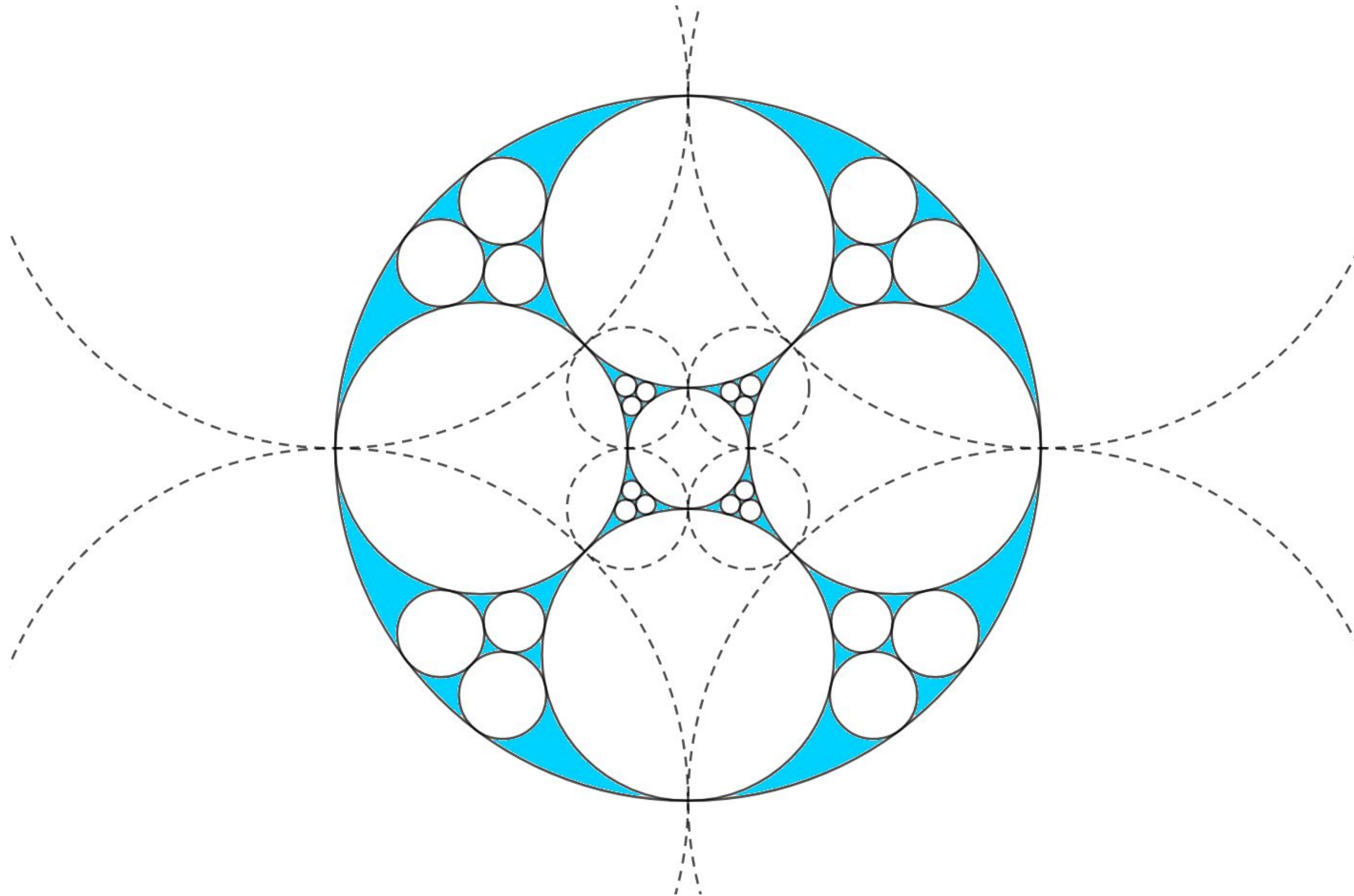


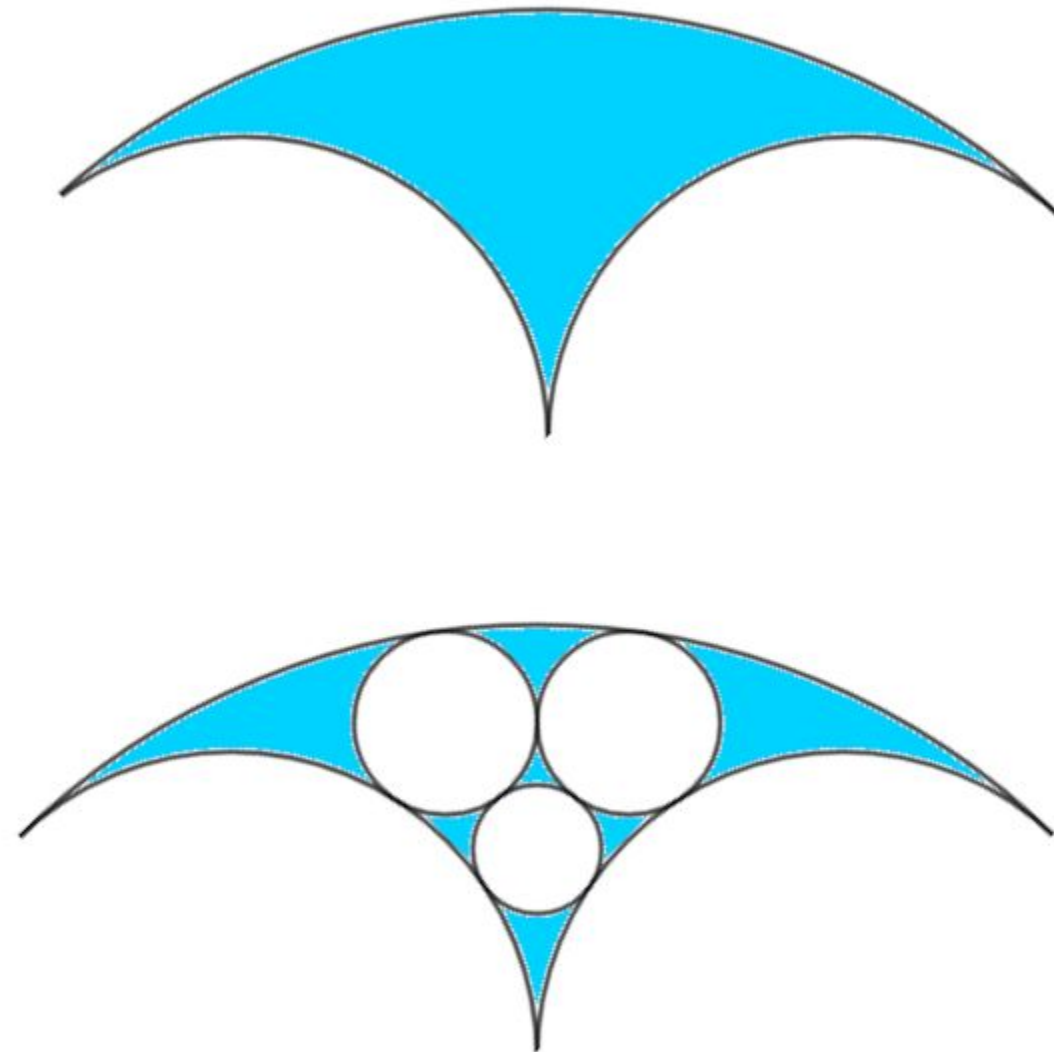












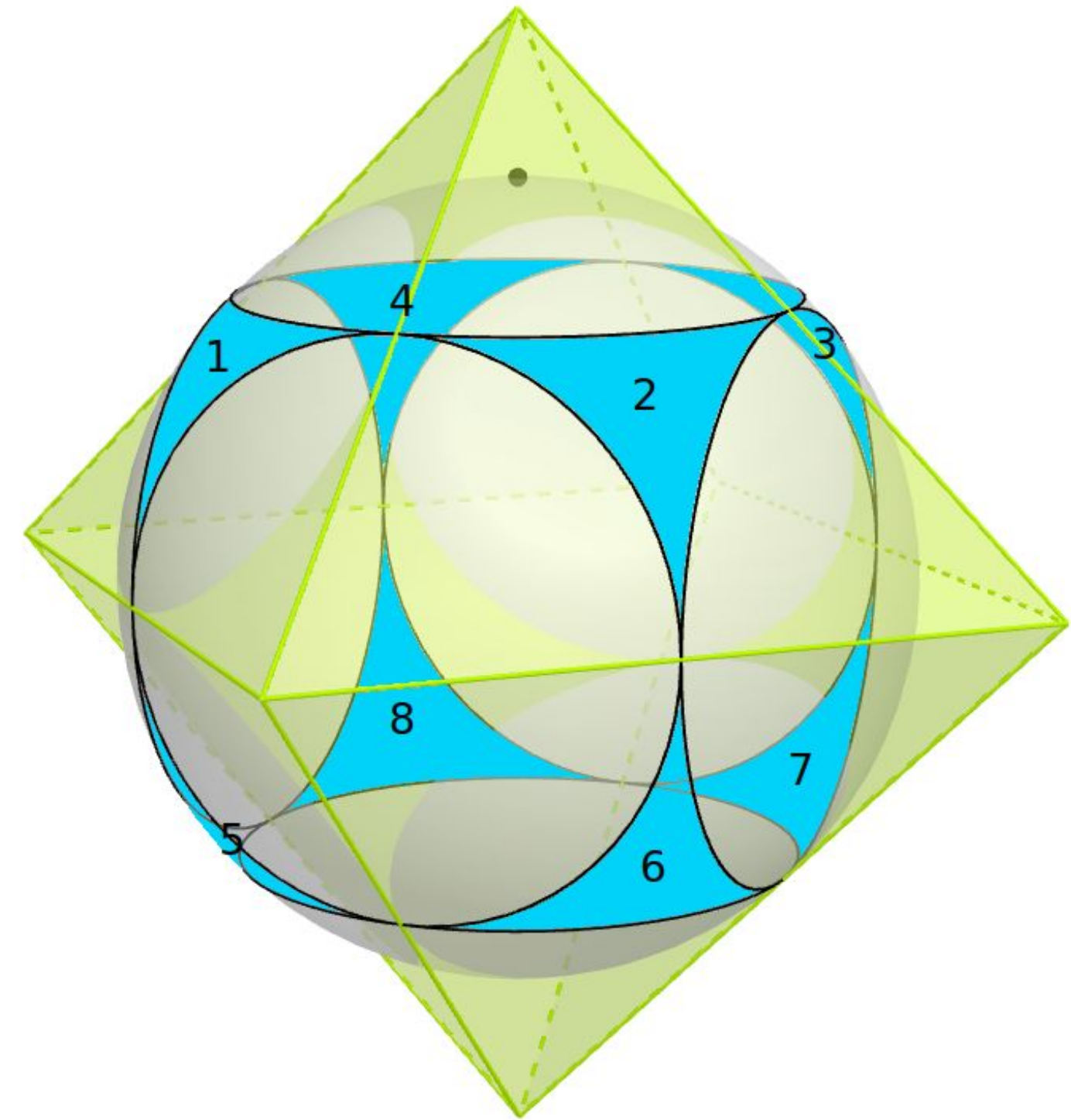
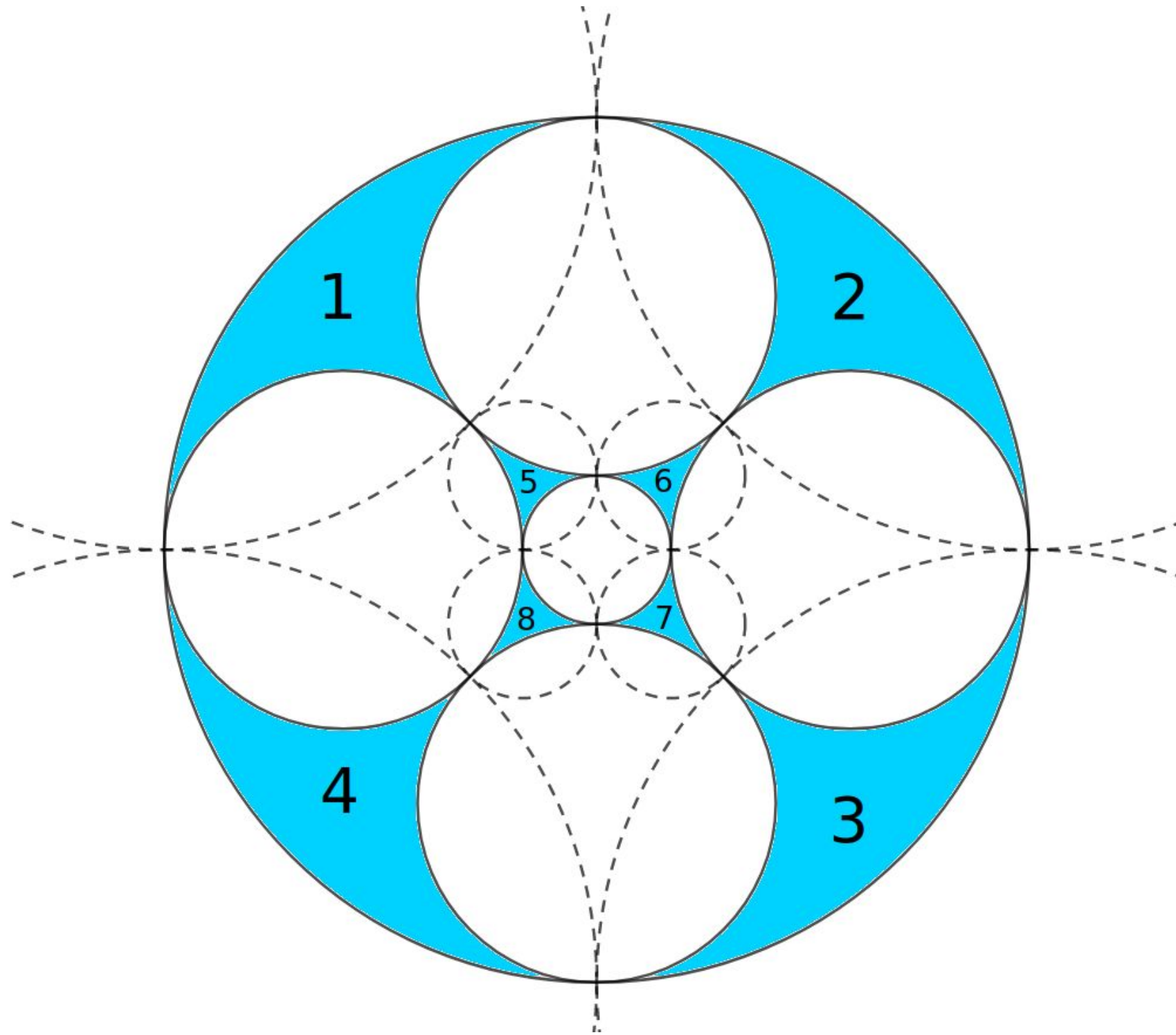
Extraction de règles de construction



- Combien de faces ?
- Comment sont-elles agencées ?
- Combien de bords par face ?
- Comment se subdivisent-elles ?
- Quelles contraintes d'adjacence ?
- Comment se subdivisent les bords ?
- Quelles contraintes d'incidence ?

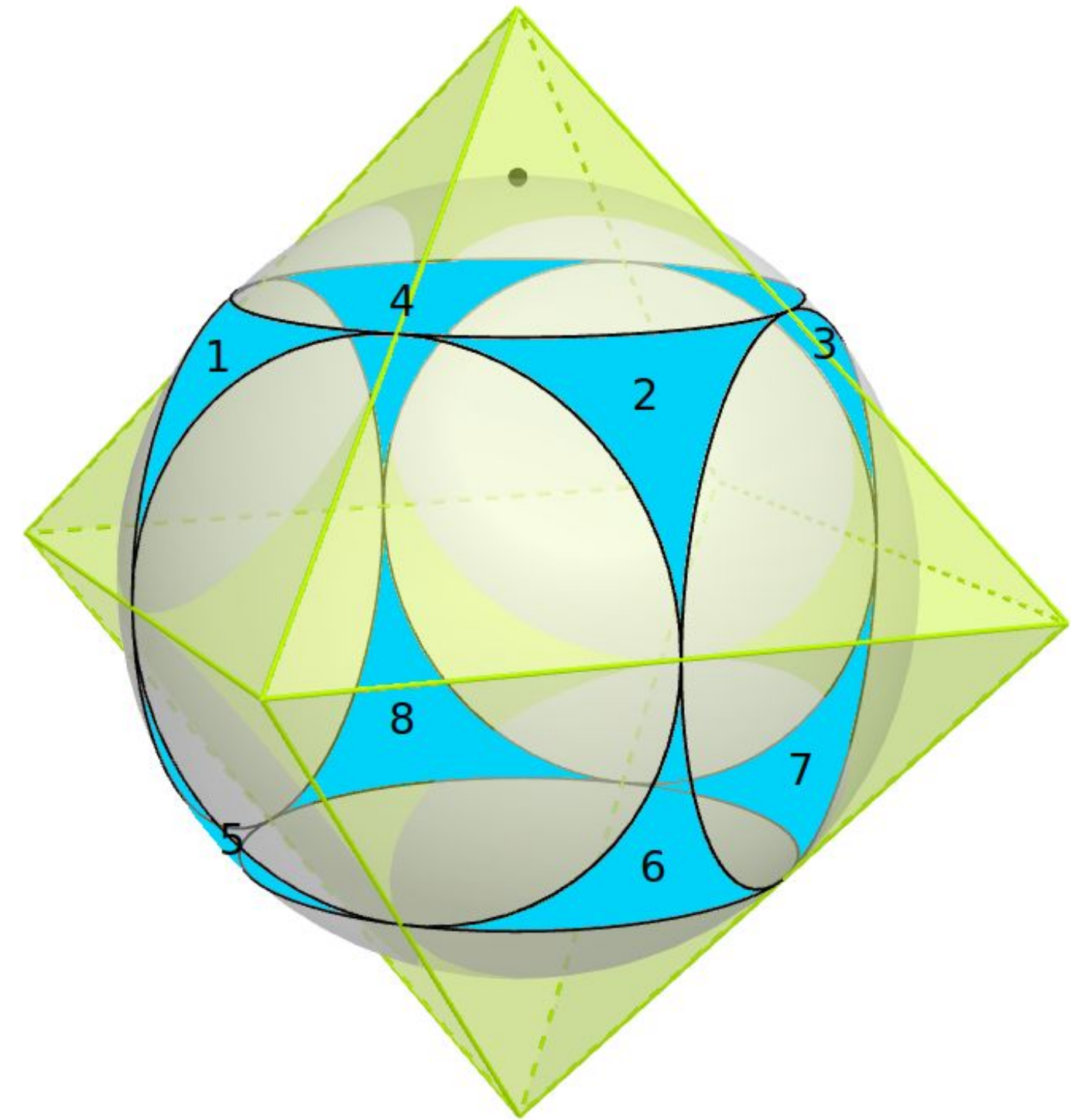
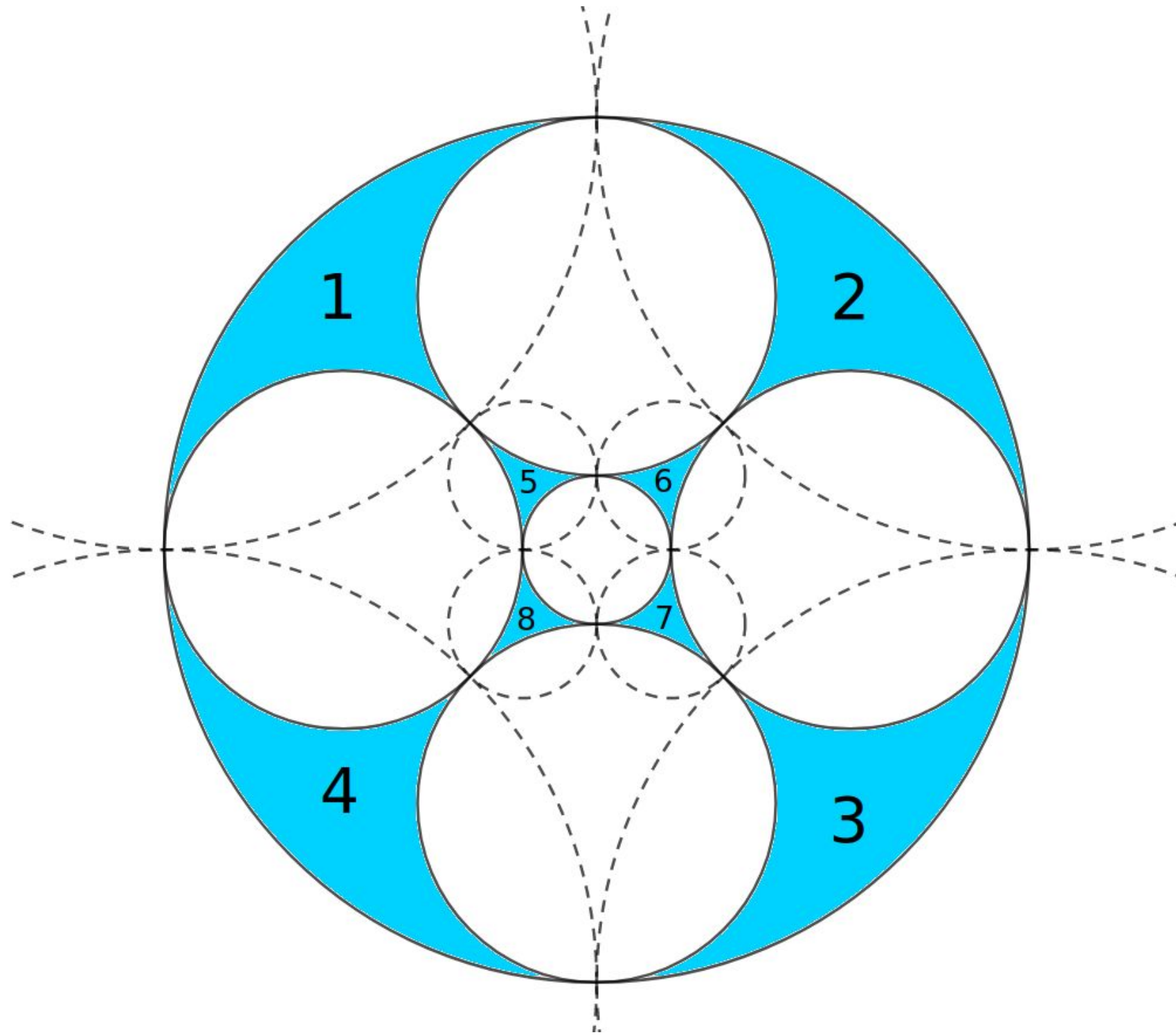
Combien de faces ?

Une face fractale par face du polyèdre



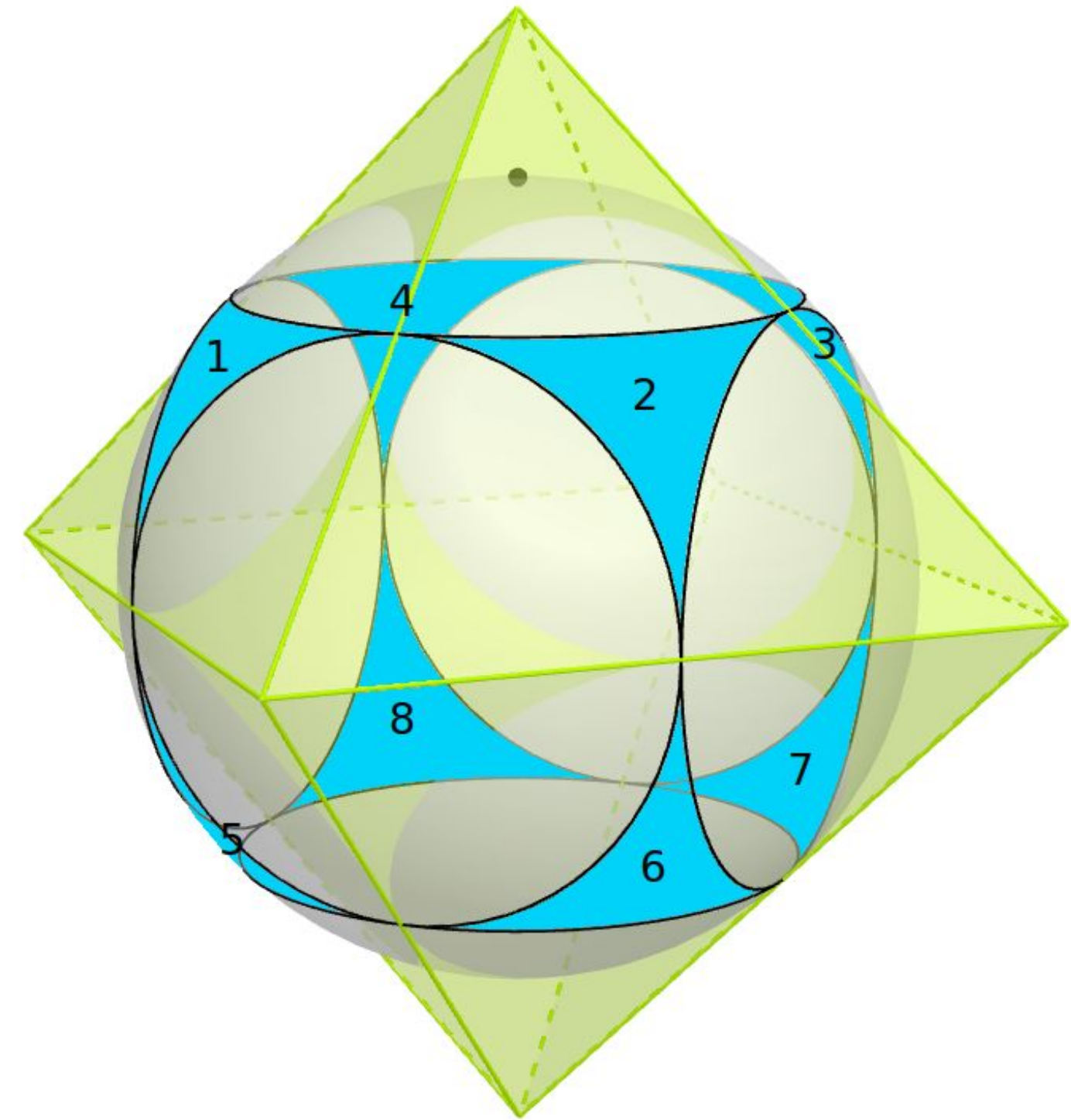
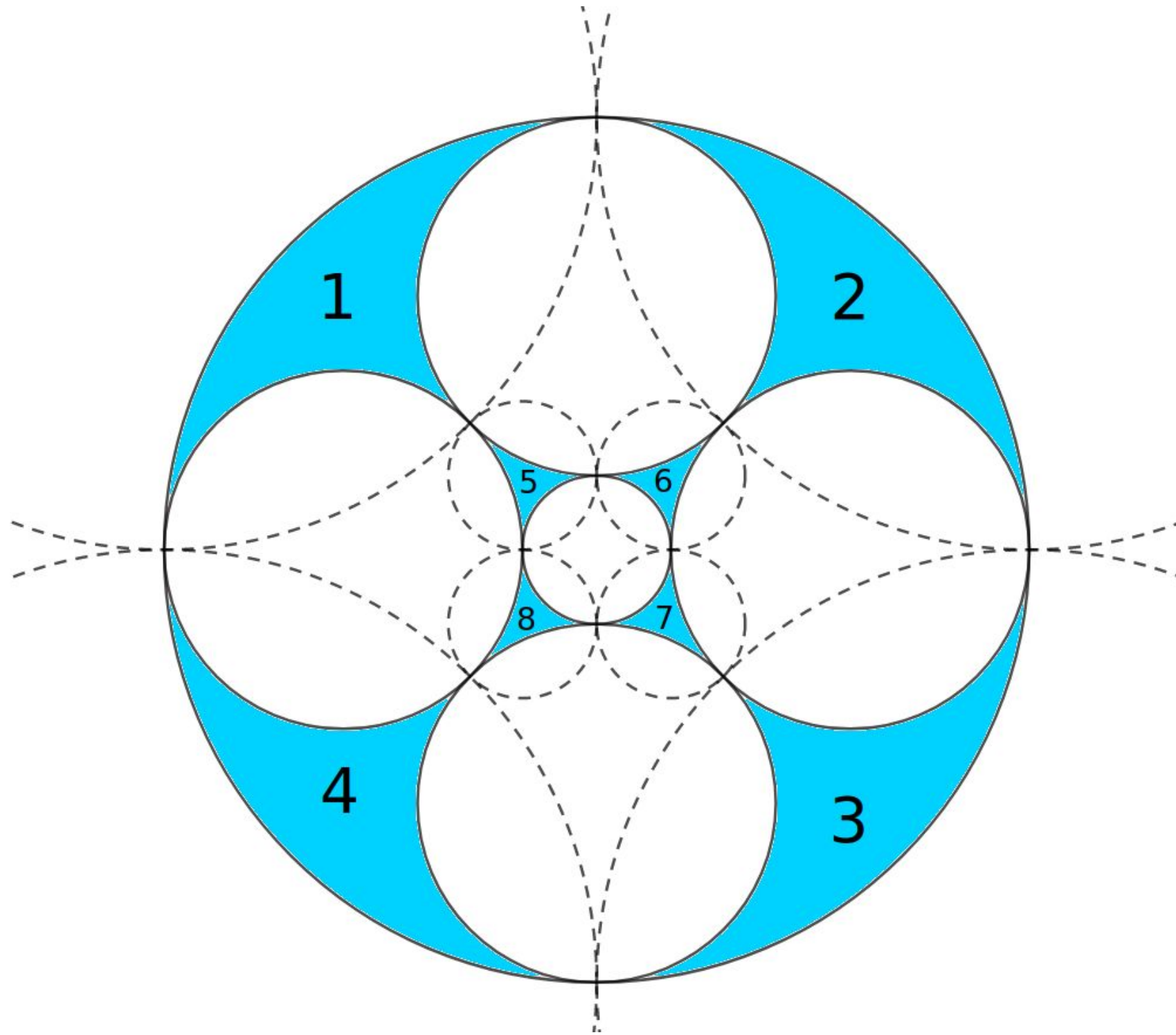
Comment sont-elles agencées ?

Adjacences des faces fractales = adjacences des faces du polyèdre



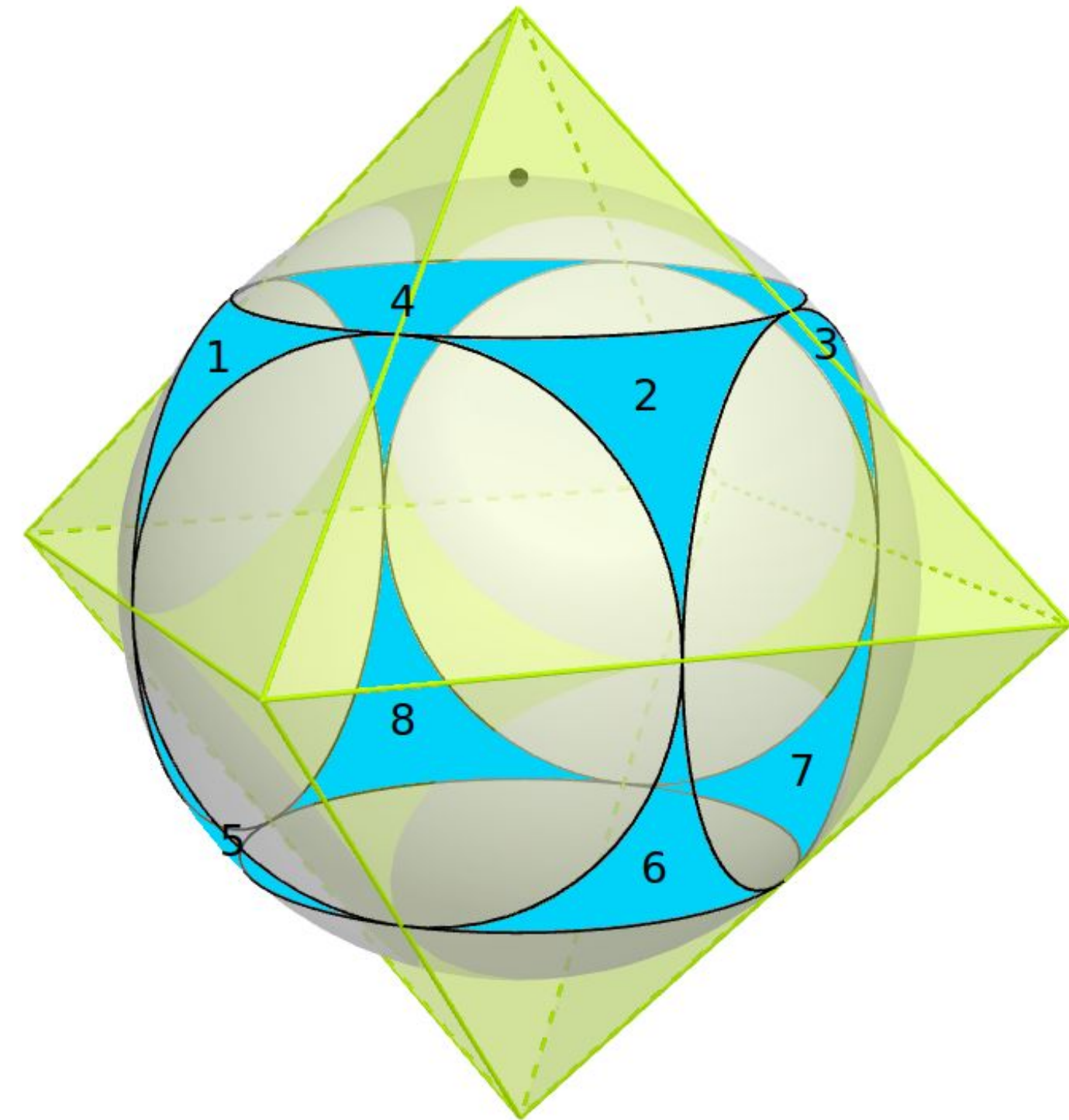
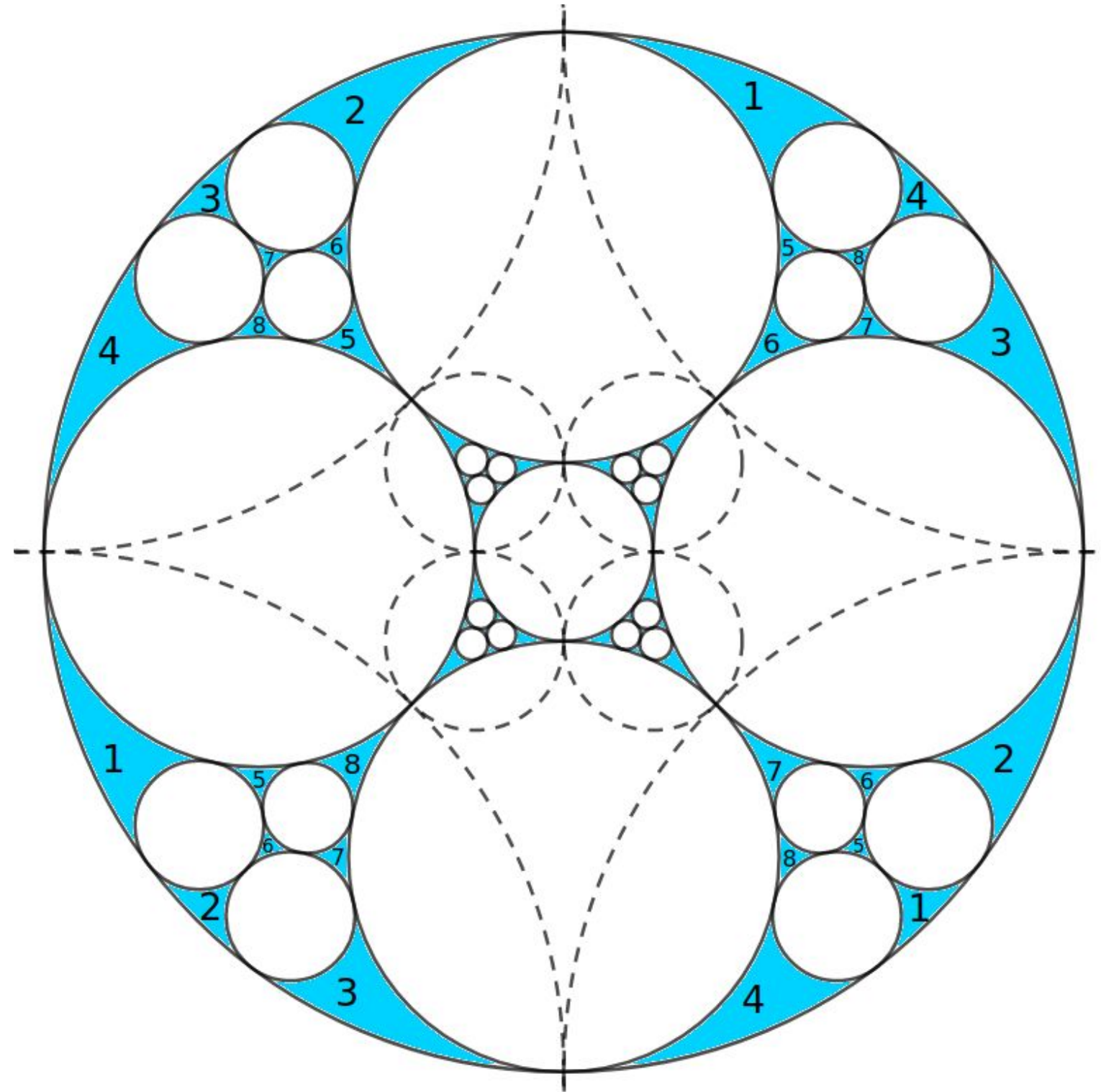
Combien de bords par face ?

Nombre de bords = nombre de sommets



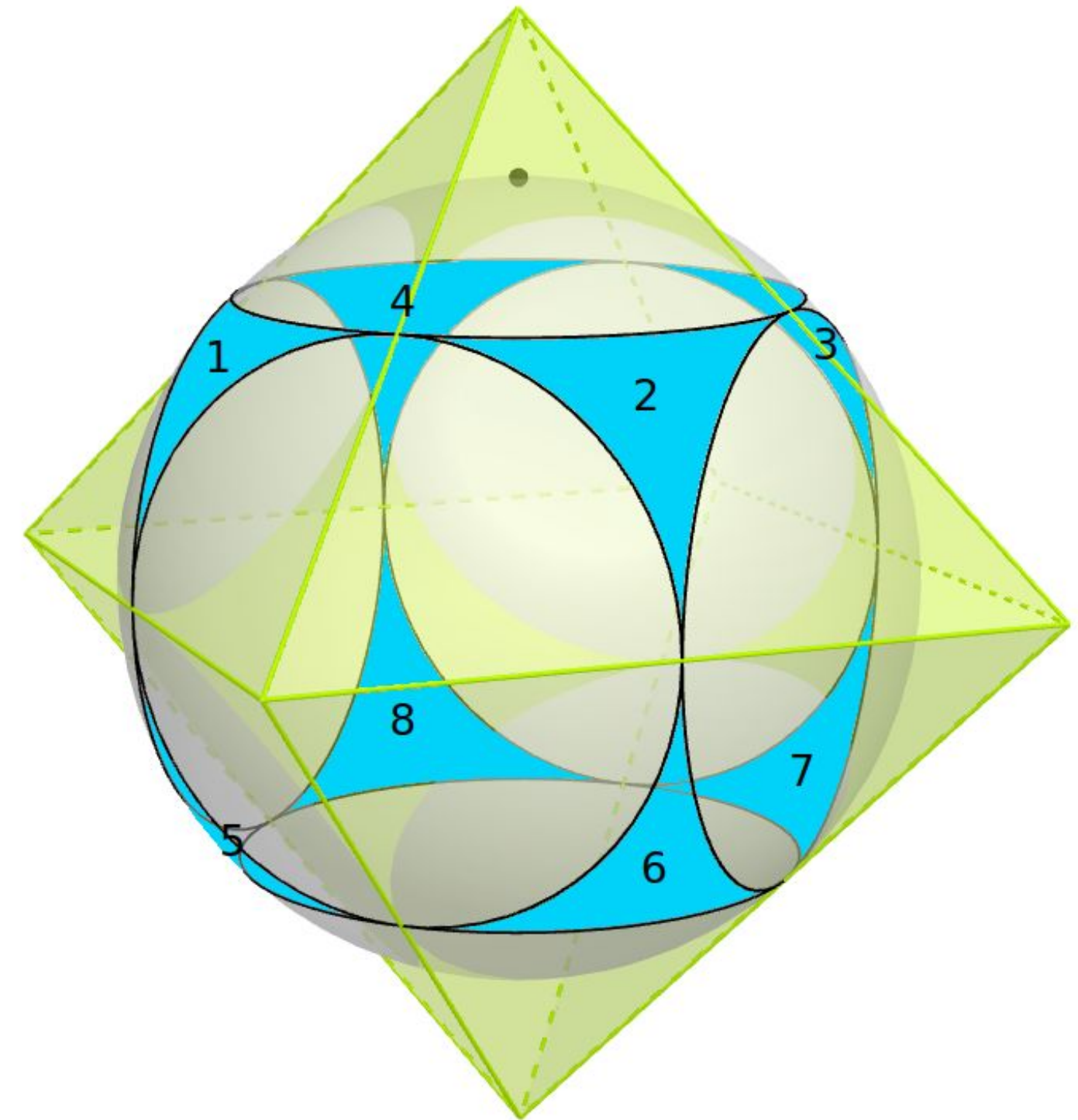
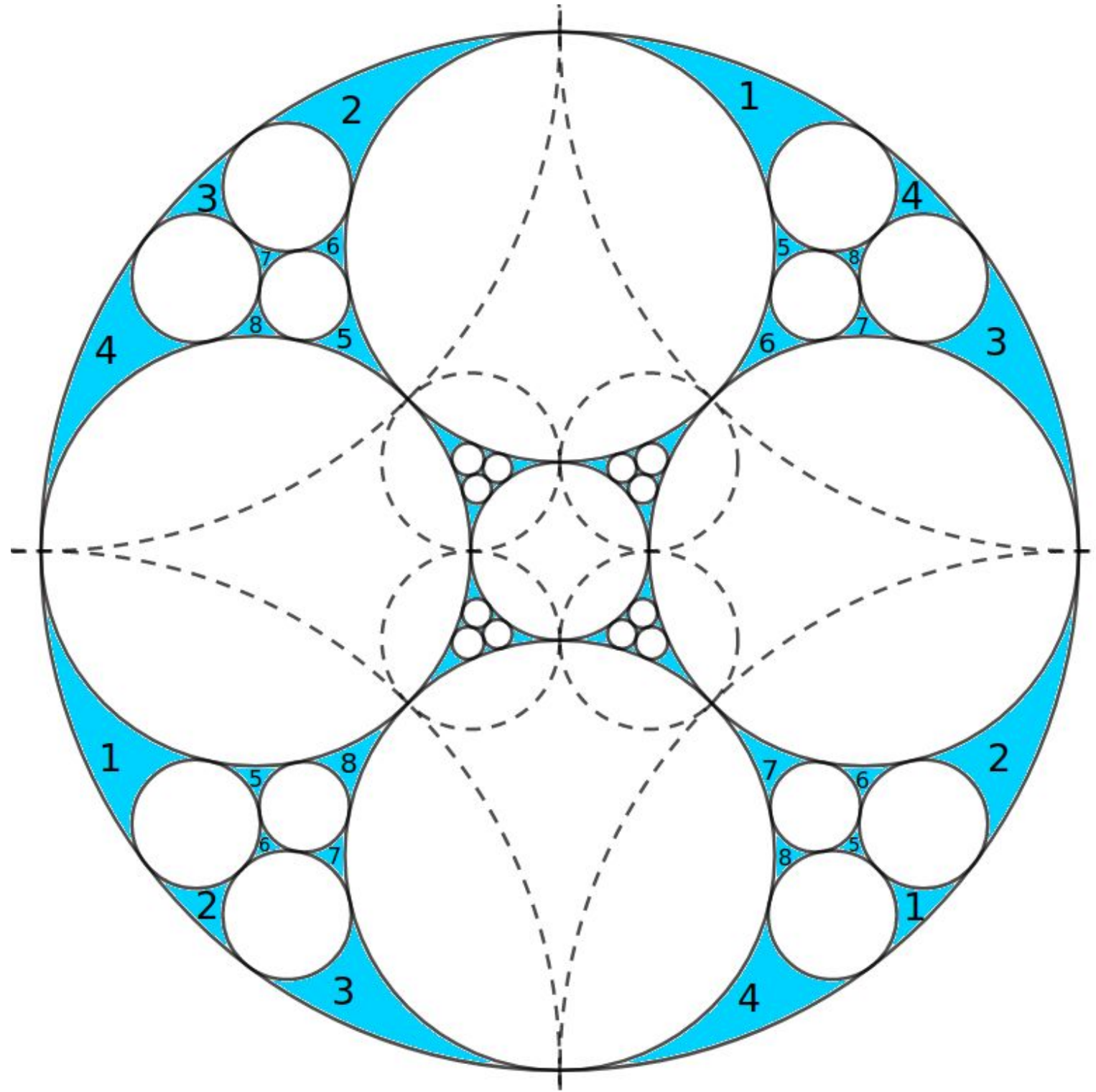
Comment se subdivisent-elles ?

Nombre de subdivisions = nombre de faces du polyèdre - 1



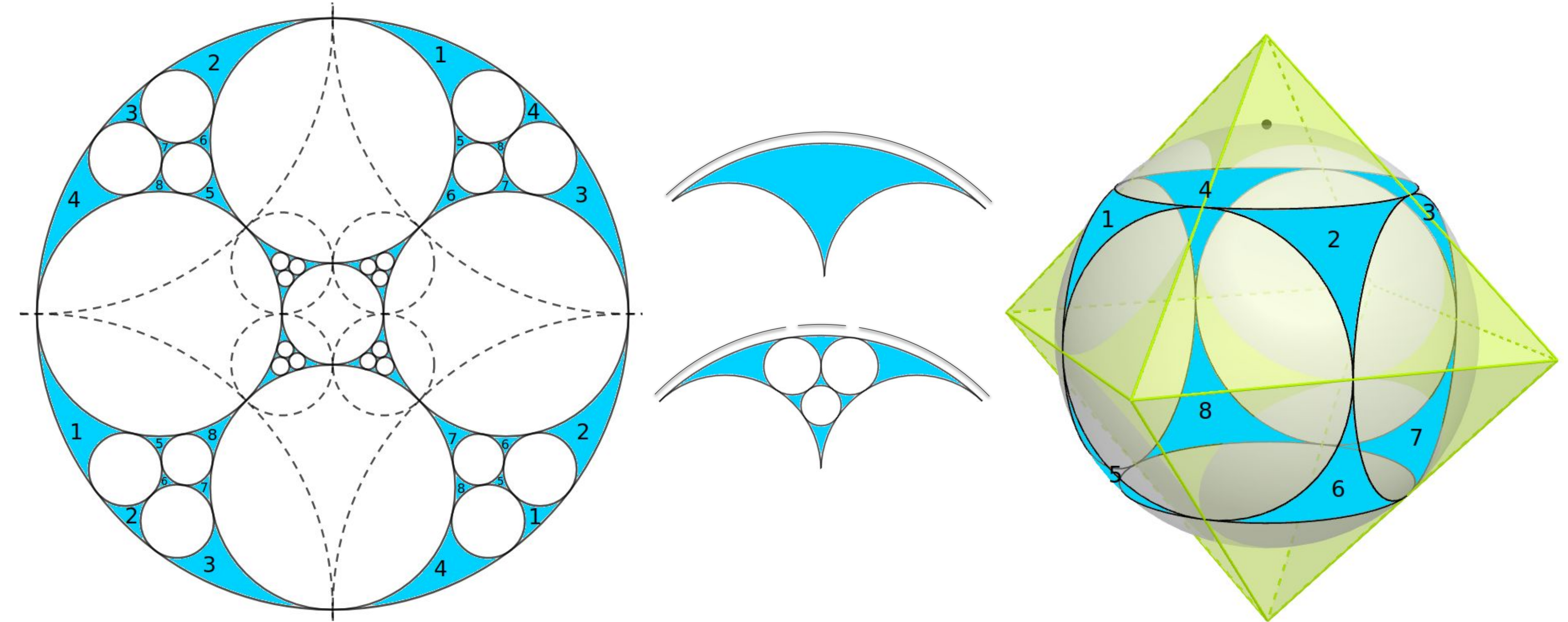
Quelles contraintes d'adjacence ?

Adjacences des subdivisions = adjacences des faces du polyèdre



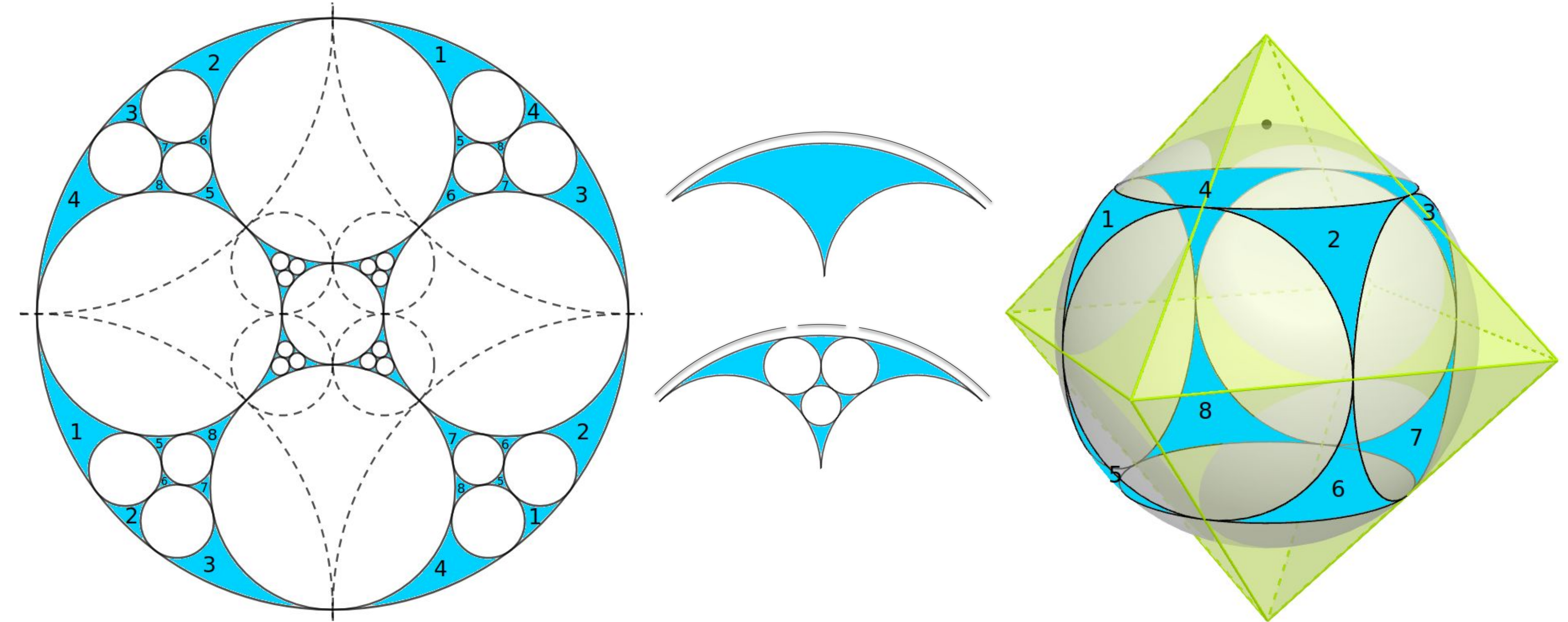
Comment se subdivisent les bords ?

Nombre de subdivisions des bords = degré du sommet associé - 1



Quelles contraintes d'incidence ?

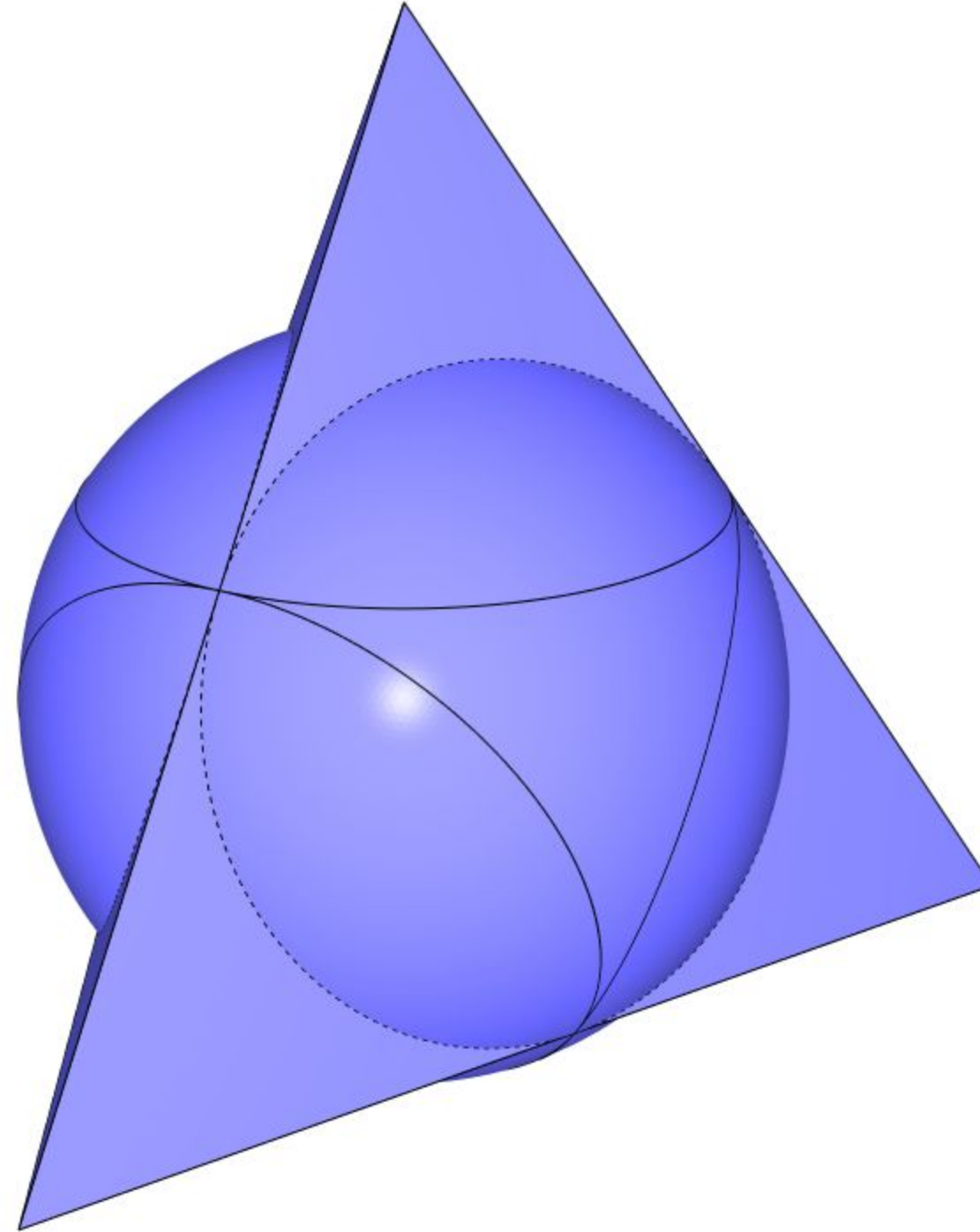
Incidences des subdivisions = mêmes sommets représentants

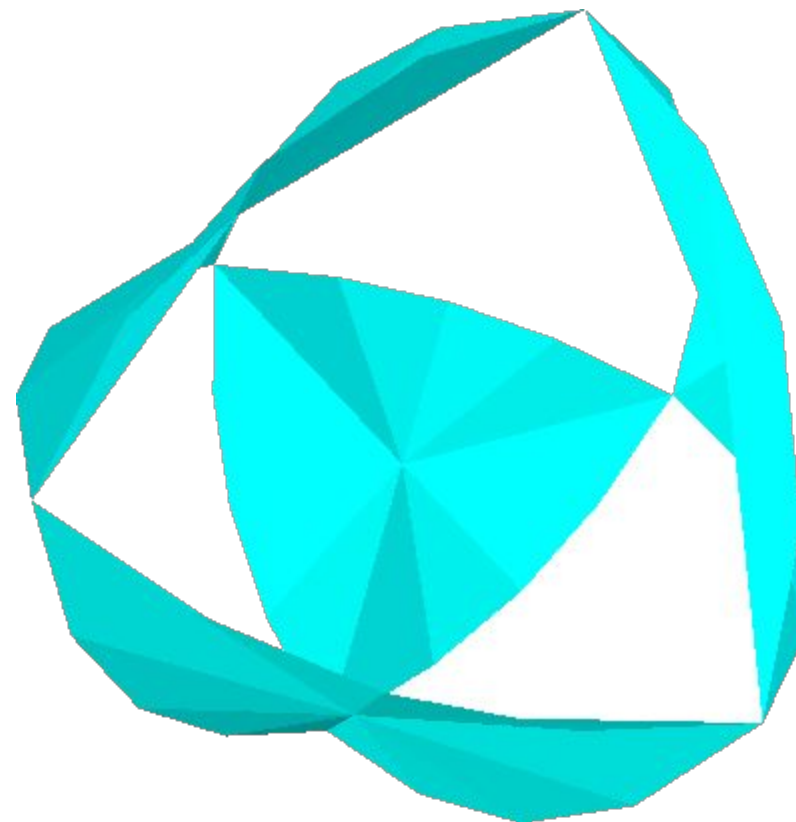


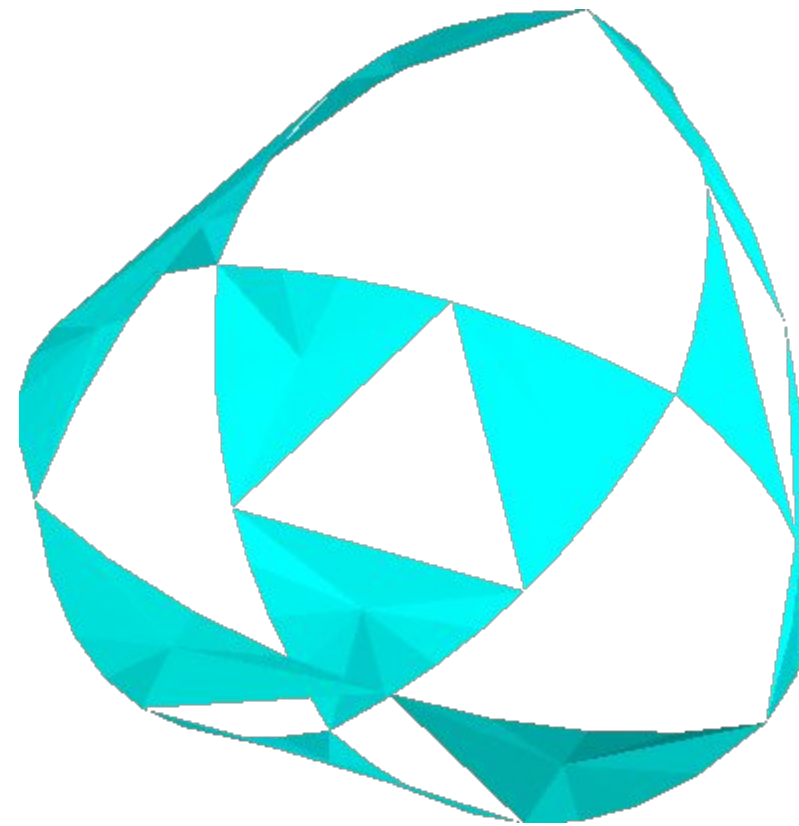
- Combien de faces ?
 - Une face fractale par face du polyèdre
- Comment sont-elles agencées ?
 - Adjacences des faces fractales = adjacences des faces du polyèdre
- Combien de bords par face ?
 - Nombre de bords = nombre de sommets
- Comment se subdivisent-elles ?
 - Nombre de subdivisions = nombre de faces du polyèdre - 1
- Quelles contraintes d'adjacence ?
 - Adjacences des subdivisions = adjacences des faces du polyèdre
- Comment se subdivisent les bords ?
 - Nombre de subdivisions des bords = degré du sommet associé - 1
- Quelles contraintes d'incidence ?
 - Incidences des subdivisions = mêmes sommets représentants

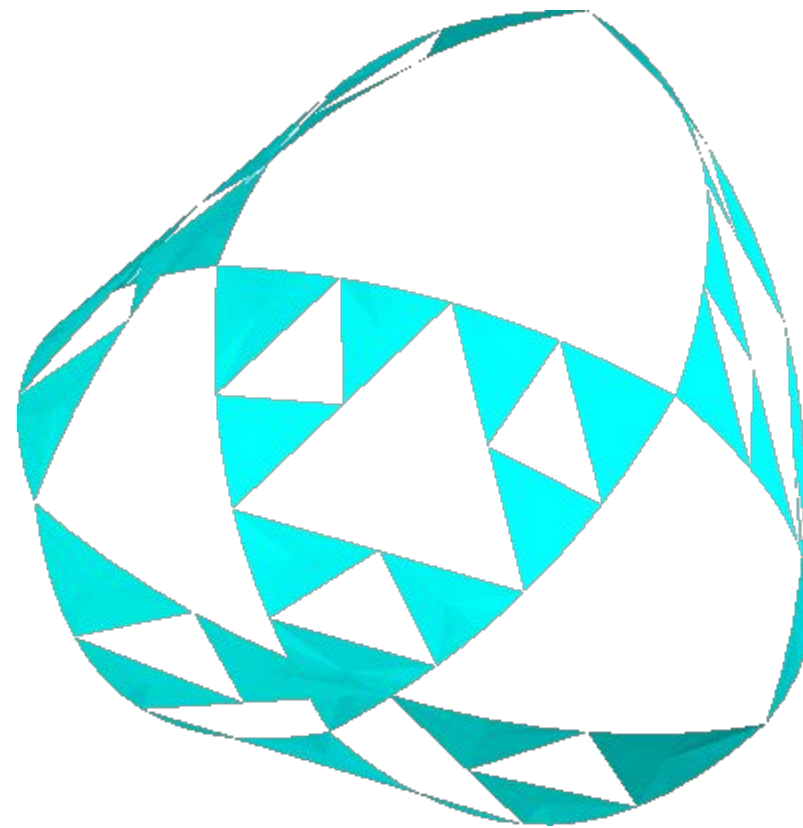
Résultats

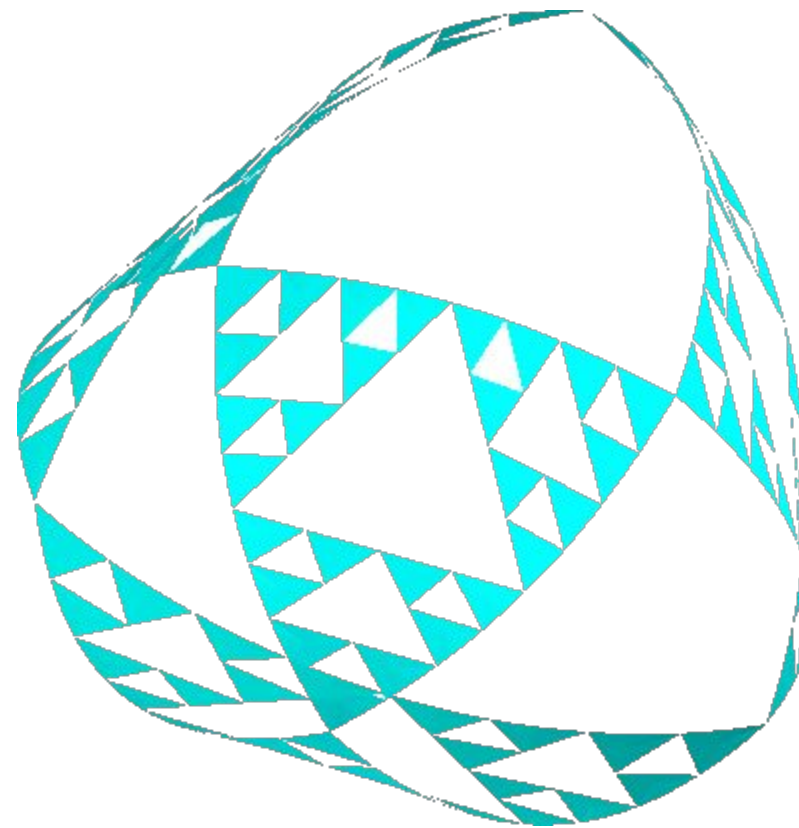


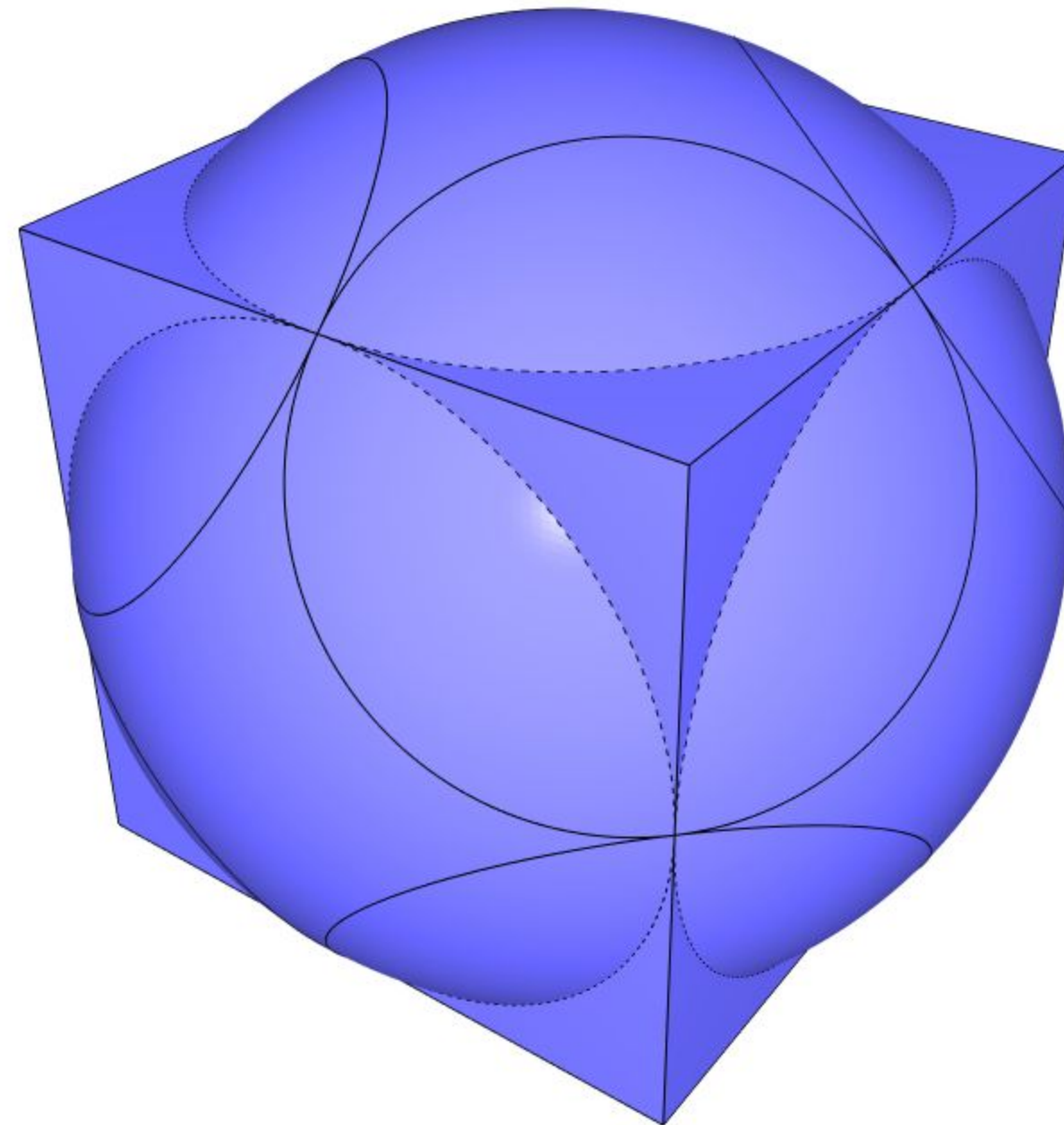


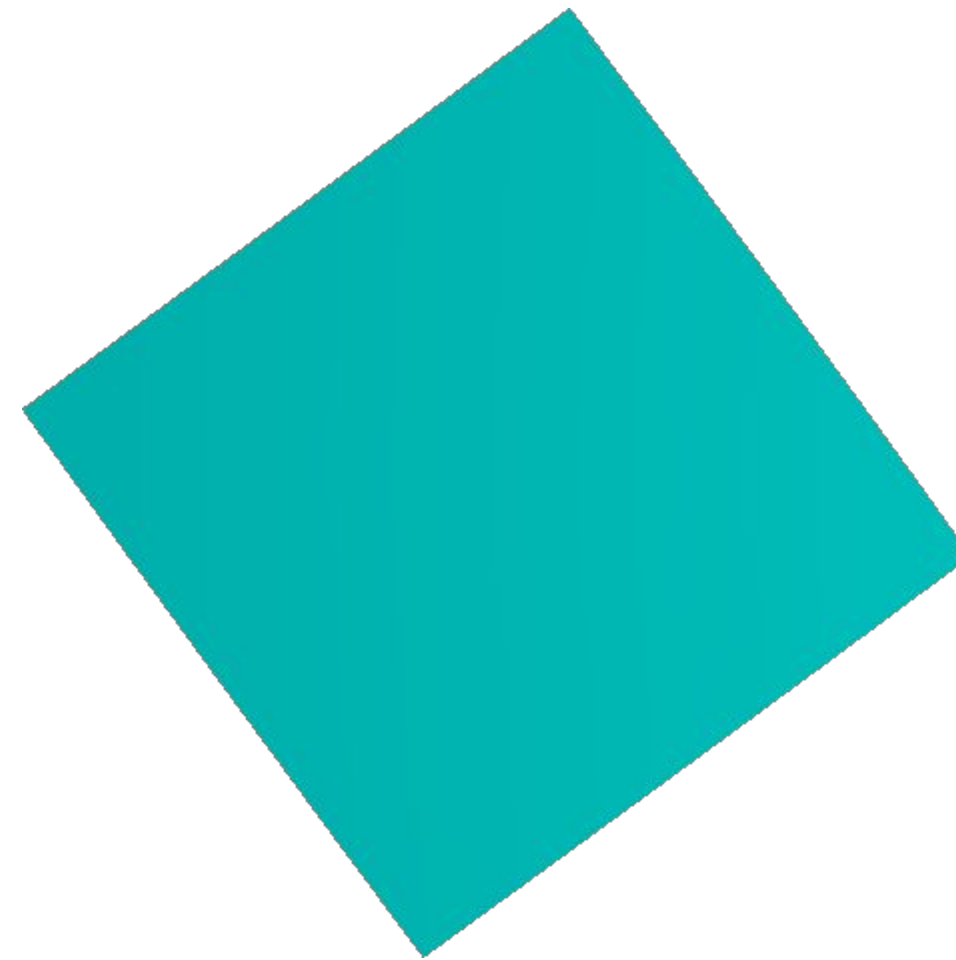


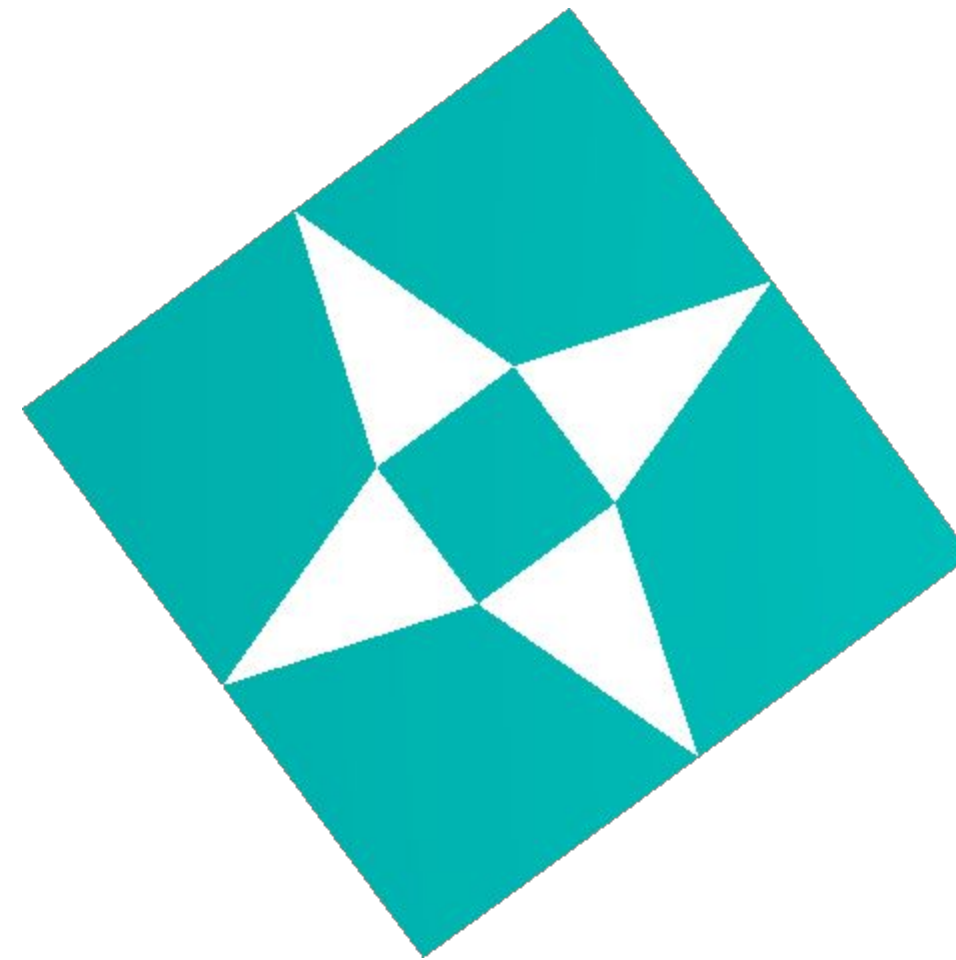


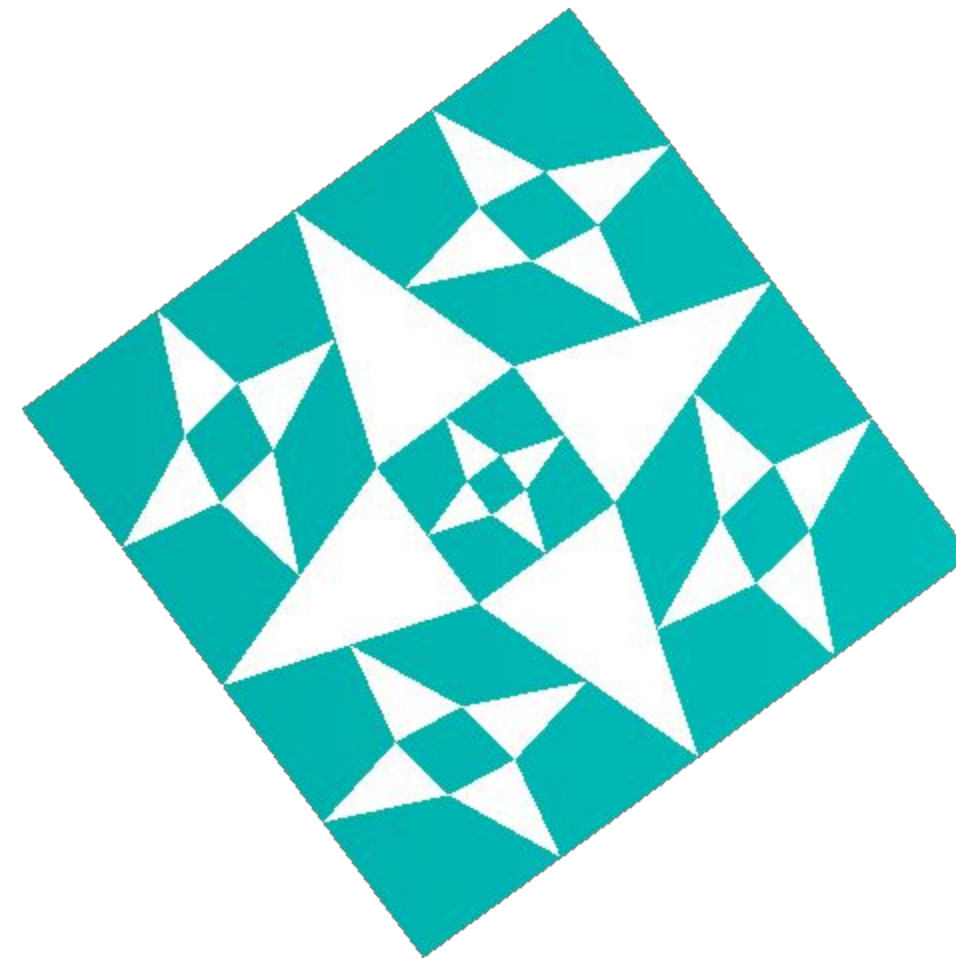


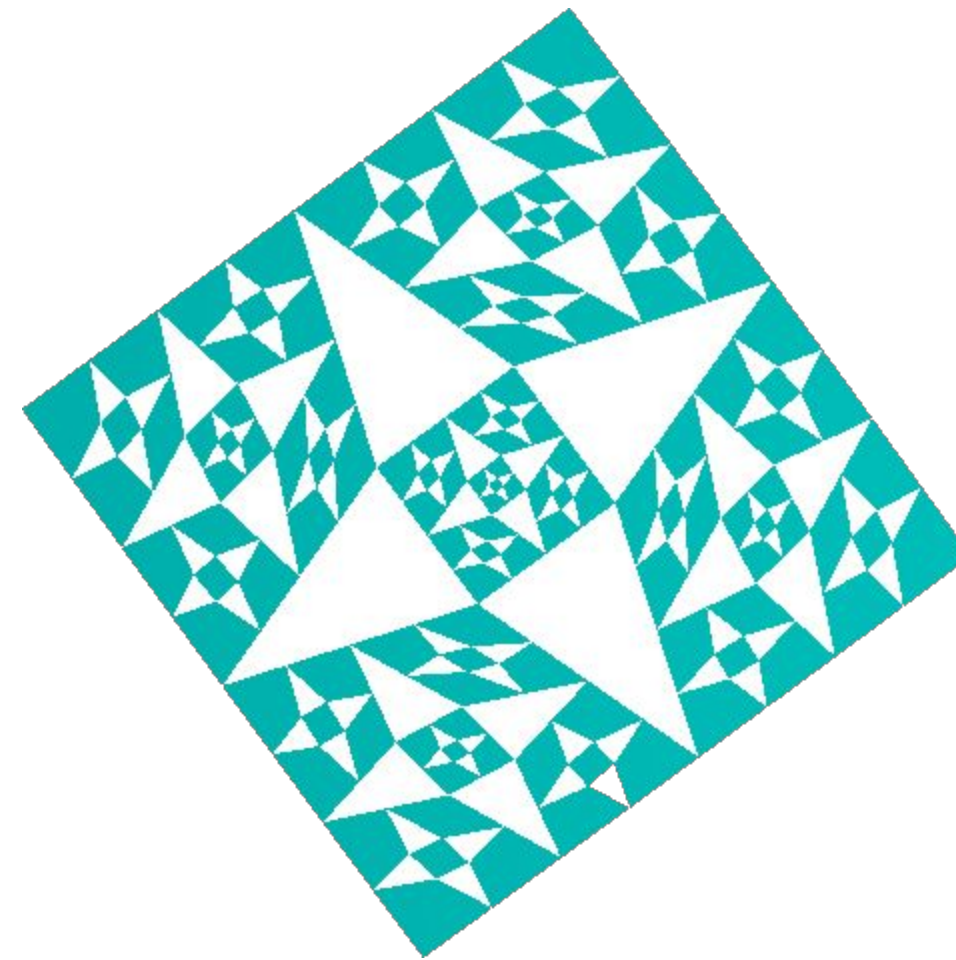


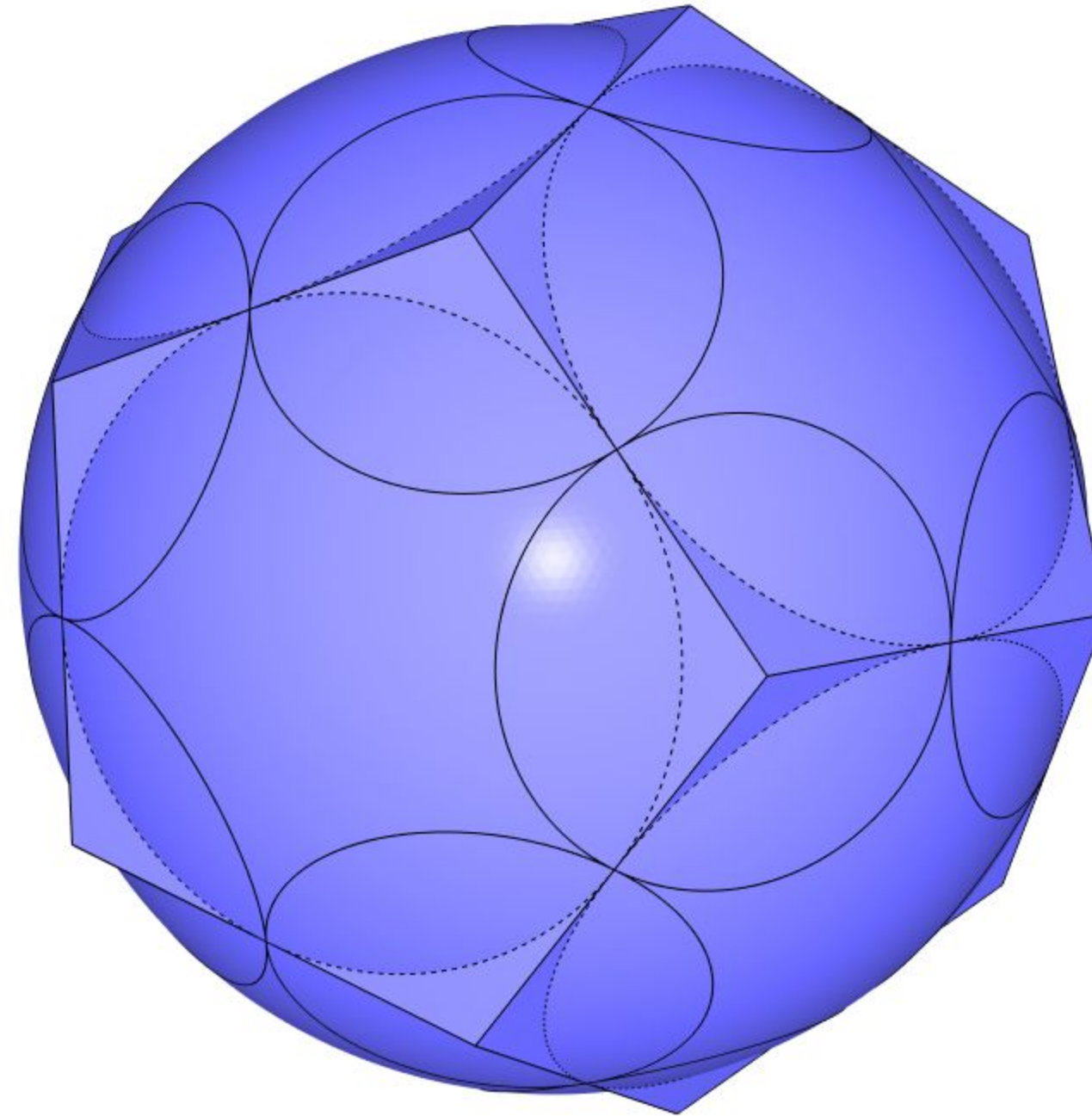


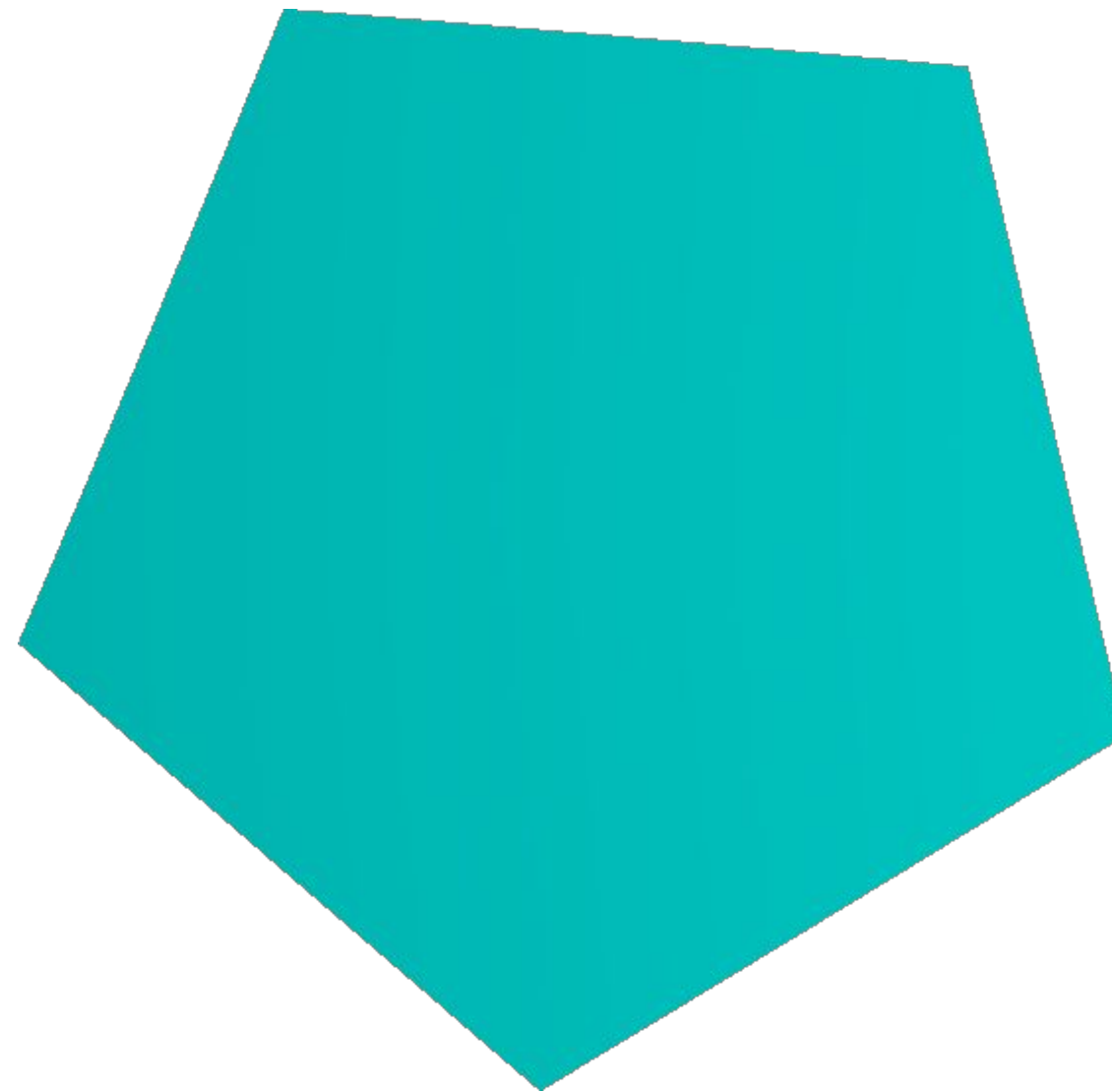


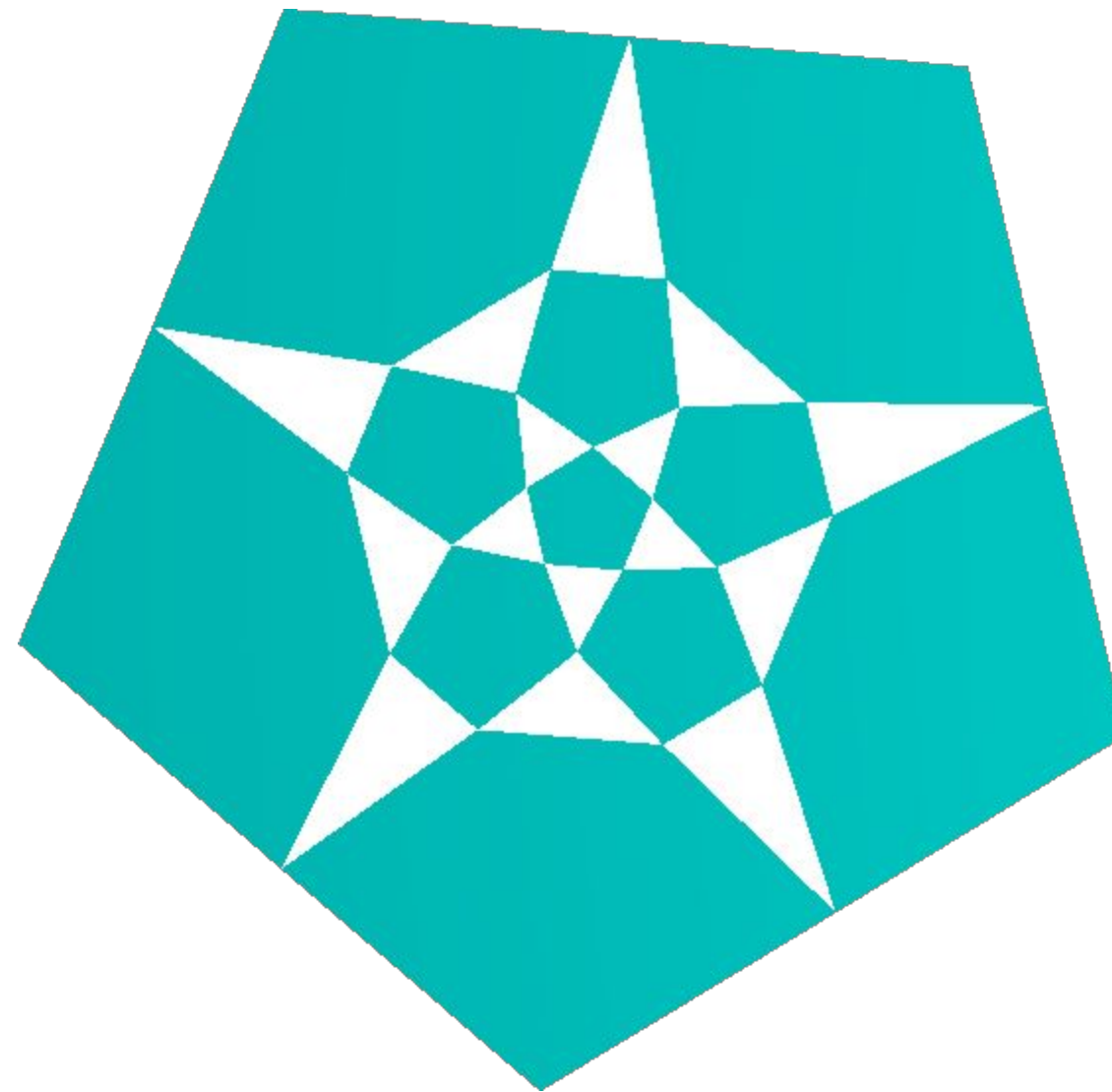


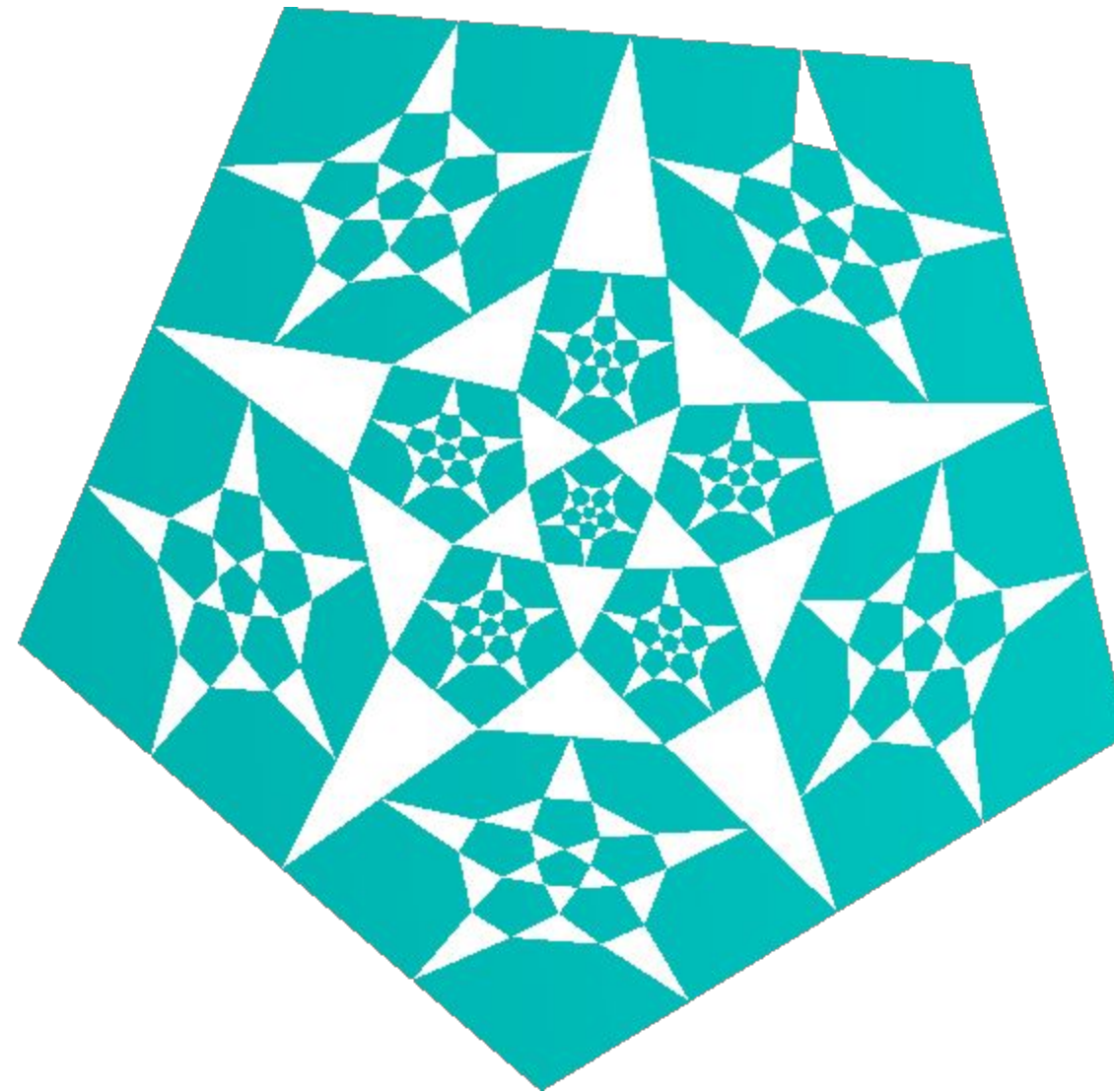


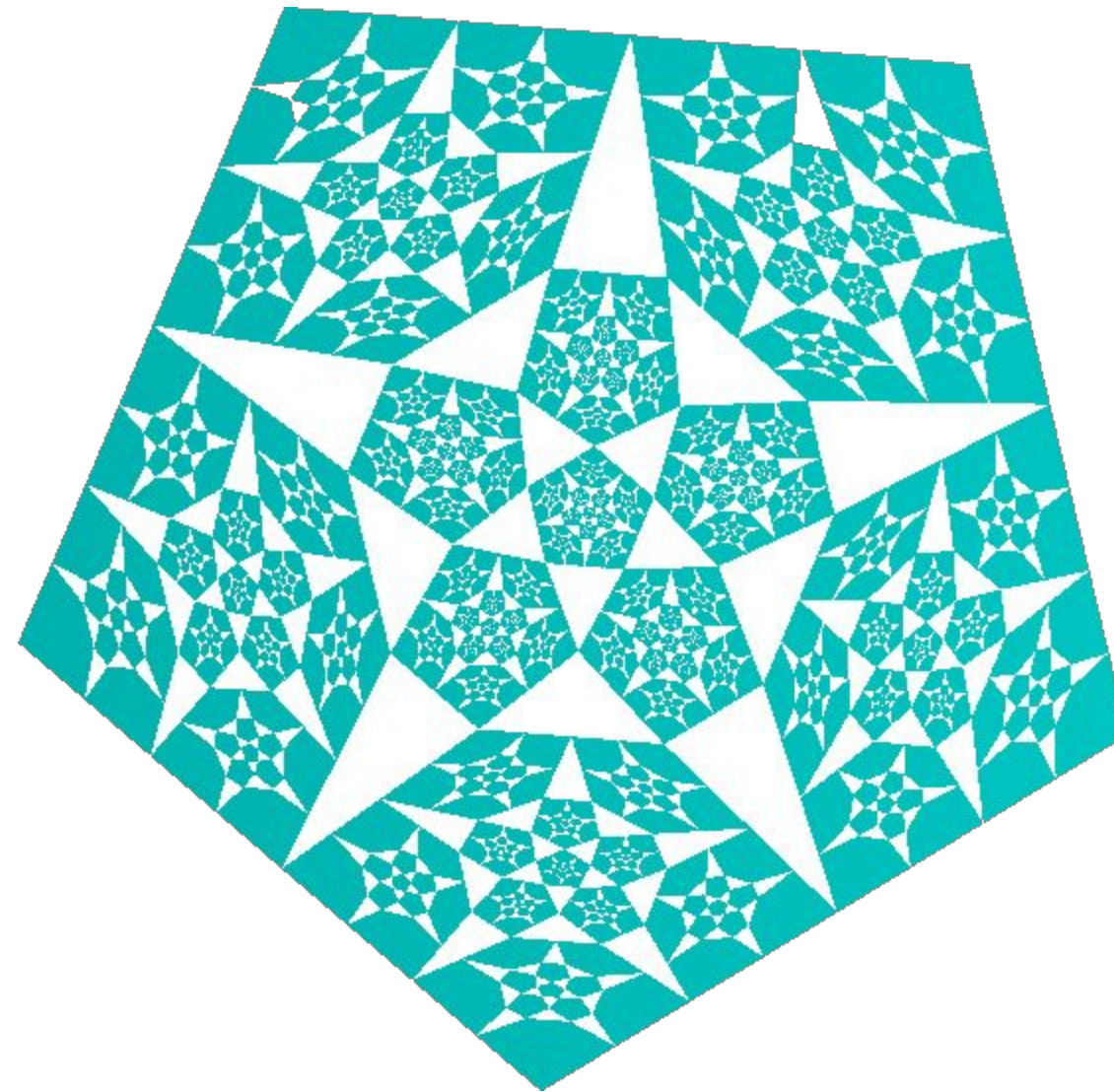


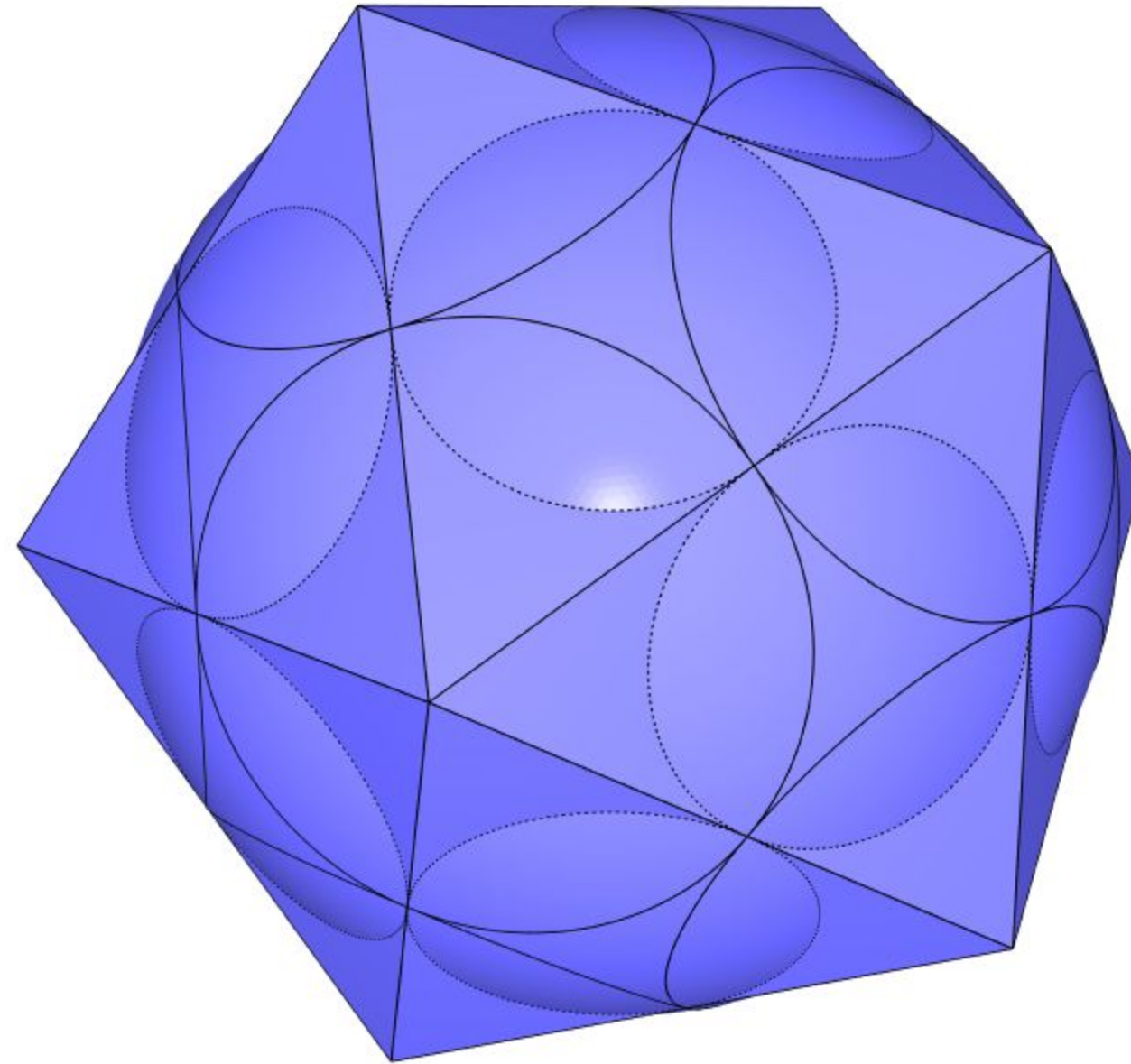


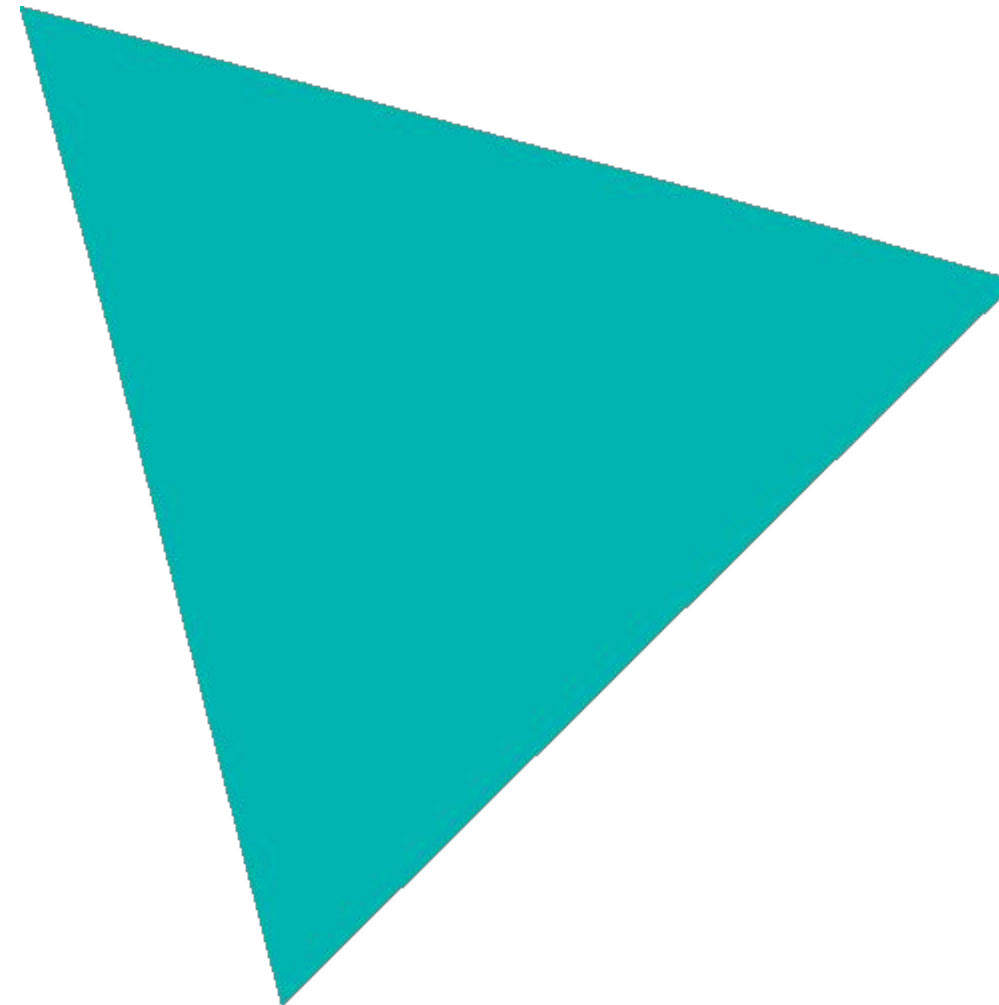


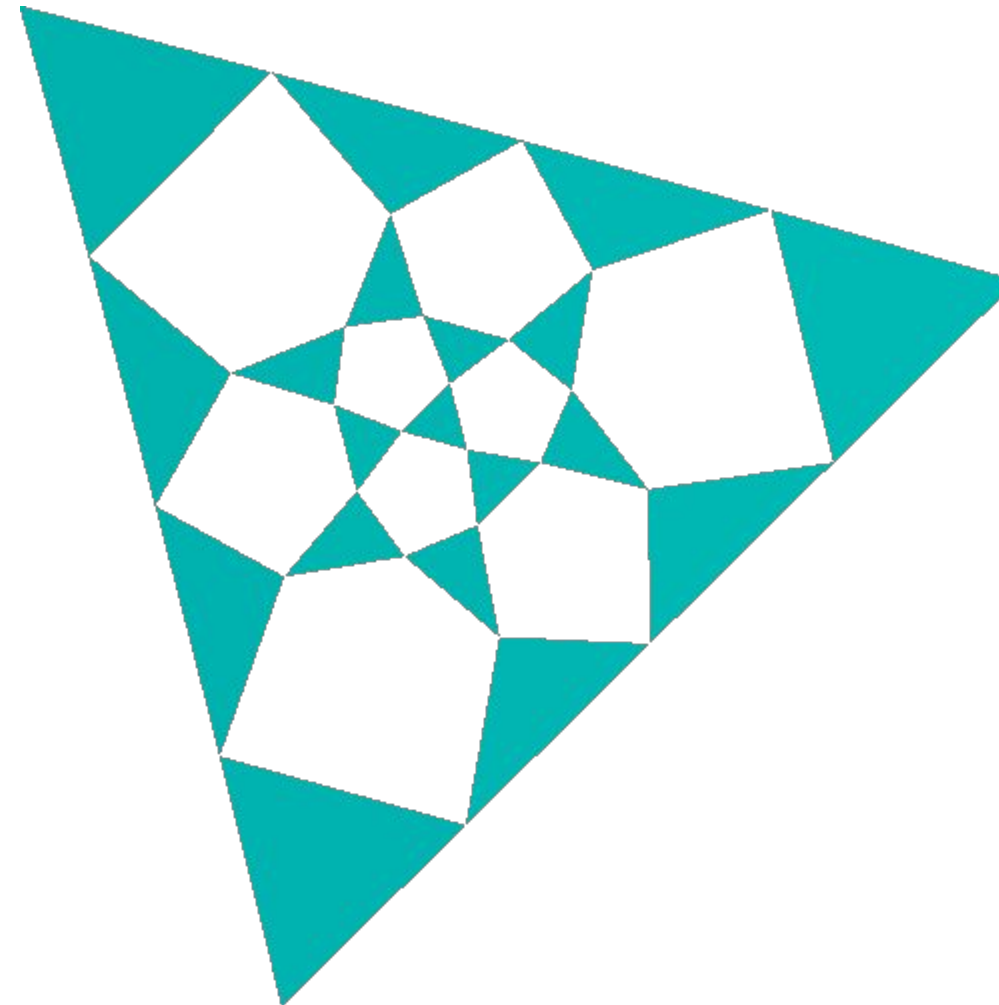


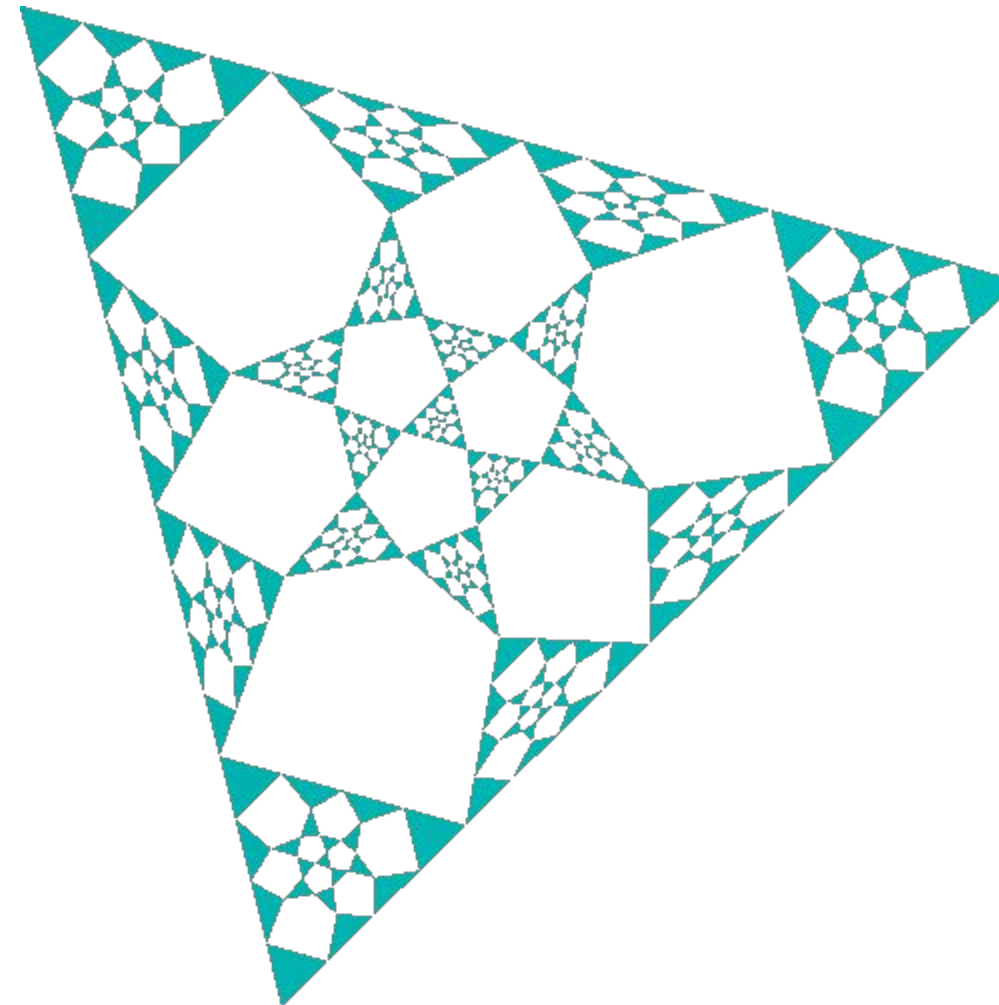


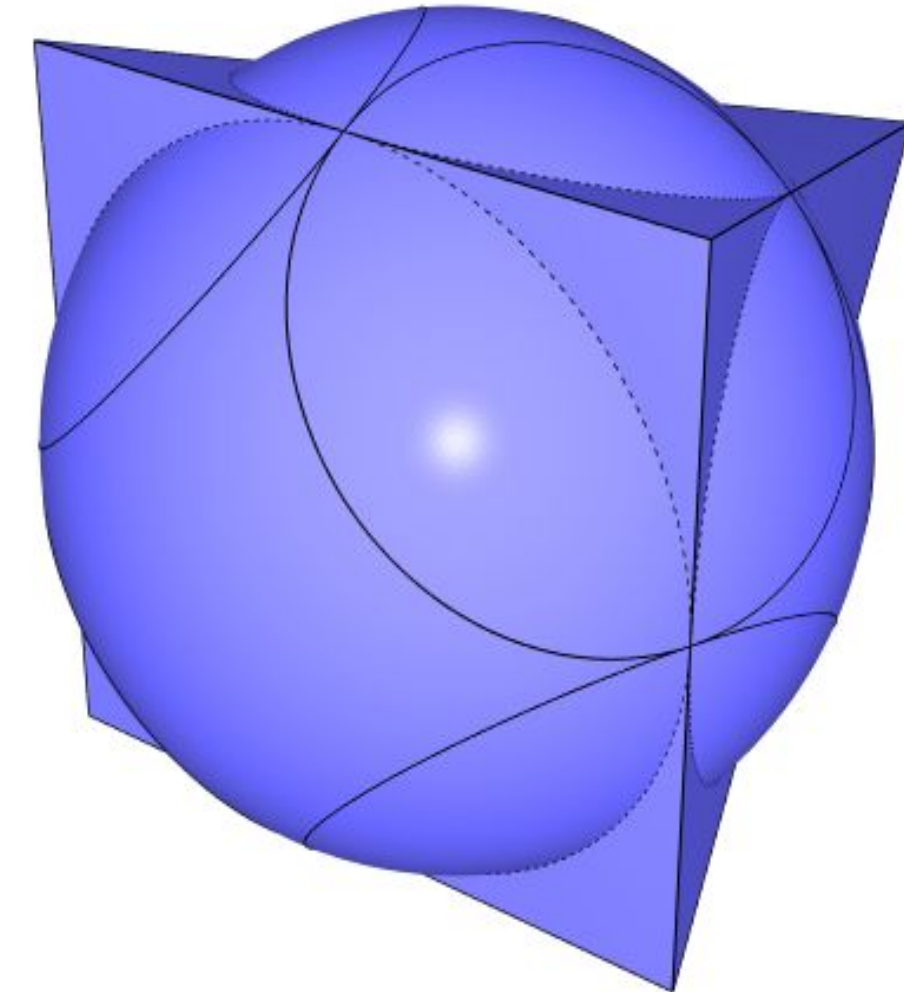
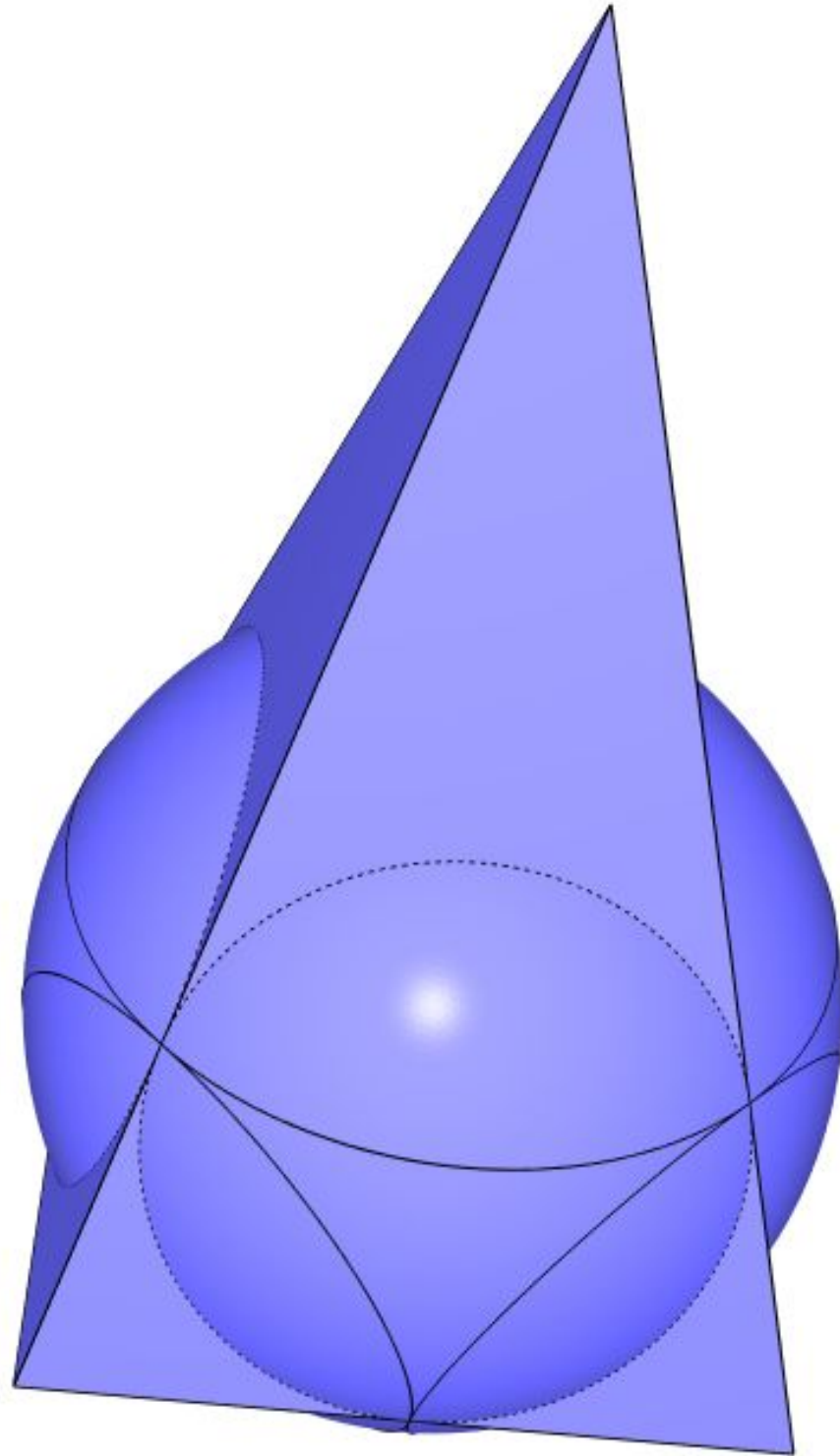


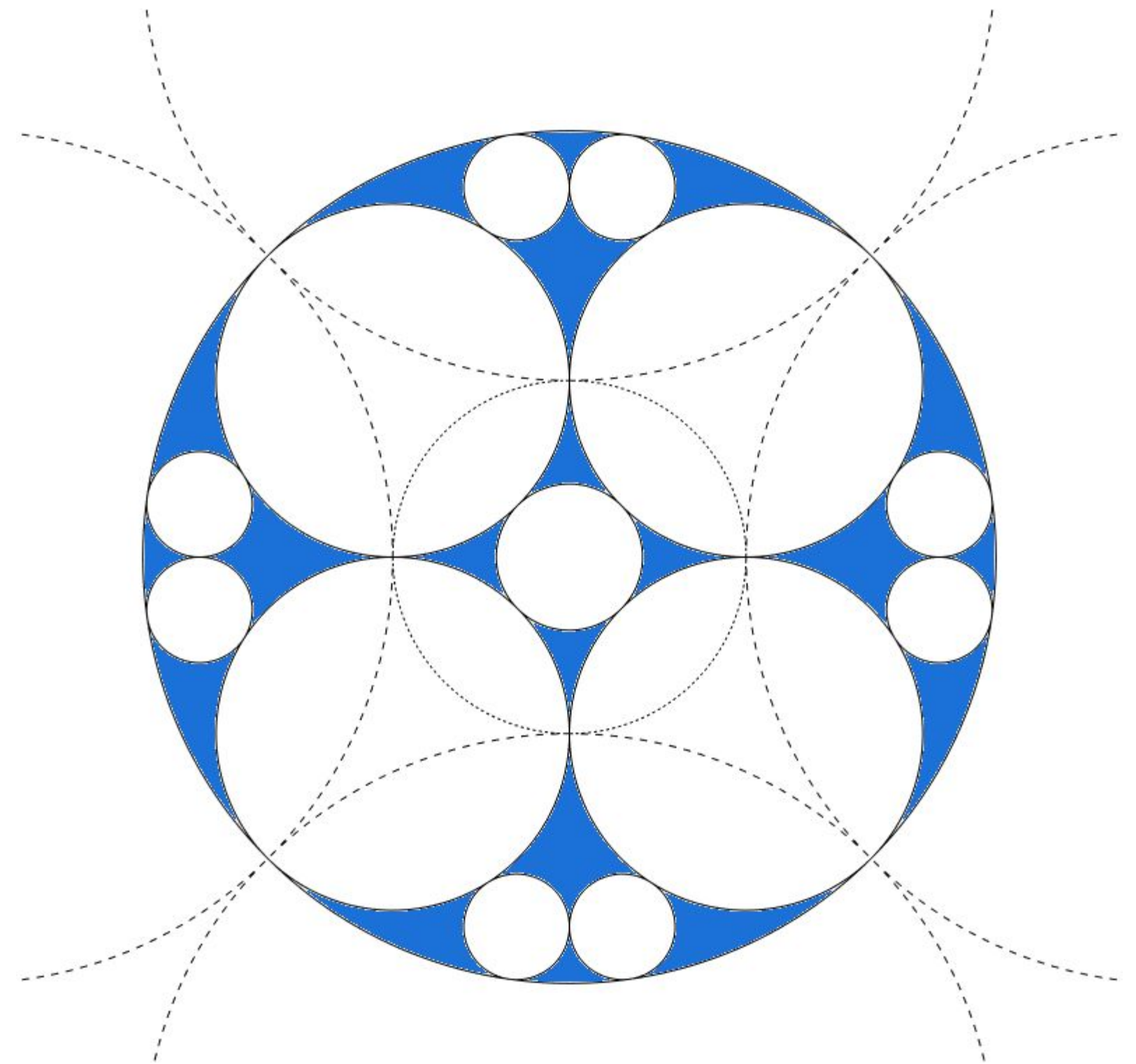
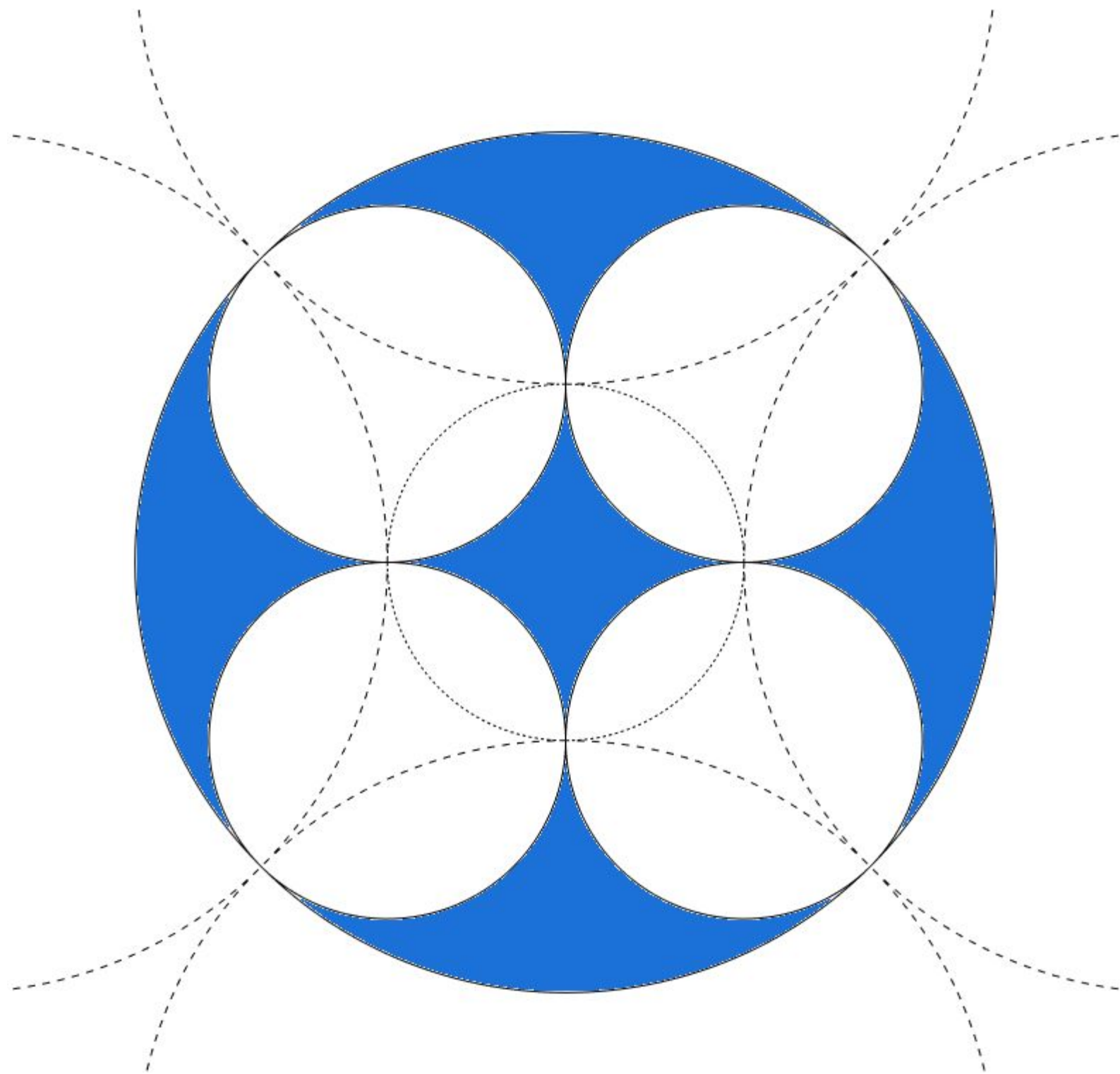


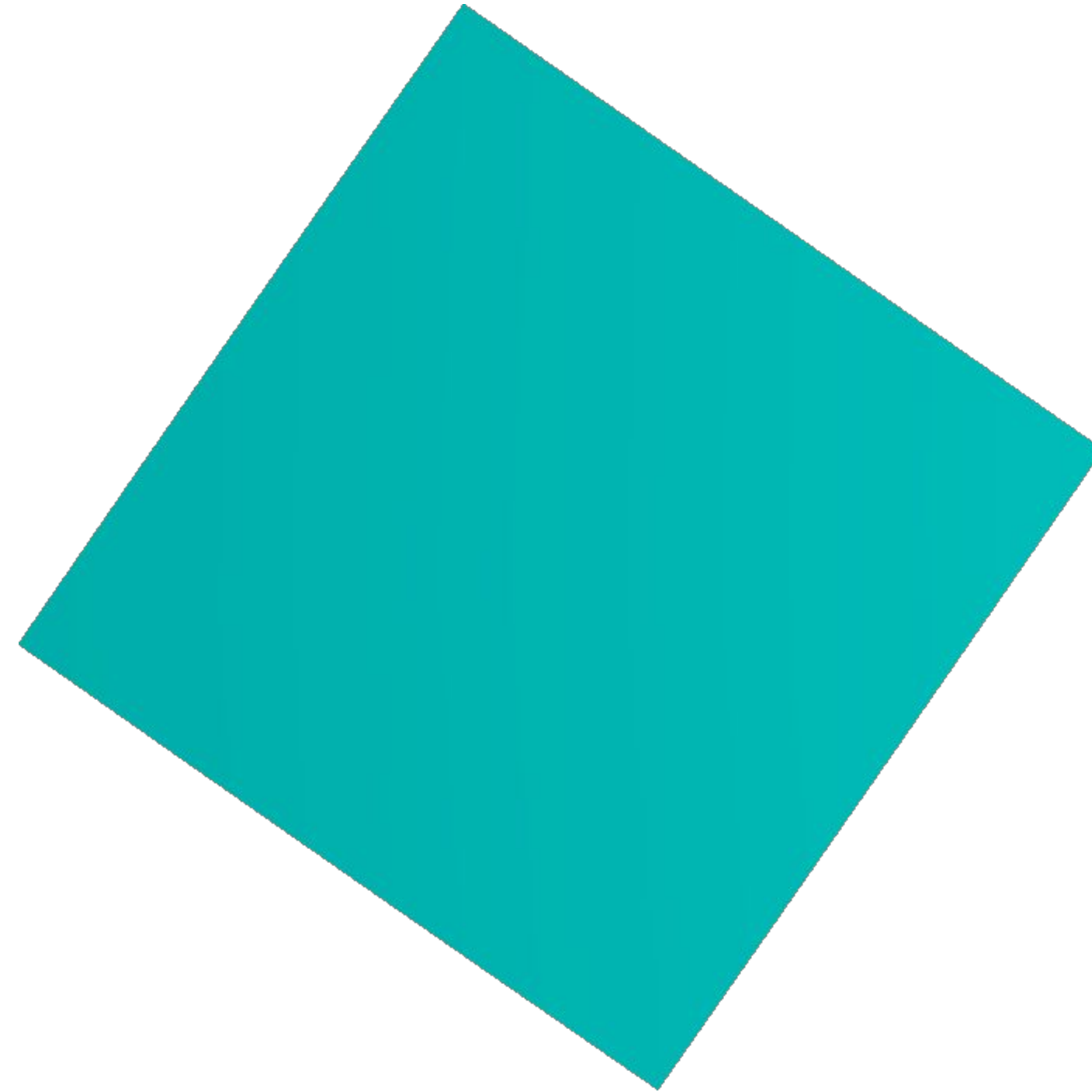






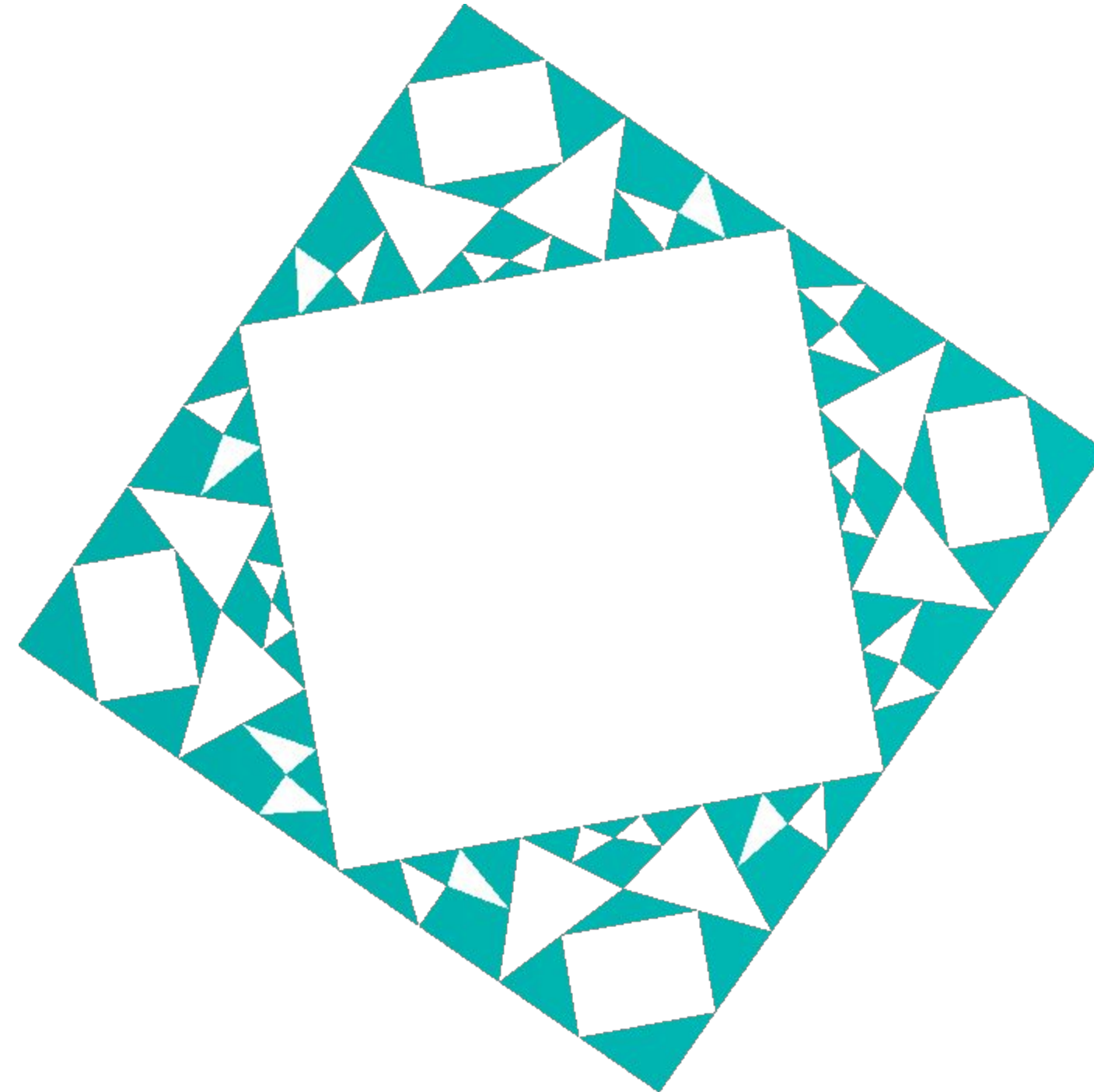


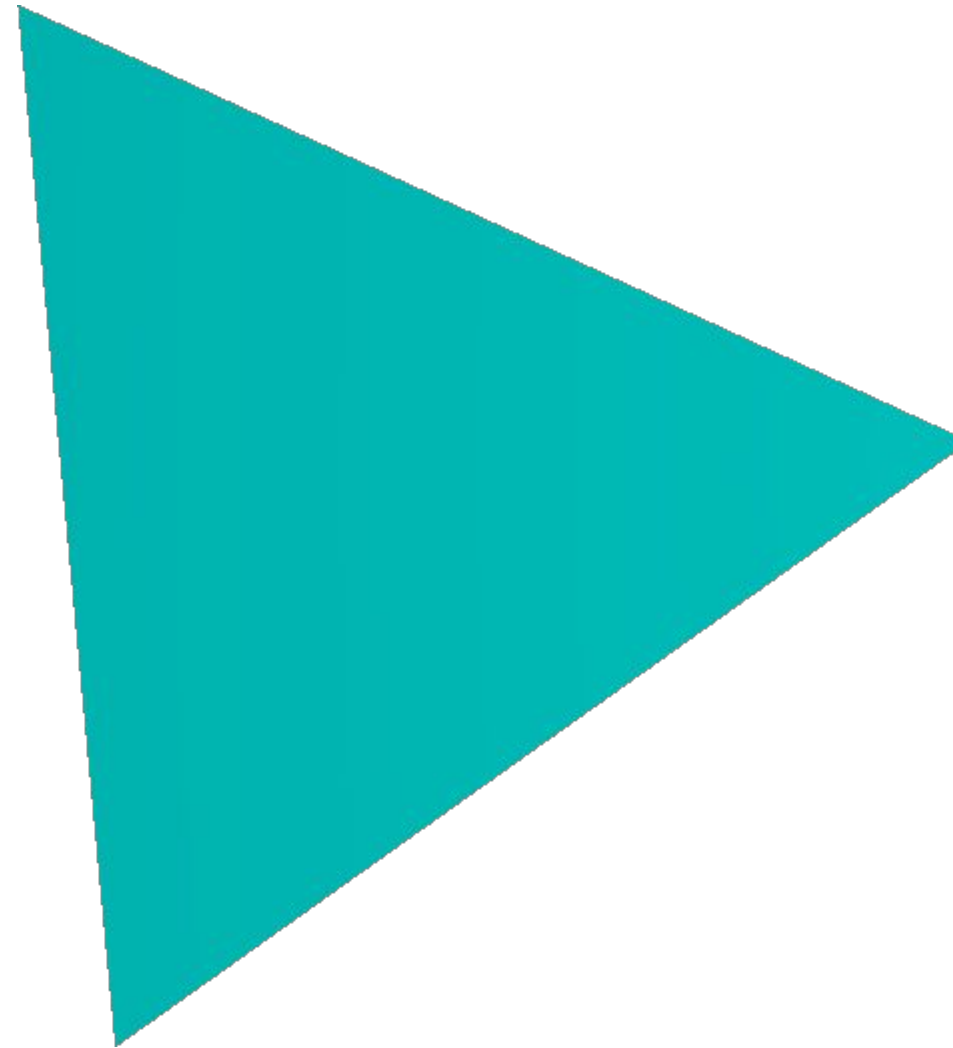


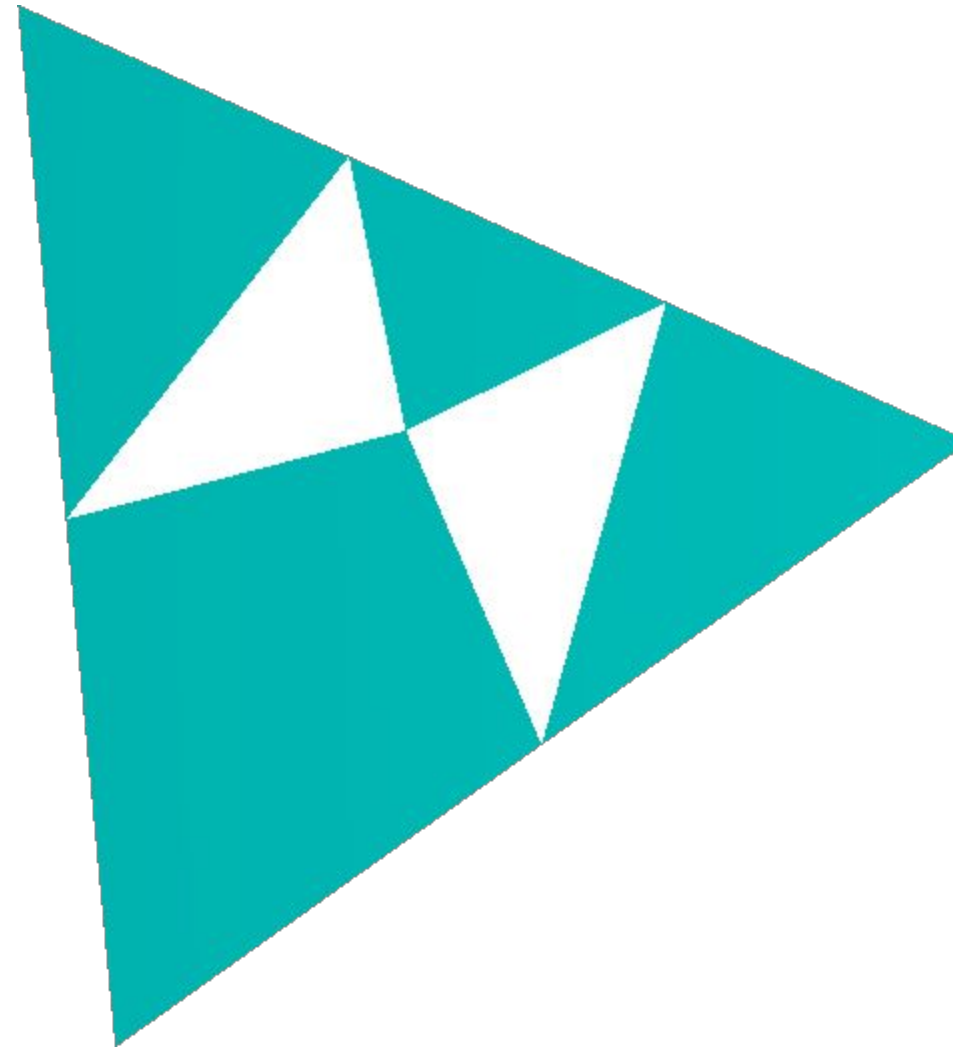




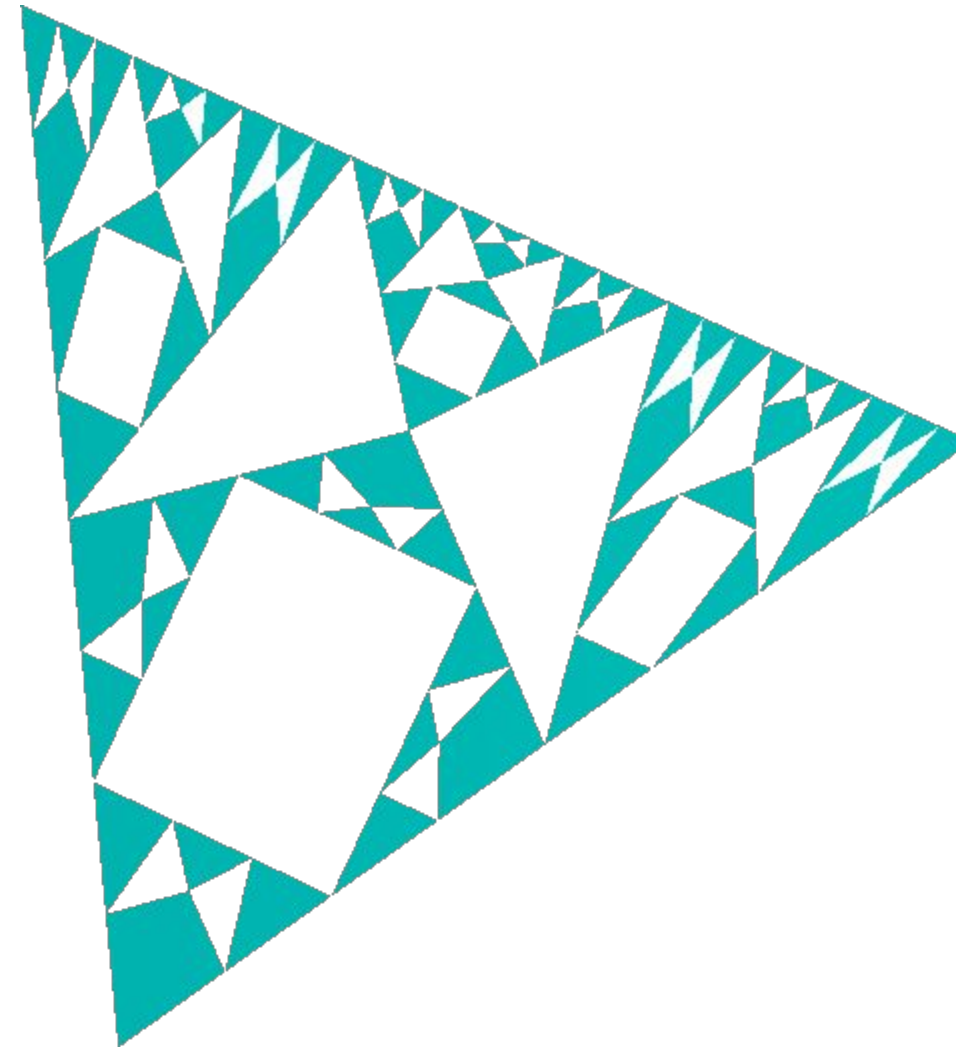








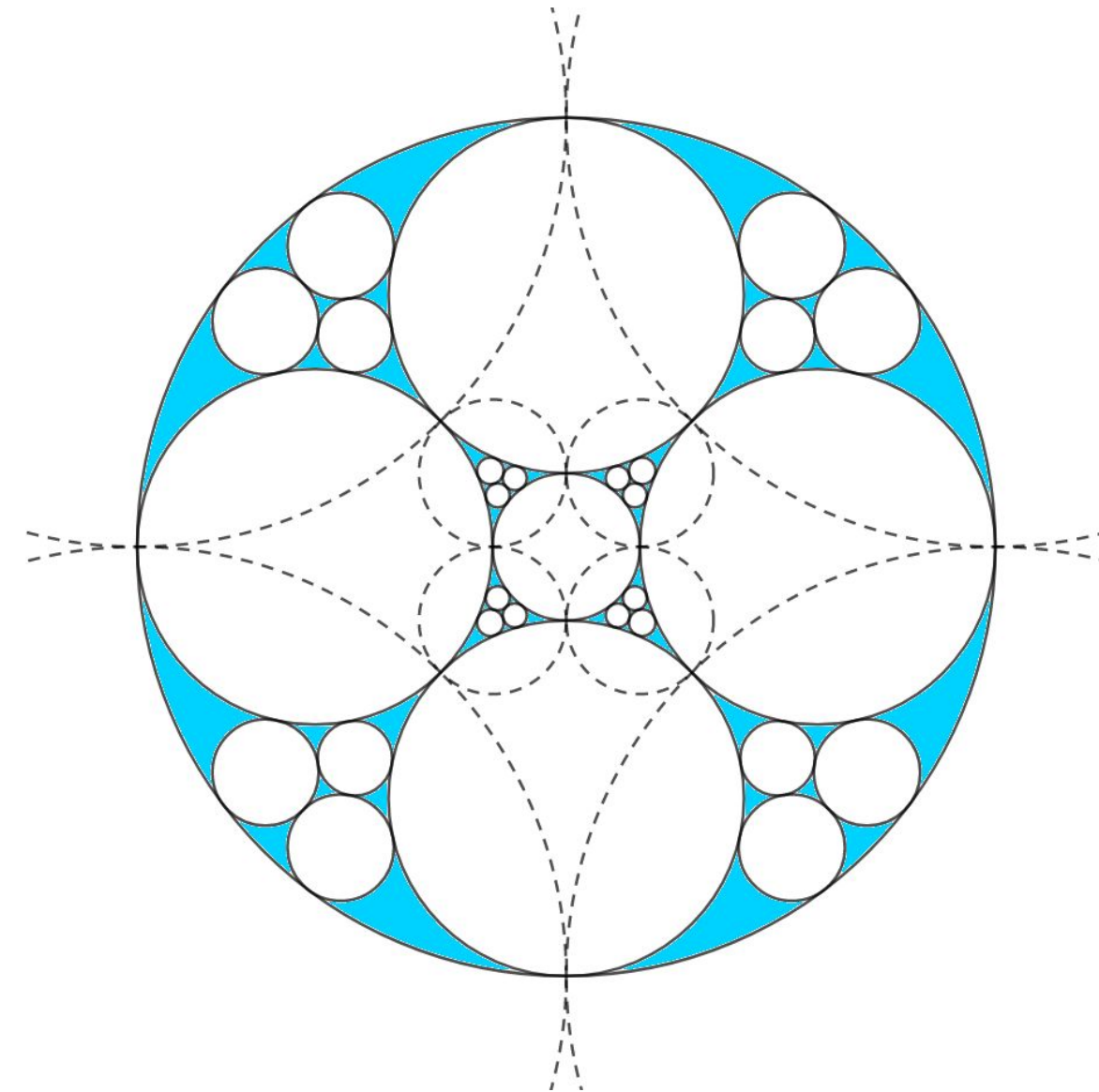
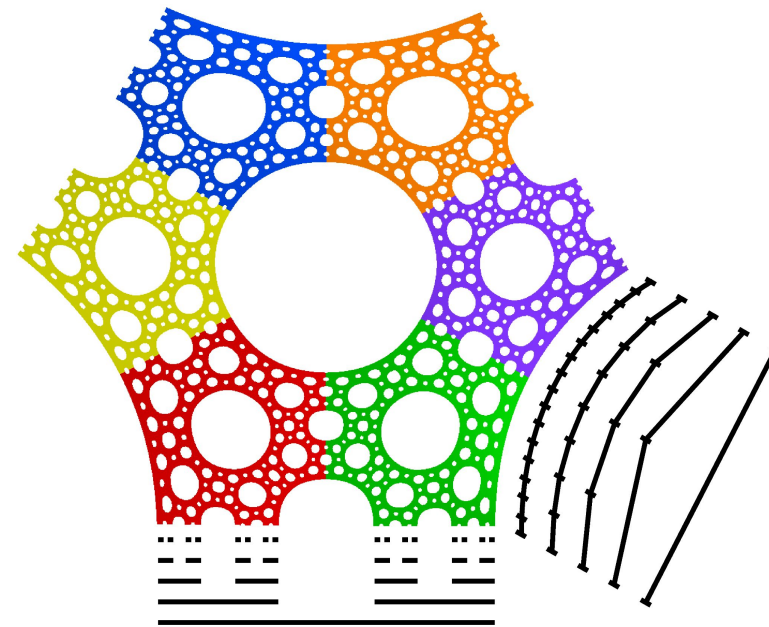




Conclusion



- Codage de la topologie fractale des empilements de sphères en 2D
- Extraction de règles de construction
 - Construction automatique
- Travaux futurs
 - Rendre imprimable les empilements 2D
 - Et en 3D ?
 - Problème inverse ?



Merci de votre attention



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